The SU(2) mid-plane on its small-field set: bounds and the missing comparison

The strongest retained result is a normalized weak-recoupling construction: under the explicit chart, rarity and $L \geq 640$ assumptions of §36, the exact dressed hard-core activities satisfy both bounds in (112). The centered quadratic moving-mean test in §37 proves the subtracted barrier source bound (177). Its Gaussian reference, source-independent coefficients and finite local mean dependence are assumptions; transfer to the actual nonlinear source law (113), physical recoupling to one and stable iteration remain open.

The argument proceeds from inverse decay and local convexity, through the flat-holonomy counterexample and covariant response, to barrier comparison, connected large-field insertions, dressing and sources. The classical value, Gaussian, barriered interacting and weak-strip results have different hypotheses. The unrestricted original normalized comparison is false under Hypothesis I (§5). No continuum construction or physical Yang–Mills gap follows here. All review checks recorded below are repository-internal, not independent peer review or evidence of literature novelty.

Take one halving of direction 1 in $1 + 2$ dimensions, with gauge group $SU(2)$, heat time $t = \lambda_3 a = \hbar g_{\mathrm{cl}}^2 a$, and

$$\varepsilon = t^{1/2-\delta}, \qquad \eta = 2\varepsilon, \qquad 0 < \delta < 1/6.$$

Here ε is the small-field chart radius, distinct from physical $\hbar$, the regulator $\hbar/S_0$, lattice spacing and a record resolution.

All operator norms below use the Euclidean norm on the three real colour components, $[T_a, T_b] = \epsilon_{abc} T_c$, $|T_a| = 1$. Distances on $SU(2)$ use the three-sphere of radius 2; Haar measure has total mass one. Inputs read at passage level: series/parallel, Proposition 1, Hypothesis P(α), Proposition 7; mid-plane order t, §§1 and 7, especially (16)–(18); exact midpoint, Theorem 1 and its image formula. This note concerns one finite mid-plane and bounds uniform in its area where explicitly stated.

1. Setting and the restriction

There are N sites, $2N$ edges and N faces on the periodic mid-plane. Use periods at least three to avoid repeated boundary incidences. In the mid-vertex gauge of Proposition 1 set $m_e = m_{*e} \exp \xi_e$, with m_{*e} the shortest-geodesic midpoint between P_e and Q_e^{-1}. Their distance is $|X_e| \leq \varepsilon$. Up to constants independent of ξ, the exponent is

$$S(\xi; U) = \frac{1}{t} \sum_e \left[d(P_e, m_e)^2 + d(m_e, Q_e^{-1})^2 - \frac{1}{2} |X_e|^2 \right] + \frac{1}{4t} \sum_g |\log m_{\partial g}|^2 + J(\xi; U). \qquad (1)$$

Here J contains the full heat-kernel and Haar amplitudes. Its local ξ derivatives are bounded uniformly on a fixed chart away from the cut locus. Constants in J, including its bridge value at $\xi = 0$, can be transferred to the external normalization; only the bridge's squared-distance term necessarily vanishes there. Count each bridge denominator once, either in S or outside the integral. Include the old-face ratios in the external normalization $B(U)$.

Use the soft-barrier definition:

$$W_\eta(\xi) = \sum_e w_\eta(|\xi_e|), \qquad \mu_U(d\xi) = Z_s(U)^{-1} e^{-S(\xi;U) - W_\eta(\xi)} d\xi, \qquad \mathcal{D}_s = -\log Z_s + B(U). \quad (2)$$

The domain is $\prod_e B(0, c_0)$; w_η is C^2, convex and radial, vanishes for $r \leq \eta$, and diverges at c_0. Choose its boundary decay so that the integrations by parts used below have vanishing

boundary terms and finite Fisher information. It is fixed independently of U in these midpoint coordinates. Its support extends beyond $\Omega_\eta = \{\max_e |\xi_e| \le \eta\}$. A hard restriction to Ω_η defines a different partition function. Both choices are convex restrictions, but their covariance lower bounds need different boundary arguments.

The interacting law includes every mid-face weight. Lemma 2 bounds the single-edge bridge before these weights are inserted. Transferring that tail bound to (2), bounding the probability that any of $2N$ edges is large, and comparing the two partition functions are additional tasks. Even for independent bridges the union bound carries a factor $2N$.

2. Lemma 1: covariant inverse decay

Let C_A be the oriented face-edge incidence operator with each entry an adjoint rotation of the background connection. Set $H_A = 4I + \frac{1}{2}C_A^* C_A$. For every background,

$$4I \le H_A \le 8I. \tag{3}$$

If $K = H_A + E$, $E = E^*$ has range one in the graph of edges sharing a face, and $\|E\| \le \kappa < 4$, then, writing $d = d(e, e')$,

$$\|(K^{-1})_{ee'}\| \le \frac{1}{4-\kappa}\left(\frac{2+\kappa}{6}\right)^d. \tag{4}$$

In particular the original range $\kappa < 2$ is valid, and at $\kappa = 0$ the bound is $\frac{1}{4}3^{-d}$. Here K denotes the perturbed Hessian; K^{-1} denotes its inverse.

Proof. For each face, $|(C_A\xi)_g|^2 \le 4\sum_{e\in\partial g}|\xi_e|^2$. Each edge belongs to two faces, so $\|C_A\xi\|^2 \le 8\|\xi\|^2$. Thus $\|K - 6I\| \le 2 + \kappa < 6$. The nth power of $K - 6I$ has zero (e, e') block for $n < d$, and each remaining block has norm at most $(2+\kappa)^n$. Sum the Neumann series about $6I$ to obtain (4). The count of faces per edge makes every constant independent of N. □

This is a bound on the sum of operator products at a given order. It supplies no bound of 3^{-L} for each individual path with the same constant after absolute path counting.

3. Lemma 2: heat kernel and bridge tail

For $0 < t \le 1$ and $0 \le \theta \le 2\pi$,

$$k_t(\theta) \le C_+ t^{-5/2} e^{-\theta^2/(2t)}; \qquad k_t(\theta) \ge c_- t^{-3/2} e^{-\theta^2/(2t)} \quad (0 \le \theta \le 1). \tag{5}$$

Here are explicit, deliberately loose constants. Put $a_n = 4\pi n$, $b_n = (4n+2)\pi$, and

$$\begin{aligned} A_0 &= \pi\left[1 + 2\sum_{n\ge1}(1 + (a_n+\pi)^2)e^{-(a_n^2 - 2\pi a_n)/2}\right], \\ A_1 &= 2\pi\sum_{n\ge0}(1 + (b_n+\pi)^2)e^{-b_n(b_n-2\pi)/2}, \\ C_+ &= e^{1/8}\sqrt{8\pi}\max(A_0, A_1), \qquad c_- = \sqrt{8\pi}. \end{aligned} \tag{6}$$

Written image check. Write $f_t(u) = ue^{-u^2/(2t)}$. The exact image formula is

$$k_t(\theta) = e^{t/8}\sqrt{8\pi}\, t^{-3/2}\frac{\sum_{w\in\mathbb{Z}} f_t(\theta - 4\pi w)}{\sin(\theta/2)}.$$

On $[0, \pi]$, pair $w = \pm n$. Their sum is $f_t(a_n + \theta) - f_t(a_n - \theta)$, which vanishes at zero. Since $|f_t'(u)| \le (1 + u^2/t)e^{-u^2/(2t)}$, its absolute value is at most

$$2\theta(1 + (a_n+\pi)^2/t)e^{-(a_n-\theta)^2/(2t)}.$$

Use $\theta/\sin(\theta/2) \le \pi$ and $a_n^2 - 2a_n\theta \ge a_n^2 - 2\pi a_n$ to obtain the A_0/t bound on the image quotient after extracting $e^{-\theta^2/(2t)}$.

On $[\pi, 2\pi]$, set $\psi = 2\pi - \theta \in [0, \pi]$. Pair **all** images as $w = -n$ and $w = n + 1$, $n \ge 0$. Each pair is $f_t(b_n - \psi) - f_t(b_n + \psi)$ and has magnitude at most

$$2\psi(1 + (b_n + \pi)^2/t)e^{-(b_n-\psi)^2/(2t)}.$$

Divide by $\sin(\psi/2) \ge \psi/\pi$ and use $(b_n - \psi)^2 - (2\pi - \psi)^2 \ge b_n(b_n - 2\pi)$. This gives A_1/t, including the antipodal limit $\psi = 0$. In particular the leading pair is

$$e^{-(4\pi^2+\psi^2)/(2t)}[(2\pi - \psi)e^{2\pi\psi/t} - (2\pi + \psi)e^{-2\pi\psi/t}],$$

as in the image calculation. Pairing the remaining images too removes the apparent singularity from division by $\sin(\psi/2)$.

For $\theta \le 1$, the absolute remainder relative to $\theta e^{-\theta^2/(2t)}$ is bounded by

$$2\sum_{n\ge1}(1 + (a_n + 1)^2/t)e^{-(a_n^2-2a_n)/(2t)} < \frac{1}{2}.$$

For completeness, $3 < \pi < 4$ bounds this sum by $2\sum_{n\ge1}(1 + 289n^2)e^{-56n^2} < 1/2$; use $t^{-1}e^{-b/t} \le e^{-b}$ for $b \ge 1$, $t \le 1$, followed by $n^2 \le 4^n$, $e^{56} > 2^{56}$ and a geometric series. Together with $\theta/\sin(\theta/2) \ge 2$ this proves the lower bound with (6), also at zero by continuity. □

Bridge conclusion. Assume $|X| \le 1$, $r \ge |X|/2$, and $0 < t \le 1$. For the normalized bridge of duration t,

$$\beta\{d(m, m_*) \ge r\} \le \min\left(1, C_2 t^{-7/2} e^{-2r(r-|X|)/t}\right), \qquad C_2 = 32C_+^2/c_-. \tag{7}$$

Indeed at distance $\rho \ge r$ both endpoint distances are at least $\rho - |X|/2 \ge 0$. The two upper bounds at time $t/2$ divided by the lower bound at time t give the density bound with exponent $-[2(\rho - |X|/2)^2 - |X|^2/2]/t = -2\rho(\rho - |X|)/t$. This exponent decreases with ρ for $\rho \ge |X|/2$; integrate against Haar measure of mass one. At $r = \eta = 2\varepsilon$ and $|X| \le \varepsilon \le 1$ the gain is $e^{-4\varepsilon^2/t}$. This proof uses the two endpoint triangle inequalities through m_*.

4. Lemma 3: local convexity with its support assumptions

The midpoint background face logarithm $Y_g = \log(m_{*\partial g})$ obeys $|Y_g| \le 3\varepsilon$. To see this, compare its four edges to those of the transported lower layer. Each midpoint changes its edge by distance at most $\varepsilon/2$, and the lower-layer plaquette has distance at most ε. Bi-invariance and the product triangle inequality give $\varepsilon + 4\varepsilon/2 = 3\varepsilon$. The bound ε in the original sketch needs an extra premise. Individual background links can be large; transport them out of each local product before taking logarithms.

Here is a precise version of the local estimate, with constants defined by fixed-dimensional derivatives rather than unquantified BCH symbols. Fix $r_0 = 1/16$. On $|b|, |z| \le r_0$ define

$$B(b, z) = d(e, e^{-b/2}e^z)^2 + d(e, e^{b/2}e^z)^2 - |b|^2/2,$$

and on $|y|, |z_i| \le r_0$ define

$$F(y, z_1, \dots, z_4) = \frac{1}{4}\left|\log(e^y e^{z_1} \cdots e^{z_4})\right|^2.$$

All these products lie in a fixed injectivity chart. Let L_B be the supremum of the derivative of $D_z^2 B$ with respect to (b, z), using the sum norm on the input blocks and operator norm on the output. Let L_F be the analogous supremum of the derivative of $D^2_{(z_1,\dots,z_4)}F$. These are finite constants on the indicated compact sets. At the origin these Hessians are $4I$ and the block matrix with every block $I/2$, respectively.

Let J_B, J_F be supremum norms of the edge and face Hessians of the corresponding local heat-kernel/Haar log amplitudes for $0 < t \le 1/2$ on the same charts; omit terms independent of ξ. They are finite: the zero-image Jacobian factors are analytic on these compact sets, and each differentiated nonzero image has a polynomial in $1/t$ times $e^{-c/t}$ with $c > 0$. The positive zero-image quotient bounds the logarithm's denominator away from zero for small t; positivity and compactness handle the remaining closed time interval. Define

$$J_* = J_B + 2J_F, \qquad C_3 = L_B + 8L_F + J_*. \tag{8}$$

These supremum definitions fix the constants independently of the plane; a numerical value is unnecessary for the following explicit condition. For $3\varepsilon \le r_0$, $\rho = |\xi|_\infty \le r_0$, $0 < t \le \min(1/2, \varepsilon)$, the **unbarriered** action satisfies

$$\|t\,\nabla_\xi^2 S - H_A\| \le L_B(\varepsilon + \rho) + 2L_F(3\varepsilon + 4\rho) + tJ_* \le C_3(\varepsilon + \rho). \tag{9}$$

Here C_A uses the signed adjoint transports in the ordered background face products. At each face the four rotations are isometries. Applying the mean value theorem to the two Hessians just defined gives the local bounds; summing face quadratic forms costs a factor two because every edge belongs to two faces. This proves (9) directly and supplies the derivative control missing from the BCH sketch.

Choose $c_0 \le r_0$ with $C_3(\varepsilon + c_0) < 4$, and $2\varepsilon < c_0$. Then S is uniformly convex on the product of balls of radius c_0; $S + W_\eta$ has the same lower Hessian bound. On Ω_η the unbarriered error is at most $3C_3\varepsilon$. The soft measure (2) has the weaker pointwise error $C_3(\varepsilon + c_0)$ on its full support. Its barrier Hessian can be arbitrarily large. These distinctions matter in §5.

5. Failure of the original normalized comparison and the surviving implication

The original Hypothesis I asked for a C^2 path $U(s)$ from 1 to U with $|X_p(s)| \le 2\varepsilon$,

$$\sum_p |\dot X_p(s)|^2 \le C_I \sum_p |X_p(1)|^2, \qquad |\ddot X_p(s)| \le C_I \sum_{p' \sim p} |X_{p'}(1)|^2. \tag{10}$$

A4 overall: REJECT, with a counterexample. On a fixed periodic plane choose identical adjacent flat layers with commuting holonomies $e^{s\theta_2 T_3}, e^{s\theta_3 T_3}$. Constant links $e^{s\theta_j T_3/N_j}$ realize the path. Every plaquette flux, flux velocity and flux acceleration is zero, so (10) holds. Nevertheless, the fixed-volume Laplace calculation in (16)–(17) of the mid-plane note gives

$$\mathcal{D}_s(U_\theta, t) - \mathcal{D}_s(1, t) = \Omega_N(\theta) + o_N(1), \qquad \Omega_N(\theta) > 0 \quad \text{for a nontrivial adjoint holonomy.} \tag{11}$$

The same leading term holds for (2): the barrier equals zero in a ball of radius η, and $\eta/\sqrt{t} = 2t^{-\delta} \to \infty$; rescaling by $\sqrt{t}$ recovers the full Gaussian determinant at fixed N. Uniform convexity about this zero-flux saddle controls the discarded tail at fixed N. The proposed right-hand side $C\varepsilon \sum_p |X_p|^2/t$ is exactly zero. Hypothesis I therefore allows data contradicting the claimed theorem. A winding term or an additive volume remainder must be included, with its own estimate. The actual P(α) allows $Ct^\alpha \sum_p 1$ and a size clause; those features distinguish it from the rejected stronger assertion.

A4(a), linear term: ACCEPT in a smooth local chart at 1. Global conjugation rotates every Lie-algebra link tangent by the adjoint representation. Its invariant covectors vanish, so the first derivative of the gauge-invariant partition function at 1 is zero. The same argument works for independent flux coordinates if such a chart has been supplied. Plaquette logarithms alone omit torus holonomies and satisfy compatibility constraints; writing a global ∇_X requires additional coordinates. Gauge invariance permits the quadratic flat holonomy response (11).

A4(b), differentiation: ACCEPT with its domain hypotheses. For a fixed coordinate domain and a U-independent barrier, differentiating under the finite integral gives, for any boundary coordinates u_i,

$$\partial_i \partial_j \mathcal{D}_s = E[S_{ij}] - \mathrm{Cov}(S_i, S_j) + B_{ij}. \tag{12}$$

Uniform local domination justifies differentiation. A moving hard boundary needs boundary terms, unless it is first represented in a fixed domain. Equation (12) alone provides no local, independent flux coordinates with plane-uniform derivative bounds.

A4(b), covariance upper bound: REFINE. Put $V = S + W_\eta$ and let κ bound the unbarriered Hessian error on the **whole support**. The Brascamp–Lieb inequality gives

$$\mathrm{Cov}_\mu(\xi) \le E[(\nabla^2 V)^{-1}] \le t(H_A - \kappa I)^{-1}. \tag{13}$$

Convex barriers preserve this upper bound. For a hard convex restriction one can obtain it by convex approximation. In (2) the available choice is $\kappa = C_3(\varepsilon + c_0)$; replacing it by $3C_3\varepsilon$ requires controlling the exterior region.

A4(b), covariance lower bound: REJECT as used. With the stated integration-by-parts hypotheses, the score identity and matrix Cauchy–Schwarz give

$$\mathrm{Cov}_\mu(\xi) \ge \left(E[\nabla^2 V]\right)^{-1}, \qquad E[\nabla V \nabla V^T] = E[\nabla^2 V]. \tag{14}$$

The right-hand side contains $E[\nabla^2 W_\eta]$. Its sign increases the Fisher information and decreases this lower bound. Discarding it has the wrong direction for the asserted lower estimate. A tail probability alone bounds neither an unbounded barrier Hessian nor the Fisher information. Hard truncation produces boundary terms; already a one-dimensional centred Gaussian truncated to a finite symmetric interval has variance strictly smaller than its untruncated variance. To recover a two-sided comparison one needs, for example,

$$\|tE[\nabla^2 S] - H_A\| \le \kappa, \qquad t\|E[\nabla^2 W_\eta]\| \le b, \tag{15}$$

in addition to the pointwise lower bound in (13). These give the conditional estimate

$$\|\,\mathrm{Cov}_\mu(\xi) - tH_A^{-1}\| \le \max\left\{\frac{t\kappa}{4(4-\kappa)}, \frac{t(\kappa+b)}{4(4+\kappa+b)}\right\}. \tag{16}$$

It follows by inversion of the Loewner bounds and $H_A \ge 4I$. The original proof established neither (15) with $b = O(\varepsilon)$ nor the required interacting tail estimate. The displayed Brascamp–Lieb inequality is the standard input (Brascamp–Lieb 1976, metadata); (14)–(16) follow by the written score calculation above.

A4(b), covariant-to-abelian comparison: REJECT as justified. The kernel $M_A H_A^{-1} M_A^T$ carries adjoint transport between distinct face frames. Gauge transformations act independently at these frames; comparison with a fixed abelian matrix needs specified identifications. Only traced closed products are gauge invariant by themselves. Even for a closed product, curvature bounds control contractible loops; a winding loop can have nontrivial holonomy with every $X_p = 0$. Finally (4) controls the total operator power, whereas taking absolute values path by path loses its cancellations and introduces path multiplicity. The proposed $\sum_L L^2 3^{-L}$ argument omitted all three issues. Needed: a gauge-covariant bulk kernel comparison with summable spatial moments, and a separate winding remainder.

A4(b), nonlinear score remainder: REFINE. If local derivatives actually satisfy $E|\nabla_\xi \langle v, R\rangle|^2 \le C(\varepsilon+\eta)^2\|v\|^2/t^2$ on a support with curvature error κ, then (13) implies $\mathrm{Var}\langle v, R\rangle \le C(\varepsilon+\eta)^2\|v\|^2/[t(4-\kappa)]$. Cauchy–Schwarz bounds the cross covariance. The Taylor bound on R requires its derivative version and a controlled boundary chart; its value bound alone supplies neither.

A4(c), Taylor step: REFINE. A precise surviving implication is as follows. Suppose that winding has been separated, a genuine C^2 coordinate path $X(s)$ satisfies (10), and $h = \mathcal{D}_s - \mathcal{D}_0 - W_N$ is a function of those coordinates. Assume $X(0) = 0$, $\nabla h(0) = 0$, and along the path

$$\|\nabla^2 h\| \le A\varepsilon/t, \qquad \sup_p |\nabla_p h| \le B\varepsilon/t. \tag{17}$$

Let q_I bound the number of p for which a fixed p' appears in $p' \sim p$. The exact chain rule and Taylor's integral formula give

$$|h(X(1)) - h(0)| \le \frac{C_I}{2}(A + Bq_I)\frac{\varepsilon}{t}\sum_p |X_p(1)|^2. \tag{18}$$

Indeed the Hessian term integrates to at most $AC_I\varepsilon\|X(1)\|^2/(2t)$ and the gradient-acceleration term to at most $BC_Iq_I\varepsilon\|X(1)\|^2/(2t)$. An operator norm estimate alone gives an ℓ^2 gradient bound; it does not imply the per-plaquette bound in (17). A uniformly summable kernel estimate would supply that missing norm. For abelian data the linear flux path is available only in a compatible coordinate sector with the harmonic data fixed. All constants in (18) are explicit in the supplied hypotheses; proving them uniformly in the plane is still part of the problem.

6. Weak comparison: tail arithmetic, winding and size conditions

(a) **Small-field comparison.** The tasks include barrier moment control, a background-covariant local response estimate, independent admissible coordinates or a replacement for the global path, and winding control. The original list of only Hypothesis I and large fields was incomplete. The formal order-t coefficients retain their status as formal coefficients; this review supplies no normalized remainder for them.

(b) **Large fields: the failed naive Peierls arithmetic is accepted.** Using the extra premise $|Y_g| \le \varepsilon$, the face loss at $|\xi_e| \le 2\varepsilon$ is $\exp[-(\varepsilon + 8\varepsilon)^2/(4t)] = e^{-81\varepsilon^2/(4t)}$ per affected face, against bridge gain $e^{-4\varepsilon^2/t}$ per large edge. Since an edge touches two faces, the crude worst-case loss is $81\varepsilon^2/(2t)$ per edge and the net exponent is $+(73/2)\varepsilon^2/t$. The comparison grows rather than decays. With only the coarse small-field assumption, §4 gives $|Y_g| \le 3\varepsilon$; replace 81 by 121 and 73/2 by 113/2. These are bounds on the exponential part; heat-kernel prefactors add powers of t, which cannot repair the wrong sign as $\varepsilon^2/t = t^{-2\delta} \to \infty$. Shared faces can reduce the loss for particular sets; the crude bound suffices to expose failure of the proposed uniform argument. A normalized connected-set or other large-field estimate is still required. The soft barrier also needs comparison with a genuine small-field partition function.

(c) **Logarithmic size: conditional, with the rate corrected.** For flat Gaussian backgrounds the explicit walk count (17) in the mid-plane note proves $0 \le \Omega_N \le (3N/n_*)3^{-n_*}$, $n_* = \min(N_2, N_3)$. More generally, if a winding estimate

$$|W_N| \le Ce^{-\gamma n_*}\sum_p (1 + |X_p|^2/t) \tag{19}$$

has been proved with $\gamma > 0$ independent of N, t, then $n_* \ge (\alpha/\gamma)\log(1/t) + C_0$ suffices for P(α), with the constant multiplied by $e^{-\gamma C_0}$. A decay factor $q = (2 + \kappa)/6$ would give $\gamma = \log(6/(2 + \kappa))$; $\log 3$ is the value at $\kappa = 0$. With fixed positive κ one must use the smaller rate. If $\kappa = O(\varepsilon)$, the $\log 3$ leading coefficient can be recovered with a bounded correction because $\varepsilon\log(1/t) \to 0$. Polynomial prefactors in n_* need absorption into a slightly smaller rate

or a $\log\log(1/t)$ correction. Lemma 1 alone proves (4); a winding estimate for the normalized interacting free energy still requires a separate argument.

7. Quasi-local response and the obstruction to plaquette telescoping

Covariance result. A covariance version of Lemma 1 gives a plane-uniform locality theorem for local perturbations of the potential, under the explicit analytic hypotheses below. Converting this to independent plaquette perturbations needs a further gauge estimate. The natural uniformly bounded local right inverse of curl already fails at the abelian linearization, as (26) proves. Thus this step leaves Hypothesis I unreplaced; it isolates the cancellation a successful replacement must establish.

7.1 Source check and scope

Helffer–Sjöstrand (1994), *On the correlation for Kac-like models in the convex case*, J. Stat. Phys. **74**, 349–409, treats dimension-dependent Gibbs measures that are controlled perturbations of harmonic potentials (reading label: **publisher metadata and abstract**; the full journal text was behind subscription access in this check). Helffer's 1993 author report explicitly presents their joint work (reading label: **passage**, §4, Theorem 4.2, and §5, printed pp. XII-13–17). It states exponential correlation decay for sufficiently small coupling in a nearest-neighbour harmonic model with a C^3 interaction having bounded derivatives through order three. Section 5 explains the elliptic vector equation and the additional weighted estimates needed for spatial decay. This checks a concrete source scope; the general finite-range statement needed here is proved below with its own hypotheses. The author report also addresses thermodynamic limits. No source passage checked here supplies gauge charts, barrier comparison or plaquette telescoping. Bibliography entries: `HelfferSjostrand1994`, `Helffer1993Kac`.

7.2 A covariance theorem with the same decay constants

Internal scope check. ACCEPT. The representation (22) is the Helffer–Sjöstrand formula; $\mathcal{B} \geq 6I$ holds because $L \geq 0$ and the on-site Hessians are nonnegative, and $\mathcal{B}^{-1}$ keeps the edge index; $\mathcal{R}$ is multiplication by a range-one matrix function of norm at most $2+\kappa$; the Neumann tail gives $\frac{1}{6}\sum_{n\geq d} q^n = q^d/(4-\kappa)$, and Cauchy–Schwarz in $L^2(\mu)$ gives (21). The application to the barrier measure is correctly left as a stated approximation condition.

Theorem 5 (local-potential response). Let E be the finite edge graph and $\mu \propto e^{-V} d\xi$ on $\mathbb{R}^{3E}$, with $V = S + \sum_e w_e(\xi_e)$ smooth and confining, w_e convex. Assume the usual closed weighted-gradient realization, so integration by parts and the differentiated Poisson equation hold; smooth potentials with bounded Hessians and a uniform positive lower bound suffice. Allow limits of such measures only when covariances and the gradient norms appearing below converge. Suppose, everywhere on the domain,

$$\|t\nabla^2 S - 6I\| \leq 2+\kappa, \qquad 0 \leq \kappa < 4, \qquad (\nabla^2 S)_{ee'} = 0 \quad \text{if } d(e,e') > 1. \tag{20}$$

For smooth f, g with gradients supported in edge sets R, T, define $a_f = \|\nabla f\|_{L^2(\mu)}$ and $a_g = \|\nabla g\|_{L^2(\mu)}$. Then, uniformly in the graph size,

$$|\operatorname{Cov}_\mu(f,g)| \leq \frac{t}{4-\kappa} q^{d(R,T)} a_f a_g, \qquad q = \frac{2+\kappa}{6} < 1. \tag{21}$$

The on-site convex Hessians can be large; the bound uses their sign, without treating them as a small perturbation.

Proof. Write $L = -\Delta + \nabla V \cdot \nabla$, nonnegative in $L^2(\mu)$. Uniform convexity gives a gap at least $(4-\kappa)/t$ for this auxiliary diffusion. Solve $Lu = g - Eg$ on the orthogonal complement of constants. Differentiating gives

$$(LI + \nabla^2 V)\nabla u = \nabla g, \qquad \mathrm{Cov}(f, g) = \langle \nabla f, (LI + \nabla^2 V)^{-1} \nabla g \rangle_{L^2(\mu)}. \tag{22}$$

The first equality is the commutator identity $\nabla L u = (LI + \nabla^2 V)\nabla u$; the second follows by integration by parts in $\langle f - Ef, Lu \rangle$. This is the covariance representation motivating the Helffer–Sjöstrand method.

On the vector Hilbert space set

$$\mathcal{B} = tLI + 6I + t \operatorname{diag}_e \nabla^2 w_e, \qquad \mathcal{R} = t\nabla^2 S - 6I.$$

The operator $\mathcal{B}$ is block diagonal in the edge **index**, although each scalar diffusion acts on all configuration variables. It satisfies $\mathcal{B} \geq 6I$; $\mathcal{R}$ has range one and norm at most $2 + \kappa$. Therefore

$$t(LI + \nabla^2 V) = \mathcal{B} + \mathcal{R}, \qquad (\mathcal{B} + \mathcal{R})^{-1} = \sum_{n \geq 0} (-\mathcal{B}^{-1}\mathcal{R})^n \mathcal{B}^{-1}.$$

Its block from T to R has zero contributions for $n < d(R, T)$. The remaining operator norm is bounded by $\frac{1}{6} \sum_{n \geq d} q^n = q^d/(4 - \kappa)$. Insert this in (22), with the factor t from the inverse scaling, to prove (21). Smooth approximation and convergence extend the inequality to the stated limits. □

For the local action in §4, (20) holds on its chart with $\kappa = C_3(\varepsilon + c_0)$, independently of the background connection. Application to the singular soft-barrier measure (2) additionally needs a realization or approximation satisfying the analytic hypotheses of Theorem 5 with the same finite-range bounds. Lemma 3 is a chart estimate; constructing that approximation remains an explicit condition here. This avoids importing a whole-space theorem into a bounded chart without specifying its boundary domain. Taking $\kappa \leq 1$ gives the concrete constants $t/3$ and decay 2^{-d} as looser bounds in (21). These rates concern mid-plane fluctuations and their auxiliary diffusion.

7.3 What locality of a free-energy increment means

Suppose $V_{u,v} = V_0 + uh_R + vk_T$, $(u, v) \in [0, 1]^2$, satisfies Theorem 5 uniformly with the same t, κ. Here the gradients of h_R, k_T have supports R, T, and let $a = \sup_{u,v} \|\nabla h_R\|_{L^2(\mu_{u,v})}$, $b = \sup_{u,v} \|\nabla k_T\|_{L^2(\mu_{u,v})}$. For $F(u, v) = -\log \int e^{-V_{u,v}}$, differentiating twice yields $F_{uv} = -\operatorname{Cov}(h_R, k_T)$. Consequently

$$|F(1,1) - F(0,1) - F(1,0) + F(0,0)| \leq \frac{tab}{4 - \kappa}\, q^{d(R,T)}. \tag{23}$$

Thus the increment caused in R depends exponentially weakly on a remote change in T. For a local observable f_T independent of u, the same argument gives

$$|E_1 f_T - E_0 f_T| \leq \frac{t}{4 - \kappa} q^{d(R,T)} \sup_u \|\nabla f_T\|_{L^2(\mu_u)} \sup_u \|\nabla h_R\|_{L^2(\mu_u)}. \tag{24}$$

These statements also hold for nonlinear parameter paths if the mixed potential derivative vanishes for disjoint supports, with h_R, k_T replaced by the parameter scores. Local external normalizations have zero mixed derivative for disjoint supports. In a fixed local link chart, changing coarse links in a region changes only incident bridge and face potentials, after enlarging the region by a fixed number of cells. Subject to the analytic condition after Theorem 5, (23) is therefore the precise localized-increment statement for $\mathcal{D}_s$.

Tails are summable uniformly on these two-dimensional edge graphs. A crude bound on the number of edges at graph distance ℓ from one edge is $2(4\ell + 3)^2$. Thus

$$\sum_{\ell \geq r} 2(4\ell + 3)^2 q^\ell \leq C(q) q^{r/2}, \qquad C(q) = 2 \sum_{\ell \geq 0} (4\ell + 3)^2 q^{\ell/2} < \infty. \tag{25}$$

Summing (23) over remote local scores with bounded gradient norms gives this tail, with their support-size and support-radius constants made explicit in a, b and the distance. The unlocalized increment itself may have size proportional to the changed region; (23) controls its dependence on the distant environment.

7.4 The estimate that fails in the proposed telescoping

Plaquette variables obey compatibility relations. A single plaquette update at fixed surrounding data generally fails those relations. Even in the linearized embedded $U(1)$ sector on an $n \times n$ torus, with harmonic data fixed, infinitesimal link changes a and flux changes x satisfy $Ca = x$ and $\sum_p x_p = 0$. At momentum $k = (2\pi/n, 0)$ the curl symbol of the mid-plane note is $c(k) = (0, e^{2\pi i/n} - 1)$. For a flux mode supported at this momentum, every solution satisfies

$$\|a\|_2 \geq \frac{\|x\|_2}{2\sin(\pi/n)} \geq \frac{n}{2\pi}\|x\|_2. \tag{26}$$

Real sine/cosine modes give the same bound. Hence a right inverse $a = \mathcal{R}x$ on admissible zero-mean fluxes cannot have norm bounded independently of n. An exponentially local kernel $\|\mathcal{R}_{ep}\| \leq Ce^{-\gamma d(e,p)}$ with C, γ independent of n would have bounded row and column sums by (25), and bounded ℓ^2 norm by the Schur test, contradicting (26). Changing two separated opposite plaquette fluxes can instead use a long string of changed links or a spatially spread representative. The supports and score norms in (23) then track that representative.

The precise missing step is a **curvature response estimate after gauge cancellations**, rather than a local right inverse. For example, in specified admissible flux coordinates with harmonic data retained and winding subtracted, one would need the normalized remainder $h = \mathcal{D}_s - \mathcal{D}_0 - W_N$ to have a Hessian kernel K_{pq} satisfying

$$\sup_p \sum_q e^{\gamma d(p,q)}\|K_{pq}\| \leq C\varepsilon/t, \tag{27}$$

together with compatible paths or local replacements whose total quadratic cost is bounded by $C\sum_p |X_p|^2$, and the required zero-flux/winding remainder. Equation (21) controls the link-score covariance. The cancellation of the large inverse-curl factors in the combined expression $E[S_{ij}] - \mathrm{Cov}(S_i, S_j) + B_{ij}$ remains to be proved. The Gaussian case exhibits such cancellations; (26) therefore diagnoses failure of the naive coordinate conversion, without ruling out (27).

With those extra estimates, (18), or a compatible telescoping version of it, and (19) would give the weak P(α) bound on the selected large-plane regime. Theorem 5 and locality alone supply neither the $O(\varepsilon)$ comparison to the Gaussian response nor the winding bound, and Part A's barrier and large-field issues remain. This is the precise obstruction reached in Part B. All claims of P(α) and the normalized expansion (18) of the mid-plane note remain conditional.

8. Cancellation in flux response

Internal scope check. §8.1 ACCEPT, checked in full: Woodbury gives $G = \frac{1}{4}(I - C^*PC)$, hence (28); the slab form (29) vanishes at constant flux and gives $-|x|^2/(4t)$ for opposite layers, both as in Proposition 2; $8I + L = 12I - \mathsf{A}$ and the walk series give the weighted row sum $1/(12 - 4e^{\gamma}) = \frac{1}{6}$ at $\gamma = \log(3/2)$, so (30) reads $\frac{1}{t}(\frac{5}{12} + \frac{1}{4}) = \frac{2}{3t}$. §8.2 ACCEPT as a formal statement: the equal-layer symbol $M(p) = -\lambda/(2(8 + \lambda))$ follows from (29), a flat neutral connection shifts it to $M(p + H)$, and the Coulomb lift $X^3(k)/(e^{i\kappa} - 1)$ of a soft flux produces the $1/\kappa$ residue in (32). The pole is a property of the fixed-frame chart (parallel transport of charged curvature), consistent with the gauge-invariant expectation of §6; it fixes that (27) must be posed covariantly. §8.3 ACCEPT.

8.1 Exact Gaussian reference

Use the unrestricted abelian Gaussian in one slab with boundary links a_0, a_1. Retain their harmonic components h_0, h_1 and impose the linear compatibility relations on all cut and transverse fluxes. Put $x = Ca_0$, $y = Ca_1$, so $\sum x = \sum y = 0$. Independent zero-mean Fourier components and harmonic data give actual coordinates. The spatial kernels below represent bilinear forms on this constrained space, by restriction of an ambient kernel; eliminating one face or inserting a nonlocal orthogonal projector would change its row sums. The three colour copies in the linearization are identical. Set $L = CC^*$, $P = (8I + L)^{-1}$ and $G = H_0^{-1}$. The mean bridge link is $b = (a_0 + a_1)/2$ up to a gradient from direction-1 links. In uncentred mid-links m the normalized exponent and external old-face term are

$$S_0 = \frac{2}{t}\|m - b\|^2 + \frac{1}{4t}\|Cm\|^2, \qquad B_0 = -\frac{\|x\|^2 + \|y\|^2}{8t}.$$

Assign half of each old-face ratio to each slab. Since $\mathrm{Cov}_0(m) = tG$, differentiation in b gives

$$E_0[S_{0,bb}] - \mathrm{Cov}_0(S_{0,b}, S_{0,b}) = \frac{1}{t}(4I - 16G) = \frac{4}{t}C^*PC, \qquad G = \frac{1}{4}(I - C^*PC). \tag{28}$$

For any lift $\mathcal{R}$ with $C\mathcal{R} = I$ on admissible fluxes, $(C\mathcal{R})^*P(C\mathcal{R}) = P$ as a restricted bilinear form. The large factors in (26) disappear. Gaussian integration consequently gives

$$\begin{aligned} \mathcal{D}_{0,\mathrm{slab}} &= \frac{1}{2t}\langle x + y, P(x + y)\rangle - \frac{\|x\|^2 + \|y\|^2}{8t} + \mathrm{const}, \\ K^0_{\sigma p,\tau q} &= \frac{1}{t}\left(P_{pq} - \frac{1}{4}\delta_{\sigma\tau}\delta_{pq}\right) I_3, \qquad \sigma, \tau \in \{0, 1\}. \end{aligned} \tag{29}$$

Cut-flux directions at fixed x, y and all harmonic directions have zero response. Summing slabs reproduces Proposition 2 of the series/parallel note (passage). On the square torus $8I + L = 12I - \mathsf{A}$, where A is nearest-neighbour adjacency. Its positive walk series yields $\sup_p \sum_q e^{\gamma d(p,q)} P_{pq} \le (12 - 4e^\gamma)^{-1}$. For $\gamma = \log(3/2)$ and the pair distance $d((\sigma, p), (\tau, q)) = d(p, q) + |\sigma - \tau|$, (29) therefore gives

$$\sup_{\sigma,p} \sum_{\tau,q} e^{\gamma d}\|K^0_{\sigma p,\tau q}\| \le \frac{1}{t}\left(\frac{1 + e^\gamma}{12 - 4e^\gamma} + \frac{1}{4}\right) = \frac{2}{3t}. \tag{30}$$

Assembly gives $4/(3t)$, independently of the periods. The Gaussian remainder h has zero kernel, hence satisfies (27) with $C = 0$; (30) bounds the reference response itself.

Mechanism. The Gaussian score identity produces the Schur complement (28), equivalently the Woodbury identity for $4I + C^*C/2$. Its two exterior curls annihilate gradients and harmonic shifts and turn every flux lift into the identity. Combining the direct Hessian and covariance before taking absolute values leaves a massive face resolvent. Normalized bridge denominators remove the cut-face response; the old-face normalization adds only the diagonal term in (29). This calculation uses the full Gaussian domain; a barrier requires its own correction.

8.2 Formal non-abelian response: a surviving soft transport term

Specify a chart before testing (27). Use face logarithms in fixed site frames, with the formal Coulomb lift $a = \mathcal{R}X + H + O(X^2)$, $\mathcal{R} = C^*L^{-1}$ on zero-mean modes; retain H and all nonlinear compatibility relations. This is a finite-volume formal chart about zero. Its cubic response already contains an inverse-curl pole. The following test concerns equal adjacent layers, zero cut flux, and $H = 0$.

Use the unrestricted zero-image expansion; barrier and image errors remain outside it. Write the normalized action as $S_0 + V_3 + V_4 + J + \cdots$, with $V_j = \mathcal{S}_j/t$, and expand the BCH logarithm as $Z_1 + Z_2 + Z_3 + \cdots$. Each squared mid-face contributes $\langle Z_1, Z_2\rangle/(2t)$ and $(|Z_2|^2 + 2\langle Z_1, Z_3\rangle)/(4t)$

to V_3, V_4 respectively. The bridge curvature contributes $-\sum_e |[X_e, \xi_e]|^2/(24t)$ to V_4; J includes the normalized Haar and heat-kernel amplitudes. With E_0 at the shifted Gaussian saddle, the first insertions and the required connected cubic pair are

$$\Delta\mathcal{D} = E_0 V_3 + E_0 V_4 - \tfrac{1}{2}\operatorname{Cov}_0(V_3, V_3) + E_0 J + \cdots . \tag{31}$$

One vertex is first order in an insertion parameter; under $[\ ,\ ] \mapsto g[\ ,\ ]$, the cubic is order g, while quartic and the connected pair are both order g^2. Gaussian Wick contraction gives the classical terms $F_3/t, F_4/t$ in (15) of the order-t note (passage), including $-\langle \partial_\xi \mathcal{S}_3, G\partial_\xi \mathcal{S}_3\rangle/(2t)$. Thus a one-vertex quartic truncation alone would miss a necessary term. The complete classical cubic can be tested without listing its vertices. For equal layers, (29) has flux Hessian $M(p)/t$, where $M(p) = -\lambda(p)/(2(8+\lambda(p)))$ and $\lambda(p) = 4 - 2\cos p_2 - 2\cos p_3$. At a constant flat neutral connection $a_j = H_j T_3$, the charged components have the exact classical quadratic symbol $M(p+H)/t$. Indeed replace C by the flat covariant curl in (28); its face Laplacian has symbol $\lambda(p+H)$. This also fixes the sum of **all** cubic transport vertices at zero incoming connection momentum. Massive Gaussian elimination makes these link vertices analytic near that momentum. For a neutral soft flux at $k = (\kappa, 0)$, the Coulomb lift has $a_3^3(k) = X^3(k)/(e^{i\kappa} - 1)$ and $a_2^3(k) = 0$; here $X(z) = \sum_k X(k) e^{ikz}$ defines the Fourier amplitudes. The resulting first-order charged flux Hessian of h, with outgoing momenta $p + k, p$, consequently contains

$$\begin{aligned} K_+^{(1)}(p+k, p) &= \frac{X^3(k)}{t}\left[\frac{\partial_{p_3} M(p)}{e^{i\kappa} - 1} + O(1)\right], \\ \partial_{p_3} M(p) &= -\frac{8\sin p_3}{(8+\lambda(p))^2}. \end{aligned} \tag{32}$$

Choose $p_3 \notin \{0, \pi\}$ and fixed nonzero p. The bounded term includes the other placements of the neutral leg; their inverse curls have momenta near p, and remain bounded. Reality pairs k with $-k$. Equation (32) has size $|X^3(k)|/(t|\kappa|)$ with a nonzero residue. A plane-uniform row-sum bound $C \sup |X|/t$ would bound every such Fourier matrix element and is incompatible with this coefficient as $\kappa \to 0$. This is a formal failure of (27) in this fixed-frame chart. It occurs in the classical cubic, before quartic terms: the latter and the cubic pair have classical boundary degree four, and bridge curvature vanishes on this equal-layer test. Loop contractions have higher powers of t than the displayed $1/t$ residue. Winding subtraction removes wrapping paths; the nonwrapping massive resolvent has the same bulk momentum shift and residue. A flat-holonomy subtraction cannot remove it.

The mechanism is parallel transport of charged curvature between distinct faces. The Gaussian Schur complement cancels the inverse curls on its two external legs; differentiating its transport introduces a third leg $\mathcal{R}X$, retained by (32). Covariantly transported derivatives or another specified flux chart could change this conclusion. Such a replacement needs its own definition and bound; (32) identifies the term that prevents promoting the Gaussian proof in fixed site frames.

8.3 Beyond first order

To pursue (27), first replace the fixed-frame Hessian by a specified covariant response that absorbs (32), and prove the corresponding compatible-path estimate. Then control all connected insertions with plane-uniform weighted sums, the normalized interacting barrier moments, the full winding remainder and the large-field comparison. The Gaussian identity and formal expansion supply no such all-order remainder bound.

9. A tree-transported flux response

A fixed tree gives a gauge-covariant finite-volume chart. Its response retains the path dependence of comparisons between different faces.

9.1 Chart and response on the periodic plane

Write the vertices as $(i,j) \in \mathbb{Z}_{n_2} \times \mathbb{Z}_{n_3}$, $n_2, n_3 \geq 3$, with root $o = (0,0)$. Choose the comb tree: all nonseam direction-2 edges in every row, and the nonseam direction-3 edges in column zero. The path τ_v goes up column zero and then along row j to v. Let $T_v = U(\tau_v)$, using the convention $U_{vw} \mapsto g_v U_{vw} g_w^{-1}$. For a face based at $v(p)$ put

$$\widehat{U}_{vw} = T_v U_{vw} T_w^{-1}, \qquad z_p = \mathrm{Ad}_{T_{v(p)}} \log U_{\partial p}, \qquad \Omega_j = U(\ell_j), \quad j = 2,3. \tag{33}$$

Here ℓ_2, ℓ_3 are the two oriented coordinate cycles through o. Tree links become identity; every displayed variable transforms by the single rotation Ad_{g_o}. Keep both Ω_j as variables. Near identity use $H_j = \log \Omega_j$; together with the fluxes these determine the linear harmonic components (equal to H_j/n_j on flat data). Larger holonomies need separate patches. Winding subtraction is performed after this retention.

The admissible set $\mathcal{M}$ consists of (z, H) produced by the $N+1$ non-tree links. This explicitly retains the nonabelian surface relation between the based face products and the cycle commutator. Locally it is a smooth embedded manifold: at identity its tangent is $\sum_p z_p = 0$ with two arbitrary harmonic variables; curl plus the two periods is injective on tree-gauge links and has dimension $3(N+1)$. The inverse function theorem supplies local reconstruction $U(z,H)$. Use separate copies on the two slab boundaries, join their roots by the fixed direction-1 path, and retain the cut-face data and their compatibility relations when extending to the full slab.

To specify the second derivative as well as the frames, equip each fixed-H admissible slice with the metric induced by $\sum_p |dz_p|^2$. Use (29) in z as the Gaussian subtraction, summed over slabs, and express W_N through the same reconstruction, keeping its holonomy data. Define $K^T = \nabla^2_{\mathcal{M},H}(h \circ U)$, the intrinsic Hessian for this metric, restricting to compatible slab variations. This is an explicit covariant response; its definition includes the derivative of the frames and the curvature of the constraint surface. Kernels are ambient face-block representatives of this bilinear form, as in §8.1, restricted to admissible tangent vectors. A weighted bound means existence of such a representative; eliminating a face changes the matrix norm being asked for. At nonzero fields all expansions below are finite-volume formal jets about $(z,H) = 0$.

9.2 Gaussian persistence and the strip term (Part 2)

Exact Gaussian statement; formal interacting interpretation, unrefereed. At zero field the tree frames are identity, the constraint linearizes to zero total flux, and (28)–(30) apply unchanged as ambient restricted forms: $\gamma = \log(3/2)$, constant $2/(3t)$ per slab, $4/(3t)$ after assembly, and zero Gaussian remainder. A fixed change of face frames conjugates every kernel block by orthogonal matrices and preserves these bounds. Differentiating field-dependent frames needs another step.

For an explicit diagnostic let $Q_{pq} = \mathrm{Ad}_{U_{pq}}$ transport between neighbouring face basepoints along the single edge, and put $\widehat{Q}_{pq} = \mathrm{Ad}_{T_p} Q_{pq} \mathrm{Ad}_{T_q}^{-1}$. Subscripts on T denote basepoints. Define A_T by these four neighbour blocks and $P_T = (12I - \mathsf{A}_T)^{-1}$. Its orthogonal walk weights give exactly the majorant in (30), for every connection, even when $P_T - P_0$ is large. It is the transported massive-resolvent sector suggested by the flat test in §8.2; identifying the full interacting Hessian requires the other cubic terms as well.

At $U(s) = \exp(sa)$ write $\theta_v = \sum_{e \in \tau_v} a_e$ with oriented signs. Then

$$\partial_s \widehat{Q}_{pq}\Big|_0 = \mathrm{ad}_{b_{pq}}, \qquad b_{pq} = \theta_p + a_{pq} - \theta_q, \qquad \dot{P}_T = P_0 \dot{\mathsf{A}}_T P_0. \tag{34}$$

For a contractible comparison loop, discrete Stokes gives $b_{pq} = \sum_{f \in \Sigma_{pq}} \pm z_f^{(1)}$; a wrapping loop also retains its period. Formula (34) absorbs a pure-gauge connection, since $a_{pq} = \theta_q - \theta_p$ then gives zero. In the comb chart, the vertical comparison at (i,j) instead encloses a strip:

$b_3(i,j) = \sum_{r=0}^{i-1} z_{rj}^{(1)}$ for nonseam rows. Thus the soft connection leg of (32) becomes a strip-flux sum.

Here is a written size test with both based cycle holonomies fixed. Let $n_2 = n$ be even, $n_3 \geq 3$, and set $f_i = \varepsilon$ for $0 \leq i < n/2$, $f_i = -\varepsilon$ otherwise. Take $a_2 = 0$, $a_3(i,j) = b_i T_3$, $b_i = \sum_{r<i} f_r$. All tree transports and both based cycles are identity, and the exact commuting face logs are $s f_i T_3$, with zero total flux. A vertical neighbour block of $\dot{\mathsf{A}}_T$ has norm $|b_{n/2}| = n\varepsilon/2$, since $\| \operatorname{ad}_{T_3} \| = 1$. The identities $\|P_0^{-1}\| \leq 16$ and (34) imply

$$\|\dot{P}_T\| \geq \frac{\|\dot{\mathsf{A}}_T\|}{16^2} \geq \frac{n\varepsilon}{512}. \tag{35}$$

This lower bound also applies to its maximal unweighted ambient row sum, by self-adjointness and the Schur bound. Conversely, writing $\|K\|_\gamma = \sup_p \sum_q e^{\gamma d(p,q)} \|K_{pq}\|$, the same example has $\|\dot{\mathsf{A}}_T\|_\gamma \leq n\varepsilon e^\gamma$ and $\|\dot{P}_T\|_\gamma \leq n\varepsilon/24$ at the stated γ. Only the two vertical neighbours contribute. The coefficient is a finite-volume derivative at $s = 0$; its growth persists however small the allowed interval in s becomes.

New obstruction term: $t^{-1} P_0 \dot{\mathsf{A}}_T P_0$ carries strip area although its two outer resolvents have uniform exponential decay. The strips above are contractible and $H = 0$; a subtraction of wrapping paths alone leaves such comparisons. Equation (35) tests this ambient sector, while (27) concerns the full restricted Hessian of h. Cancellation with the remaining cubic vertices and the admissible response, or a different response using short pairwise paths, remains to be established. Consequently this tree chart supplies the Gaussian constants and an explicit new term obstructing the proposed proof of $C\varepsilon/t$; a plane-uniform first-order constant for h remains open. The calculation asserts no failure theorem for every covariant chart.

9.3 Higher orders (Part 3)

After resolving the strip term, (27) still requires summable connected BCH insertions, including the cubic pair in (31), with derivatives of the tree frames and the admissibility constraints included. Short pairwise transports would require their own compatible Hessian and path-cost construction. Uniform barrier moments, holonomy-dependent winding control and the large-field comparison must then transfer the formal coefficients to the normalized h; iteration remains a further step.

10. The complete classical cubic on the strip

Internal scope check. ACCEPT as a formal classical statement. Checked: $D_n \in [1/98, 1/50]$ (numerator mean 2, $10 \leq 12 - 2\cos k \leq 14$); (41) from $M = -\lambda/(2(8+\lambda))$, $dM/d\lambda = -4/(8+\lambda)^2$, $\partial_p \lambda = 2\sin p$; the leading value $-4D_n H/t$ per layer and $H = n\varepsilon/2$. Two scope qualifications. (i) The surviving term is carried by a delocalized tangent: V occupies a whole row and wraps the j-cycle, and $H = n\varepsilon/2$ is the Aharonov–Bohm phase of that cycle's holonomy at the middle row, a gauge-invariant non-local datum (a non-based cycle). The obstruction therefore belongs to the same family as the winding functional, now for cycles that the fixed-cycle slice leaves free. (ii) The exact dependence on H enters through $\cos(p + H)$ and saturates, so the value change along V is at most $O(\min(n\varepsilon, 1))|u|^2/t$ with pointwise amplitude $|u|/\sqrt{n} \leq \varepsilon$: of order $\varepsilon^2 n/t$, against P(α)'s additive term $t^\alpha N \simeq \varepsilon n^2$. Under the size clause this crude count gives no value counterexample, consistent with §10.2's caution.

Work on the equal-layer, zero-cut-flux slice containing §9.2's strip, with both based cycles fixed at identity. The calculation below includes every classical cubic term on this slice, hence tests the full Hessian on compatible equal-layer vectors. The classical coefficient means the stationary-action term of order t^{-1} in the unrestricted zero-image expansion. Barrier, image and loop estimates remain separate from this formal coefficient.

10.1 Elimination, frames and the intrinsic derivative

Use the comb-tree gauge itself, so its transports are identically one throughout the coordinate family. Let a be the boundary link logarithms and u the uncentred mid-link logarithms. For the four oriented link logs l_1, l_2, l_3, l_4 around a face define

$$Q(a)_p = \frac{1}{2}\sum_{r<s}[l_r(a), l_s(a)], \qquad q(c,d) = \frac{Q(c+d) - Q(c) - Q(d)}{2}, \qquad R = I - C^*PC = 4G. \tag{36}$$

Thus $\log U_{\partial p} = (Ca)_p + Q(a)_p + O(a^3)$. Ad-invariance gives $\langle u - a, [a,u]\rangle = 0$: the bridge squared distance has zero cubic term. Gaussian elimination gives $u_0 = Ra$; stationarity annihilates the quadratic action's pairing with the second-order correction to u_0. Including the old-face ratio, the quadratic and cubic stationary actions, with their common factor t^{-1} removed, are

$$F_2 = 2\langle Ca, PCa\rangle - \tfrac{1}{4}\|Ca\|^2, \qquad F_3^{\mathrm{link}} = \tfrac{1}{2}\langle CRa, Q(Ra)\rangle - \tfrac{1}{2}\langle Ca, Q(a)\rangle. \tag{37}$$

The bridge denominator is constant at zero cut flux. Its classical contribution therefore vanishes; Haar and heat-kernel amplitudes start at order t^0. Expressing the Gaussian subtraction in the actual tree flux $z = Ca + Q(a) + O(a^3)$ contributes $4\langle Ca, PQ(a)\rangle - \frac{1}{2}\langle Ca, Q(a)\rangle$. Consequently, for the linear tree lift A with $CAz = z$ on $\sum z = 0$,

$$\boxed{h_3^{\mathrm{cl}}(z) = \frac{4}{t}\langle Pz, Q(RAz) - Q(Az)\rangle.} \tag{38}$$

This subtraction includes the second-order reconstruction of links: its contribution to F_2 cancels the corresponding Gaussian chain-rule term. In another gauge the same cancellation includes derivatives of the tree frames; working in tree gauge makes their value exactly zero. Equation (38) is the full cubic, including the old-face normalization.

At the origin the normal space of the fixed-cycle admissible manifold consists of constant face fields. The equal-layer Gaussian Hessian $M/t = (4P - I/2)/t$ annihilates this space. Thus its second-fundamental-form term and its first-order change under intrinsic parallel transport vanish. The remainder has zero quadratic jet, so its Christoffel term starts at second order in the background. For $z = sf$, $a = Af$ and compatible v, w with $c = Av, d = Aw$, the complete linear-in-s classical Hessian correction is therefore $K_1 = D^2 h_3^{\mathrm{cl}}(f)$:

$$\begin{aligned} tK_1(v,w) = 8\{&\langle Pf, q(Rc, Rd) - q(c,d)\rangle \\ &+ \langle Pv, q(Ra, Rd) - q(a,d)\rangle \\ &+ \langle Pw, q(Ra, Rc) - q(a,c)\rangle\}. \end{aligned} \tag{39}$$

The same coefficient belongs to $\nabla^2\mathcal{D}_s$ after its Gaussian Hessian is removed, using intrinsic parallel identification of tangent spaces. This specifies the derivative independently of a chosen acceleration of a compatible path.

10.2 Proposition 6: a surviving compatible strip coefficient

Formal classical verdict. On the square strip family of §9.2 take $n \in 4\mathbb{N}$, $n \geq 8$, and $m = n/2$. There are compatible charged tangents V of unit norm in the two-boundary metric for which

$$K_1(V,V) = -\frac{n\varepsilon}{t}D_n + \mathcal{E}_n, \qquad D_n = \frac{1}{n}\sum_{r=0}^{n-1}\frac{2 - 2\cos(2\pi r/n)}{(12 - 2\cos(2\pi r/n))^2}, \qquad |\mathcal{E}_n| \leq C_*\varepsilon/t, \tag{40}$$

where $1/98 \leq D_n \leq 1/50$ and, as one deliberately loose explicit choice, $C_* = 10^6[\sum_{\ell\geq 0}(\ell+1)^4 3^{-\ell}]^3$. Thus a term of order $n\varepsilon/t$ survives in the complete intrinsic Hessian. The coefficient refers to differentiation at $s = 0$ for each finite n; the small logarithm chart may shrink with n.

Written proof. Put $e_j = T_1\cos(\pi j/2)+T_2\sin(\pi j/2)$ and choose $c_2 = 0$, $c_3(i,j) = \delta_{i,m}e_j/\sqrt{2n}$, $v = Cc$. Then $\|c\|^2 = 1/2$, $\|v\|^2 = 1$, and each row of v sums to zero. Its tree lift is exactly c. Varying the vertical boundary links by $U_3(i,j;s,u) = \exp(uc_3(i,j))\exp(sb_iT_3)$ and keeping horizontal links equal to one fixes the tree and both based cycles exactly. The induced face logs supply the nonlinear compatibility corrections. At zero, take $V = (v,v)/\sqrt{2}$; intrinsic parallel continuation gives the coefficient (39), independently of the displayed path's acceleration.

Split the neutral link field into $a = h + r$, with $h_3 = (n\varepsilon/2)T_3$, $h_2 = 0$ and $r_3(i,j) = -\varepsilon d_{\mathbb{Z}_n}(i,m)T_3$. This is an algebraic splitting of the cubic (39); the comparison field h supplies a flat symbol while the actual strip retains zero cycles. For a constant neutral link HT_3, the full quadratic elimination in charged face variables replaces $M(k,p)$ by $M(k,p+H)$, exactly as in §8.2: the covariant curl has face Laplacian symbol $2-2\cos k+2-2\cos(p+H)$. Consequently

$$\partial_H M(k,p+H)|_{H=0} = -\frac{8\sin p}{(12-2\cos k-2\cos p)^2}. \tag{41}$$

For $p = \pi/2$, the normalized v has horizontal spectral weights $(2-2\cos k)/(2n)$. Equation (41) gives $-4D_nH/t$ per common-layer unit vector v, and hence $-2D_nH/t$ on V. Substituting $H = n\varepsilon/2$ proves the leading term in (40), including its sign.

The remainder comes from inserting r in (39). Its curl is f, with $\|Pf\|_\infty \le \varepsilon/8$. The other two terms contain one localized factor c or Rc and one factor growing at most as $\varepsilon d(i,m)$. For an explicit summation bound use $\|R_{ee'}\| \le 3^{-d(e,e')}$, $|P_{pq}| \le 3^{-d(p,q)}/8$, at most $2(4\ell+3)^2$ edges in a shell, and $|q(c,d)_p| \le \frac{1}{4}\sum_{r<s}(|l_r(c)||l_s(d)| + |l_r(d)||l_s(c)|)$. Move the distance weight across each kernel using the triangle inequality, then apply Cauchy–Schwarz to the two localized factors. The resulting three kernel moments, four face incidences and the coefficients in (39) are bounded by C_* above, uniformly in n. Finally $10 \le 12-2\cos k \le 14$ and the mean numerator is 2, which prove the stated bounds on D_n. □

The contractible-walk part has the same limiting coefficient: wrapping terms have length at least n and their differentiated massive walk tails are a polynomial in n times 3^{-n}. Thus winding subtraction leaves this obstruction. Equation (40) rules out a plane-uniform $C\varepsilon/t$ bound for the formal first-order comb-tree Hessian, even on admissible vectors. Weak P(α) remains a value estimate; its truth on large planes requires another argument, since a growing Taylor coefficient alone gives neither a uniform finite-field remainder nor a counterexample to that value inequality.

11. Short-path classical value comparison

Short transports give compatible local increments and a uniform covariant classical comparison. Comparing that value to the fixed-frame Gaussian subtraction leaves the explicit transport term (46). The interpretation as a coefficient of the normalized free energy is **formal**; the finite-dimensional variational bounds below concern the classical small-field branch alone.

Internal scope check. Proposition 7 ACCEPT at the level of structure: the contraction for the stationary equation in the ball $\|\xi\|_\infty \le \varepsilon$, with Hessian error $K\varepsilon$ in operator and block row-sum norm and $\nabla^2 E \ge 3I$, yields a unique interior minimum and the plane-uniform bound (43) on an actual finite-dimensional minimum. The open part is exactly the transport term $\mathcal{T}_T$ of (46), currently bounded by $\frac{1}{2}\|z\|^2$; by the remark on §10, it is expected to carry the Aharonov–Bohm phases of cycles left free by the slice. The covariant replacement is defined in §12.

Covariant reference. Proposition 7 bounds the classical value uniformly against the covariant short-path reference; what resists is only the comparison (46) with the abelian fixed-frame reference, which carries the holonomy phases of §10. This suggests stating weak P(α) with a covariant background-field reference $\mathcal{D}_0^{\mathrm{cov}}(U)$, the Gaussian defect of the mid-plane fluctuations in the background connection of U, gauge-invariant and holonomy-dependent, in place of the abelian $\mathcal{D}_0$. Two questions decide whether this is the right formulation: whether $\mathcal{D}_0^{\mathrm{cov}}$ reduces to $\mathcal{D}_0$ plus terms admissible in P(α) on smooth fields (so that the coupling bookkeeping of

the iteration survives), and whether the one-loop and barrier parts obey the same covariant comparison.

11.1 Compatible increments (Part 1)

Keep equal boundary layers U, zero cut flux and the two based cycles fixed. Write $z_p = \log U_{\partial p}$ in its own face frame, $\|z\|_\infty \le \varepsilon$. A dual path of length ℓ avoiding the edges of the fixed cycles gives an exact boundary family: multiply its crossed links by $\exp(sv_e)$, with the signed v_e transported successively along the path. Choose these transports so that the left-trivialized face holonomy velocities cancel at each interior face. The endpoints then have velocities v and $-Q_\gamma^{-1}v$. Differentiating logarithms applies $d\log$ at each endpoint; its norm is at most 2 for $|z| \le 1/16$. Thus the link cost is $\ell|v|^2$, and the flux-velocity cost is at most $8|v|^2$. Both layers use the same family. The untouched cycle links fix both cycles exactly; the actual link products enforce all nonlinear surface relations. Shortest paths are taken in this cut dual graph. Their lengths can grow across the cut. This constructs local transfers; a uniform decomposition of an arbitrary boundary contraction remains a separate requirement.

For the value comparison use compatible **midpoint** increments instead: $m_e = U_e \exp \xi_e$, keeping all boundary links fixed. These increments have unconstrained midpoint cycles, as required by the original integral. Let C_U be the signed rotated incidence obtained by moving each inserted ξ_e to the basepoint of its face product. Then $C_U C_U^* = 4I - \mathsf{A}_U$, with four orthogonal neighbour blocks, and

$$P_U = (8I + C_U C_U^*)^{-1} = (12I - \mathsf{A}_U)^{-1}, \qquad \xi_0 = -C_U^* P_U z, \qquad \|\xi_0\|_\infty \le \varepsilon/4, \quad \|\xi_0\|_2^2 \le \|z\|_2^2/8. \tag{42}$$

Indeed $P_U = \sum_{\ell \ge 0} 12^{-\ell-1} \mathsf{A}_U^\ell$: the absolute row sum is at most $1/8$, each edge meets two faces, and $\|C_U\| \le \sqrt{8}$. Every term transports between successive neighbouring faces; its weight includes the full multiplicity 4^ℓ. The actual links $U_e \exp(s\xi_{0e})$ realize the increment and all its induced face relations, with integrated link cost at most $\|z\|_2^2/8$. The quadratic action $2\|\xi\|^2 + \|z + C_U\xi\|^2/4 - \|z\|^2/4$ is minimized at (42), with value $F_U^{\mathrm{cov}} = 2\langle z, P_U z\rangle - \|z\|^2/4$.

11.2 Proposition 7: value bound and the surviving transport (Part 2)

Put $K = 40(1 + L_F)$, with L_F defined in §4, and assume $\varepsilon \le \min(1/16, 1/(4K))$. Let $F^{\mathrm{cl}}(U)$ be the minimum, on $\|\xi\|_\infty \le \varepsilon$, of the classical action $t(S - J) - \|z\|^2/4$ on this equal-layer slice. Then

$$|F^{\mathrm{cl}}(U) - F_U^{\mathrm{cov}}| \le K\varepsilon \|z\|_2^2. \tag{43}$$

This is a plane-uniform bound for an actual finite-dimensional minimum. Its t^{-1} contribution to $\mathcal{D}_s$ remains a formal classical identification, with barrier and amplitude estimates separate.

Proof. The bridge term is exactly $2\sum_e |\xi_e|^2$. Ad-invariance makes the face gradient at zero exactly $C_U^* z/2$. The local Hessian bound defining L_F, summed over the two faces at each edge, bounds the Hessian error from $4I + C_U^* C_U/2$ by $K\varepsilon$, both in operator norm and in block row-sum norm, throughout this ball. Indeed each face costs at most $L_F(\varepsilon + 4\varepsilon)$ in operator norm; conversion to a four-block row sum costs at most 2, and each edge meets two faces. Also $H_U^{-1} = (I - C_U^* P_U C_U)/4$ has block row sum at most $1/2$. The stationary equation is a contraction $\xi = \xi_0 - H_U^{-1}(\nabla E(\xi) - \nabla E(0) - H_U \xi)$, where $E = t(S - J) - \|z\|^2/4$. It maps the ball into radius $\varepsilon/4 + K\varepsilon^2/2 < \varepsilon$, with Lipschitz constant at most $1/8$. Convexity gives the unique interior minimum ξ_*. Since $\nabla^2 E \ge 3I$ and $\|\nabla E(0)\| \le \sqrt{2}\|z\|$, $\|\xi_*\| \le \sqrt{2}\|z\|/3$. Taylor's formula bounds the value error at either minimizer by $K\varepsilon\|\xi\|^2/2$; evaluating each functional at the other's minimizer proves (43). □

For each unordered face pair choose a shortest dual path σ_{pq}, with $\sigma_{qp} = \sigma_{pq}^{-1}$, and its transport Q_{pq} from C_U. Retain walks $\gamma : p \to q$ with the same lifted displacement as σ_{pq}; their comparison loops are contractible. Denote their resolvent sums by P_U^c and, with every rotation replaced by 1, p_{pq}^c. All remaining walks define $P_U^w = P_U - P_U^c$. Each incidence transport uses at most three physical edges, so a loop $\gamma\sigma_{pq}^{-1}$ for a length-ℓ walk has length at most 16ℓ. Cancelling backtracks and commuting perpendicular steps fills its lift with at most $256\ell^2$ plaquettes, counted with multiplicity. The exact ordered plaquette product and telescoping orthogonal matrices therefore give

$$\|Q_\gamma - Q_{pq}\| \le 256\ell^2\varepsilon, \qquad \sup_p \sum_q \|(P_U^c)_{pq} - p_{pq}^c Q_{pq}\| \le \frac{256\varepsilon}{12} \sum_{\ell\ge 0} \ell^2 3^{-\ell} = 32\varepsilon. \tag{44}$$

This uses enclosed curvature, permits self-intersections, and includes walk multiplicity. Reverse paths give the same column bound. Consequently $F_{\mathrm{short}} = 2\sum_{pq} p_{pq}^c \langle z_p, Q_{pq} z_q\rangle - \|z\|^2/4$ satisfies the explicit quadratic-cost estimate

$$|F^{\mathrm{cl}} - 2\langle z, P_U^w z\rangle - F_{\mathrm{short}}| \le (K+64)\varepsilon\|z\|^2. \tag{45}$$

To compare with §§9–10's Gaussian value, set $I_{pq}^T = \mathrm{Ad}_{T_p}^{-1}\mathrm{Ad}_{T_q}$ and define

$$F_{0,T}^c = 2\sum_{pq} p_{pq}^c \langle z_p, I_{pq}^T z_q\rangle - \|z\|^2/4.$$

The precise surviving term is

$$\begin{aligned} F^{\mathrm{cl}} - 2\langle z, P_U^w z\rangle - F_{0,T}^c &= \mathcal{T}_T + \mathcal{R}, \\ \mathcal{T}_T &= 2\sum_{pq} p_{pq}^c \langle z_p, (Q_{pq} - I_{pq}^T) z_q\rangle, \\ |\mathcal{R}| \le (K+64)\varepsilon\|z\|^2, &\qquad |\mathcal{T}_T| \le \tfrac{1}{2}\|z\|^2. \end{aligned} \tag{46}$$

The last bound uses $\|Q - I\| \le 2$ and row sum $\le 1/8$. Short-path loops in (44) have area controlled by walk length; the tree comparison in (46) can enclose §9's long strips even for neighbours. Equation (46), divided by t, is the verdict for the classical value: the covariant short-path reference has a uniform $O(\varepsilon)$ cost, while the prescribed Gaussian reference retains $\mathcal{T}_T/t$. The subtraction here removes the explicitly defined quadratic winding walks; identifying it with the full normalized W_N requires its separate estimate. An $O(\varepsilon\|z\|^2)$ bound on $\mathcal{T}_T$ for admissible fields, or a value counterexample, remains open; (40) alone decides neither.

11.3 Implication for weak P(α) (Part 3)

Proposition 7 gives the classical value estimate $|F^{\mathrm{cl}}/t - F_U^{\mathrm{cov}}/t| \le Kt^{1/2-\delta}\|z\|^2/t$ on the equal-layer, zero-cut-flux slice, hence the weak P(α) error budget with $\alpha = 1/2 - \delta > 2\delta$ for $0 < \delta < 1/6$ against this covariant reference, with classical coupling shifts zero. Equation (46) leaves $\mathcal{T}_T/t$ against the prescribed fixed-frame reference after the displayed quadratic winding subtraction; its current bound $\|z\|^2/(2t)$ lacks the required small factor. The unresolved value comparison differs from the strip Hessian coefficient: that coefficient alone decides neither the value inequality nor its failure. The covariant reference is defined in §12 and its normalized barriered comparisons in §§13–15. Fixed-frame equivalence, full winding and large-field transfer and stability under perturbed actions remain distinct.

12. A covariant reference and its flux sectors

The covariant replacement gives a classical comparison on the equal-layer, zero-cut-flux slice.

Internal scope check. ACCEPT, with one constant corrected in place. The old-face subtraction in the quadratic action of §11.1, in E of Proposition 7 and in (47) was printed as

$-\frac{1}{2}\|z\|^2$. The old-face ratio of (29) and (37) at $x = y = z$ is $-\frac{1}{4}\|z\|^2$, and only that value gives the stated minimum $F_U^{\mathrm{cov}} = 2\langle z, P_U z\rangle - \frac{1}{4}\|z\|^2$, since the regularized least-squares minimum of $2\|\xi\|^2 + \frac{1}{4}\|z + C_U\xi\|^2$ is $2\langle z, P_U z\rangle$. With $-\frac{1}{2}$, (43) would fail by $\frac{1}{4}\|z\|^2$. The four places now read $-\frac{1}{4}$; the proofs use only gradients and Hessians and stand as written. Checked: (47) by completing the square with Hessian $H_U = \frac{1}{2}(8 + C_U^* C_U)$; (48) from Sylvester's identity $\det(8 + C_U^* C_U) = 8^{3N}\det(8 + L_U)$ with $8 + L_U = 12 - \mathsf{A}_U$, the trace series with $\|\mathsf{A}_U\|/12 \le \frac{1}{3}$, $3 - \operatorname{tr} R = \frac{1}{2}\|R - I\|_F^2$ for a rotation, at most $N4^\ell$ rooted walks and $\sum_\ell \ell^3 3^{-\ell} = \frac{33}{8}$, so $B = 202\,752$; (49) from the row bound $\frac{1}{8}$ and $a = t/\lambda_3$; (50) from $I - P_U L_U = 8P_U$; (51), where the twisted sector $m = 1$ has no adjoint zero mode; the electric projection, normalized by $|\mathbb{Z}_2^2|^{-1} = \frac{1}{4}$ and applied to partition functions; (52) from (43) with $\varepsilon = t^{1/2-\delta}$ and $d_U \le Bt^{1-2\delta}N$, admissible since $\frac{1}{2} - \delta \le 1 - 2\delta$. Two readings for STATE. (i) On the smooth family, any quadratic reference with a bounded kernel obeys a bound of the form (49): $\|z\|^2/t \le M^2 N t^3/\lambda_3^4$ is the classical action of the slab, which lies below the additive term $t^\alpha N$ for $\alpha \le 3$. Smooth-field admissibility thus protects the coupling bookkeeping without selecting $\mathcal{D}_0^{\mathrm{cov}}$. The selection comes from (52), which holds on the whole small-field slice for $\mathcal{D}_0^{\mathrm{cov}}$ and is open for $\mathcal{D}_0$ through $\mathcal{T}_T$. (ii) By (50) the classical defect is $-\frac{1}{32}\|C_U^* z\|^2/t$ at leading order, of dimension six, so the classical step carries no F^2 term, as in the exact tree-level statement of G1; the $O(t)$ coupling shift comes from the determinant d_U, the superrenormalizable pattern of $D = 3$. Correction to my §10 remark: H is the rotation per link, and the row cycle's Aharonov–Bohm phase is $n_3 H$ modulo 2π, as §12.3 states.

12.1 Finite-plane definition

Keep all midpoint links, including their cycles, free as in §11.1. For each background U with principal face logs z, define $L_U = C_U C_U^*$, $H_U = 4I + C_U^* C_U/2$ and $P_U = (8I + L_U)^{-1}$. Normalize the Gaussian integral by its value at the identity background:

$$\begin{aligned} \mathcal{D}_0^{\mathrm{cov}}(U) &= -\log \frac{\int_{\mathbb{R}^{6N}} e^{-E_U^{(2)}(\xi)/t}\, d\xi}{\int_{\mathbb{R}^{6N}} e^{-E_1^{(2)}(\xi)/t}\, d\xi} = \mathcal{D}_{0,\mathrm{cl}}^{\mathrm{cov}}(U) + d_U, \\ E_U^{(2)}(\xi) &= 2\|\xi\|^2 + \tfrac{1}{4}\|z + C_U\xi\|^2 - \tfrac{1}{4}\|z\|^2, \\ \mathcal{D}_{0,\mathrm{cl}}^{\mathrm{cov}} &= F_U^{\mathrm{cov}}/t, \qquad d_U = \tfrac{1}{2}\log\frac{\det H_U}{\det H_1}. \end{aligned} \tag{47}$$

Completing the square proves (47) exactly for this finite Gaussian model. Gauge changes rotate the edge and face spaces orthogonally, preserving both terms. The bound $H_U \ge 4I$ includes harmonic modes and makes the integral finite without a zero-mode deletion. The walk series retains all holonomies, including winding paths. This is a frozen-background quadratic model; the exact classical Hessian, heat-kernel/Haar amplitudes and the barrier supply corrections. The fixed-frame $\mathcal{D}_0$ on this slice means (29) with $x = y = z$ transported to the tree frame and its field-independent constant removed. At identity background the kernels coincide; expanding U and z together recovers its classical quadratic jet. The term d_U is explicitly a Gaussian one-loop contribution.

12.2 Smooth fields and the coupling budget

Finite bounds; continuum derivative interpretation formal. Set $n = \min(n_2, n_3)$ and compare to the full periodic fixed-frame kernel. For a smooth family assume explicitly $\max_p |z_p| \le Ma^2$, with M independent of a, N, and $t = \lambda_3 a \le 1$. The row bound $1/8$ gives $|\mathcal{D}_{0,\mathrm{cl}}^{\mathrm{cov}} - \mathcal{D}_0| \le \|z\|^2/(2t) \le M^2 N t^3/(2\lambda_3^4)$, including tree seams. The determinant lemma and its absolutely convergent trace series give

$$d_U = -\frac{1}{2}\sum_{\ell\ge 1}\frac{\operatorname{Tr}(\mathsf{A}_U^\ell - \mathsf{A}_1^\ell)}{\ell 12^\ell}, \qquad |d_U| \le B\varepsilon^2 N + \tfrac{9}{2}N3^{-n}, \quad B = \frac{3\cdot 256^2}{4}\sum_{\ell\ge 1}\ell^3 3^{-\ell}. \tag{48}$$

Indeed a contractible closed walk has adjoint holonomy R with $\|R-I\| \le 256\ell^2\varepsilon$ by (44), and $3 - \operatorname{tr} R = \|R-I\|_F^2/2 \le 3\|R-I\|^2/2$. There are at most $N4^\ell$ rooted walks. Wrapping walks have $\ell \ge n$; using $|\operatorname{tr}(R-I)| \le 6$ gives the last term in (48). For the smooth family replace ε by Ma^2. Hence

$$|\mathcal{D}_0^{\mathrm{cov}} - \mathcal{D}_0| \le \frac{M^2}{\lambda_3^4}(t^3/2 + Bt^4)N + \tfrac{9}{2}N3^{-n}. \tag{49}$$

Under P(α)'s $n \ge c_0/t$, (49) is an admissible additive $C_{M,\lambda_3,c_0,\alpha}t^\alpha N$ for $0 < \alpha \le 3$; $3^{-n} \le [\alpha/(ec_0\log 3)]^\alpha t^\alpha$ makes the constant explicit. This proves smooth-family admissibility with its stated M dependence. For coupling bookkeeping, spectral calculus also gives exactly

$$F_U^{\mathrm{cov}} = -\tfrac{1}{4}\langle z, L_U(8+L_U)^{-1}z\rangle = -\tfrac{1}{32}\|C_U^* z\|^2 + r_U, \qquad 0 \le r_U \le \tfrac{1}{256}\|L_U z\|^2. \tag{50}$$

For a fixed smooth connection $z = a^2F_{23} + O(a^3)$ and $C_U^* z = a^3(D_3F_{23}, -D_2F_{23}) + O(a^4)$ up to orientations. Thus the classical bulk starts at dimension six, $O(a^2)$ relative to F^2/t; (48)'s contractible determinant starts at curvature squared, allowing $O(t)$ relative coupling shifts. These formal local orders and $\sum_j t_j^\alpha < \infty$ for $t_j = 2^{-j}t_0$ preserve the proposed smooth ultraviolet bookkeeping. The full small-field class still leaves $\mathcal{T}_T/t$ in (46) outside the established remainder bound: (49) uses Ma^2, while that class allows $t^{1/2-\delta}$. Holonomy-dependent transport is the unresolved value term; a value counterexample remains unproved. Stability for perturbed actions remains an extra theorem.

12.3 't Hooft sectors and the free-cycle phases (Part 2b)

Source reading and its finite-plane application. 't Hooft, *A property of electric and magnetic flux in non-Abelian gauge theories*, Nucl. Phys. B **153** (1979), 141–160 (`tHooft1979Flux`): **Crossref metadata verified 2026-09-29; passage**, §§2–5, pp. 143–148, original-paper text transcription. His transition functions close up to centre elements; spatial twists label magnetic flux, while electric flux labels characters of the centre-periodic gauge transformations. Temporal twists select those characters by a finite Fourier transform of partition functions.

Here is the resulting $1+2$ formulation, with direction 1 interpreted as Euclidean time for this sector discussion. Write $m = n_{23} \in \mathbb{Z}_2$, $k = (n_{12}, n_{13}) \in \mathbb{Z}_2^2$. Spatial transition functions obey $\Omega_2(x+L_3)\Omega_3(x) = (-1)^m\Omega_3(x+L_2)\Omega_2(x)$. For each fixed m, construct $C_{U,m}$ using the adjoint transition functions at seams, and use (47) with H_U replaced by $H_{U,m}$. Take z from near-identity plaquette lifts after removing the prescribed seam-centre factors. Keep the common denominator $\det H_1$: sector-dependent constants then remain in relative sector weights. Ordinary periodic SU(2) backgrounds here have $m = 0$; $m = 1$ requires an explicitly twisted boundary problem, or an external centre-flux insertion. Summing magnetic sectors would be a further choice of theory.

For an explicit flat representative choose constant transitions $\Gamma_2\Gamma_3 = (-1)^m\Gamma_3\Gamma_2$. Their commuting adjoint rotations have joint phases (ϕ_2, ϕ_3), and the eigenvalues are

$$L_m(r,s;\phi) = 4 - 2\cos\frac{2\pi r + \phi_2}{n_2} - 2\cos\frac{2\pi s + \phi_3}{n_3}, \qquad P_m = (8+L_m)^{-1}. \tag{51}$$

For $m = 0$, commuting $\Gamma_j = \exp(\theta_j T_3)$ give phases $(0,0)$ and $\pm(\theta_2,\theta_3)$. For $m = 1$, take $\Gamma_2 = i\sigma_1$, $\Gamma_3 = i\sigma_2$ (Pauli matrices): conjugation fixes its own axis and reverses the other two, giving phase pairs $(0,\pi), (\pi,0), (\pi,\pi)$. Formula (51) follows by translating a plane wave once around each cycle. Curved backgrounds use $C_{U,m}$ directly and retain their continuous Wilson-loop data within the sector. In §10's strip, H in $\cos(p+H)$ is the rotation **per link**; the free

row cycle is $\exp(n_3 H T_3)$, with adjoint phase $n_3 H$ modulo 2π. Fixing two based cycles, or fixing $m = 0$, leaves those row holonomies curvature-dependent. Discrete flux labels alone therefore leave the transport term of (46) present.

With a temporal closure the electric-sector prescription is $Z_{e,m} = \frac{1}{4}\sum_{k\in\mathbb{Z}_2^2}(-1)^{e\cdot k} Z_{k,m}$, $e \in \mathbb{Z}_2^2$. Apply it to the assembled reference weights with their temporal seam data retained, before taking a logarithm. Electric flux is a quantum sector of the assembled kernel; a single classical background U specifies neither e nor a Fourier transform of log references. The adjoint local bounds survive either spatial twist; relative electric-sector estimates additionally require control of cancellation in this signed projection.

12.4 Classical comparison and the normalized-scope boundary

Proposition 7 gives, for $\alpha = 1/2 - \delta > 2\delta$, $0 < \delta < 1/6$,

$$|F^{\mathrm{cl}}/t - \mathcal{D}^{\mathrm{cov}}_{0,\mathrm{cl}}| \le K t^{\alpha} \|z\|^2/t. \tag{52}$$

This proves classical weak $\mathrm{P}(\alpha)$ on the stated equal-layer slice, uniformly in plane size; (48) also permits the full reference (47) in (52) after adding $Bt^{1-2\delta}N + \frac{9}{2}N3^{-n}$, admissible under the size clause. Classical coupling shifts may be zero. Identification with the normalized free energy remains **formal**. The Gaussian determinant bound settles only that frozen model's one-loop term; the actual saddle Hessian, Haar/heat-kernel amplitudes, normalized loop remainder, barrier moments, image/large-field comparison and general boundary layers remain. The normalized barriered extension is treated in §13; unequal layers and response/dressing require the subsequent estimates. Equivalence to the original fixed-frame $\mathrm{P}(\alpha)$ on the full small-field set still requires the transport-value estimate, and electric-sector projection requires its own relative bound.

13. The normalized covariant comparison

Throughout this section the layers are equal, cut flux is zero, and all free-cycle holonomies remain in C_U. Put $\alpha = 1/2 - \delta$, $0 < \delta < 1/6$, $\varepsilon = t^{\alpha}$, and impose Proposition 7's smallness conditions.

Internal scope check. ACCEPT (53)–(58) as finite-plane written proofs under the stated chart and time conditions. Checked: (53) from the block-ℓ^1 contraction with $\|H_U^{-1}\|_{1\to 1} \le \frac{1}{2}$ and $\|\xi_0\|_{1,b} \le \sum_p |z_p|$; (54) from rank 12 per face, $\sum_p \sum_{e\in\partial p} |\xi_{*e}| = 2\|\xi_*\|_{1,b}$ and $\frac{1}{2}\cdot\frac{1}{3}\cdot 12 L_F(1+\frac{16}{7}) = \frac{46}{7}L_F$; the moment bound (55) by integration by parts against $|y_e|^{k-2}y_e$, whose divergence in $\mathbb{R}^3$ is $(k+1)|y_e|^{k-2}$, with the barrier term nonnegative because $|\xi_{*e}| \le 2\varepsilon/7 < \eta$, and diagonal dominance $(5-\kappa_r)-(3+\kappa_r) = 2-2\kappa_r \ge \frac{3}{2}$ at the maximizing edge; the cubic Taylor bound with 16 products per face; Šidák's rectangle, which stays in $|\xi_e| \le 9\varepsilon/7 < \eta = 2\varepsilon$, with coordinate variances at most $t/3.75$; the assembly of C_0 from (43), (54), (56) and $e^{-t^{-2\delta}/2} \le B_\delta t^{\alpha}$; (57) with the counts 2 and 3 and $\frac{38}{7}$; and the $k = 2$ bound $M_2^2 \le 3t + 3D_J t M_2$ with net dominance at least 1 when $\kappa_r, 9tD_J \le \frac{1}{4}$. The condition $\kappa_r \le \frac{1}{4}$ shrinks the chart to $c_0 \lesssim 1/(32L_F)$, which moves more of the integral into the large-field comparison that §13 leaves open. One remark for the iteration: the remainder $2A_L N\sqrt{t}$ in (56) is additive and depends on the whole field through the interpolated measure, including flat holonomies. The stated form of $\mathrm{P}(\alpha)$ admits it. A multi-step argument will need it as a sum of quasi-local terms; the uniform moments (55) and Lemma 1's covariant decay are the ingredients for that rewriting.

13.1 Actual saddle determinant (Part 1)

Write $A_U = \nabla^2 E_U(\xi_*)$, $\Delta_U = A_U - H_U$ and $\|\xi\|_{1,b} = \sum_e |\xi_e|$. The contraction proof of Proposition 7 also works in block ℓ^1: symmetry gives the column bounds from its row bounds, and $\|C_U^* z/2\|_{1,b} \le 2\sum_p |z_p|$. Hence

$$\|\xi_*\|_{1,b} \le \frac{\sum_p |z_p|}{1 - K\varepsilon/2} \le \frac{8}{7} \sum_p |z_p|. \tag{53}$$

Each face Hessian difference has rank at most 12 and operator norm at most $L_F(|z_p| + \sum_{e \in \partial p} |\xi_{*e}|)$, by §4's mean-value bound with the orthogonal incidence maps held fixed. The bridge Hessian is exactly $4I$. Thus the trace norm satisfies $\|\Delta_U\|_{\mathrm{tr}} \le 12 L_F(\sum_p |z_p| + 2\|\xi_*\|_{1,b})$. Every $H_U + s\Delta_U \ge 3I$, $0 \le s \le 1$, so integrating $\frac{1}{2}\operatorname{Tr}[(H_U + s\Delta_U)^{-1}\Delta_U]$ proves

$$\left| \frac{1}{2} \log \frac{\det A_U}{\det A_1} - d_U \right| \le C_D \sum_p |z_p|, \qquad C_D = \frac{46}{7} L_F. \tag{54}$$

Here $A_1 = H_1$ exactly. More generally $z = 0$ gives $\xi_* = 0$ and $A_U = H_U$, even with nontrivial flat holonomies. This is a local field-dependent bound; the trace norm replaces any need to discard transports in the resolvent. Both (53) and (54) are uniform in N. For $t \le 1$ and $\alpha \le 1/2$, $|z_p| \le (t^\alpha + |z_p|^2/t^\alpha)/2 \le t^\alpha(1 + |z_p|^2/t)/2$, giving weak $\mathrm{P}(\alpha)$ with constant $C_D/2$. The coarser bound $6NK\varepsilon/4$ also follows from $\|\Delta_U\| \le K\varepsilon \le 1/4$ and $|\log(1+x)| \le |x|/(1-|x|)$; it is admissible in the additive budget, while (54) additionally vanishes at zero curvature.

13.2 Laplace remainder and the soft barrier (Part 2)

Choose $2\varepsilon < c_0 \le 1/16$ still smaller if necessary so that $\kappa_r = 8L_F(\varepsilon + 4c_0) \le 1/4$. Define $y = \xi - \xi_*$, $S_2 = (E_U(\xi_*) + y^T A_U y/2)/t$, $R = E_U/t - S_2$, and $V_\lambda = S_2 + \lambda R + W_\eta$, $0 \le \lambda \le 1$. The partition function $Z_E = \int e^{-E_U/t - W_\eta}$ equals $e^{\|z\|^2/(4t)} Z_s^0$, where Z_s^0 is (2) with J removed. Thus its normalization includes precisely the classical old-face ratio. On the whole chart, the unbarriered Hessians have diagonal blocks $\ge (5 - \kappa_r)I/t$, off-diagonal block majorants summing to $(3 + \kappa_r)/t$, and lower operator bound $(4 - \kappa_r)I/t$. Indeed H_U has diagonal $5I$ and six off-diagonal incidences of norm $1/2$; the eight local Hessian blocks per row give the stated error majorant. Convex interpolation preserves these bounds. The fixed convex barrier preserves the lower bound, hence (13), along the entire path.

Here is the additional saddle-centering argument. Proposition 7 gives $|\xi_*|_\infty \le 2\varepsilon/7 < \eta$, so $y_e \cdot \nabla w_\eta(\xi_e) \ge 0$. Integration by parts against $|y_e|^{k-2} y_e$, followed by Hölder at an edge maximizing $M_k = (\langle |y_e|^k \rangle_\lambda)^{1/k}$, gives $(2 - 2\kappa_r) M_k^k \le t(k+1) M_k^{k-2}$. Thus, for $k \ge 2$,

$$\sup_{\lambda, e} \langle |y_e|^k \rangle_\lambda \le M_k^{\mathrm{bd}} t^{k/2}, \qquad M_k^{\mathrm{bd}} = [2(k+1)/3]^{k/2}. \tag{55}$$

The barrier's boundary decay in (2) justifies these integrations; regularizing $|y_e|^{k-2} y_e$ at zero gives the same inequality. In particular the probability of $|\xi_e| \ge r > \varepsilon$ is at most $M_k^{\mathrm{bd}} t^{k/2}/(r - \varepsilon)^k$, including layers near c_0. This controls ordinary local moments; expectations of the unbounded barrier Hessian require separate hypotheses, and are unused here.

Taylor's integral remainder gives $|R| \le L_F \sum_p (\sum_{e \in \partial p} |y_e|)(\sum_{e \in \partial p} |y_e|^2)/(6t)$. There are 16 products per face, each bounded in expectation by $M_3^{\mathrm{bd}} t^{3/2}$ from (55), with $M_3^{\mathrm{bd}} = (8/3)^{3/2}$. Put $A_L = (8/3) L_F (8/3)^{3/2}$. Integrating $\partial_\lambda \log Z_\lambda = -\langle R \rangle_\lambda$ yields $|\log Z_E - \log Z_{2,W}| \le A_L N \sqrt{t}$. Expansion about the actual Hessian absorbs the ε-quadratic term into (54); the remaining Taylor error starts cubically.

Let $G_U = \int_{\mathbb{R}^{6N}} e^{-S_2}$ and $q = 2e^{-\varepsilon^2/(2t)} \le 1/2$. The centered Gaussian has scalar variances at most $t/3$. The rectangle $|y_{e,a}| \le \varepsilon/\sqrt{3}$ lies inside $W_\eta = 0$. Šidák's Gaussian rectangle inequality gives $Z_{2,W}/G_U \ge (1 - q)^{6N}$, while $Z_{2,W}/G_U \le 1$. Input: Šidák 1967, Corollary 1, p. 628 (**passage**, original paper; Crossref metadata verified 2026-09-29). Consequently the identity-normalized remainder obeys

$$\left| \log \frac{Z_E(U)}{G_U} - \log \frac{Z_E(1)}{G_1} \right| \le 2A_L N \sqrt{t} + 48 N e^{-t^{-2\delta}/2}. \tag{56}$$

This includes the barrier cost for every barrier satisfying (2), regardless of its growth near c_0. Put $B_\delta = (\alpha/(\delta e))^{\alpha/(2\delta)}$; maximizing $x^{\alpha/(2\delta)}e^{-x/2}$ gives $e^{-t^{-2\delta}/2} \le B_\delta t^\alpha$. Combining (43), (54) and (56) proves the normalized, amplitude-free bound $|-\log[Z_E(U)/Z_E(1)] - \mathcal{D}_0^{\mathrm{cov}}(U)| \le C_0 t^\alpha \sum_p (1+|z_p|^2/t)$, with $C_0 = K + C_D/2 + 2A_L + 48B_\delta$. The normalized field dependence is controlled by the allowed additive term; (56) permits a flat-holonomy remainder as well as curvature.

13.3 Haar/heat-kernel amplitudes (Part 3)

Include the external old-face amplitude once: write $\mathcal{J} = J + B_{\mathrm{amp}}$ as one local edge term, one mid-face term and one old-face term per cell, with background-independent constants removed. Define $D_J \ge 1$ as the maximum of their first block derivatives (in z, ξ_i) and second ξ-block derivatives on §4's compact charts, uniformly for $0 < t \le 1/2$. Section 4's analytic amplitude/image argument makes this supremum finite. At $z = \xi = 0$ all local values equal their identity-background values, regardless of the transports. Incidence counting and (53) give

$$|\mathcal{J}(U,\xi_*) - \mathcal{J}(1,0)| \le D_J(2\sum_p |z_p| + 3\|\xi_*\|_{1,b}) \le \frac{38 D_J}{7}\sum_p |z_p|. \tag{57}$$

For interpolation by $s\mathcal{J}$, impose also $9tD_J \le 1/4$. Its Hessian block row majorant is $9D_J$, and its saddle score is at most $3D_J$ per edge. The proof of (55), now with $k = 2$, gives $M_2^2 \le 3t + 3D_J t M_2$, hence $M_2 \le (\sqrt{3}+3D_J)\sqrt{t}$ for $t \le 1$. Thus the interpolated potential stays convex, and $|\langle \mathcal{J} - \mathcal{J}(U,\xi_*)\rangle_s| \le 6D_J(\sqrt{3}+3D_J)N\sqrt{t}$. Integrating in s, identity-normalizing, and using (57) proves, with $Z_E^J = \int e^{-E_U/t - W_\eta - \mathcal{J}}$,

$$\left| -\log \frac{Z_E^J(U)}{Z_E^J(1)} - \mathcal{D}_0^{\mathrm{cov}}(U) \right| \le C t^\alpha \sum_p (1+|z_p|^2/t), \qquad C = C_0 + \frac{19 D_J}{7} + 12 D_J(\sqrt{3}+3D_J). \tag{58}$$

14. Unequal layers and a quasi-local remainder

The cut-face identity is $P_e Q_e = U_{f_e}$ (series/parallel, Proposition 1, **passage**). Consequently zero cut flux means $P_e = Q_e^{-1}$: after transporting both layers to the mid-vertex frames, they coincide. Unequal layers in those frames require cut flux. We treat $|X_e| \le \varepsilon$ below, and recover the requested zero-cut-flux case by setting $X = 0$.

Internal scope check. ACCEPT (59)–(62) and (63); ACCEPT (64) as the conditional statement it is. The opening observation fixes the coordinate convention: in mid-vertex frames the equal-layer slice *is* the zero-cut-flux slice, so unequal layers carry temporal plaquettes, and (62) rightly budgets them with $\sum_e (1+|X_e|^2/t)$. Checked: (59) against (29), with minimum $2\langle w, Pw\rangle - S/8$ and value $2\langle z, Pz\rangle - \frac{1}{4}\|z\|^2$ at $x = y = z$; the bridge Hessian $4\Pi_b + 4h_b\Pi_b^\perp$ from $SU(2) \cong S^3$ of radius 2, with deficit $4(1-h_b) \approx |b|^2/12$; $|a_p| \le c_M \sum |b_e|^2$ from the second difference of the face-log map and Cauchy–Schwarz over four edges; $\|a\|^2 \le 8c_M^2\varepsilon^2 T$ with each edge in two faces; (60) from $2\langle a, P(2w+a)\rangle$ and $\|a\|\|w\| \le c_M\varepsilon(S+T)$; (61) with bridge trace norm $3\beta T$ against the lower bound $3I$; the rectangle inside $|\xi_e| \le 13\varepsilon/7 < 2\varepsilon$; the $31/7$ count in the amplitude bound; and in (64) the constants of (21) at $\kappa = \frac{1}{2}$, $t/(4-\kappa) = 2t/7$, $q = 5/12$, with (20) satisfied because the spectrum of H lies in $[4,8]$. One condition is missing from the list: $|Y_p| \le 3\varepsilon$ uses $|a_p| \le 4c_M\varepsilon^2 \le 2\varepsilon$, so add $2c_M\varepsilon \le 1$. On Part 2: (63) already makes the remainder face-local in size, $|\ell_p| \le 2A_u t^\alpha$ within each face's budget; what stays open is the locality of its *dependence* on U, which an effective action for the next step needs. The three premises named at the end of §14.2 are the precise gap, and the Gaussian barrier cost $b(U)$, of size $O(Ne^{-t^{-2\delta}/2})$, looks the easiest of the three: the same interpolation applied to sW_η with (21)'s decay should localize it.

14.1 Unequal layers: reference and comparison (Part 1)

Write the endpoints as $m_{*e}\exp(\pm b_e/2)$, $|b_e| = |X_e|$, and put $Y_p = \log m_{*\partial p}$, $w = (x+y)/2$, $a = Y - w$. Here x, y are the two transverse face logs in their common basepoint frames; assume $|x_p|, |y_p|, |b_e| \le \varepsilon$. Keep all midpoint cycles free. With $C = C_{m_*}$, $P = (8 + CC^*)^{-1}$, $H = 4 + C^*C/2$, the covariantization of (29), including its determinant, is

$$\begin{aligned} E_w^{(2)}(\xi) &= 2\|\xi\|^2 + \tfrac{1}{4}\|w + C\xi\|^2 - \tfrac{1}{8}(\|x\|^2 + \|y\|^2), \\ \mathcal{D}_{xy}^{\mathrm{cov}} &= \tfrac{1}{2t}\langle x+y, P(x+y)\rangle - \tfrac{1}{8t}(\|x\|^2 + \|y\|^2) + \tfrac{1}{2}\log(\det H/\det H_1). \end{aligned} \tag{59}$$

The actual energy is $E = \sum_e B(b_e, \xi_e) + \sum_p F(Y_p, O_{pe}\xi_e) - (\|x\|^2 + \|y\|^2)/8$, with the four signed rotations of §4. Define β as half the maximum of the suprema of $\|D_b^2 D_\xi^2 B\|, \|D_b^2 D_\xi^3 B\|$ on §4's compact chart. Evenness in b and $B(0,\xi) = 2|\xi|^2$ give $\|D_\xi^2 B - 4I\|, \|D_\xi^3 B\| \le \beta|b|^2$ there. At zero the exact Hessian is $4\Pi_b + 4h_b\Pi_b^\perp$, $h_b = (|b|/4)\cot(|b|/4)$, with the continuous value at $b = 0$; the bridge gradient vanishes. Thus $\nabla E(0) = C^*Y/2$ exactly.

Let M be the supremum of the second derivative of the local face-log map in its four insertion blocks, using their sum norm, on the same chart; set $c_M = M/2$. Taylor expansion of the two endpoint products at insertion $\pm b/2$ cancels the linear terms and gives $|a_p| \le c_M \sum_{e\in\partial p} |b_e|^2$. Consequently, writing $S = \|x\|^2 + \|y\|^2$ and $T = \sum_e |b_e|^2$, $\|a\|^2 \le 8c_M^2\varepsilon^2 T$ and $\|Y\|^2 \le S + 16c_M^2\varepsilon^2 T$; also $|Y_p| \le 3\varepsilon$.

Put $K_u = \beta + 56L_F + 1$ and impose $3\varepsilon \le 1/16$, $K_u\varepsilon \le 1/4$, $2\varepsilon < c_0 \le 1/16$ and $\kappa_u = \beta\varepsilon^2 + 8L_F(3\varepsilon + 4c_0) \le 1/4$. The proof of (43), with Y replacing z, has Hessian row error $\le K_u\varepsilon$ on the radius-ε ball and initial iterate $\|\xi_0\|_\infty \le 3\varepsilon/4$. Its contraction gives $\|\xi_*\|_\infty \le 6\varepsilon/7$, $\|\xi_*\|_{1,b} \le (8/7)\sum_p |Y_p|$ and $\|\xi_*\|_2 \le \sqrt{2}\|Y\|_2/3$. The classical comparison is

$$|\min E - \min E_w^{(2)}| \le C_{\mathrm{cl}}\varepsilon(S+T), \qquad C_{\mathrm{cl}} = K_u(1 + 16c_M^2) + c_M + 2c_M^2. \tag{60}$$

Indeed comparison first to $E_Y^{(2)}$ costs $K_u\varepsilon\|Y\|^2$; the remaining difference is $2\langle Y, PY\rangle - 2\langle w, Pw\rangle$, bounded by $\|w\|\|a\|/2 + \|a\|^2/4$ since $\|P\| \le 1/8$. For $A = \nabla^2 E(\xi_*)$, (54)'s trace proof now gives

$$\left|\tfrac{1}{2}\log\frac{\det A}{\det A_1} - \tfrac{1}{2}\log\frac{\det H}{\det H_1}\right| \le C_D\sum_p |Y_p| + \tfrac{\beta}{2}T, \qquad C_D = 46L_F/7. \tag{61}$$

The bridge contributes trace norm $3\beta T$; all interpolating Hessians exceed $3I$. The moments (55) survive with κ_u. The Gaussian rectangle still lies in the barrier-free region because $6\varepsilon/7 + \varepsilon < 2\varepsilon$. Equation (56) holds with A_L replaced by $A_u = A_L + (\beta/3)(8/3)^{3/2}$, accounting for the cubic bridge remainder on $2N$ edges. Enlarge $D_J \ge 1$ to include the first derivatives in b, Y, x, y and the same second insertion derivatives of the local amplitudes. Then (57)'s right side becomes $D_J[\sum_e |b_e| + \sum_p(|x_p| + |y_p|) + (31/7)\sum_p |Y_p|]$. With $9tD_J \le 1/4$ the centered amplitude cost remains $12D_J(\sqrt{3} + 3D_J)N\sqrt{t}$ after identity normalization.

Thus, for $\alpha = 1/2 - \delta$, $0 < \delta < 1/6$, $t \le 1/2$ and $2e^{-t^{-2\delta}/2} \le 1/2$, the full barriered normalized bound is

$$\left|-\log\frac{Z_E^J(U)}{Z_E^J(1)} - \mathcal{D}_{xy}^{\mathrm{cov}}\right| \le C_u t^\alpha\left[\sum_p\left(1 + \frac{|x_p|^2 + |y_p|^2}{t}\right) + \sum_e\left(1 + \frac{|X_e|^2}{t}\right)\right], \tag{62}$$

where an explicit loose choice is $C_u = 1000(1 + C_{\mathrm{cl}} + C_D + \beta + c_M + c_M^2 + A_u + B_\delta + D_J + D_J^2)(1 + c_M)$. Use $\sum|Y| \le \frac{1}{2}\sum(|x| + |y|) + 2c_M T$ and §13's scalar Young bound to assemble it. This is weak $\mathrm{P}(\alpha)$ including the cut faces, with zero extra coupling shifts. Setting $X = 0$ gives $x = y = Y$ and (58) (the extra $2N$ is absorbed in its constant). For genuinely unequal layers, the transverse-only target additionally needs control of the explicit midpoint term $[2\langle Y, PY\rangle - 2\langle w, Pw\rangle]/t$ and the cut contributions in (60)–(61) by that smaller budget.

14.2 Local densities and the remaining decay hypothesis (Part 2)

Exact decomposition; spatial response conditional. Taylor-expand each face and bridge term at the actual saddle, assigning half of each edge remainder to each adjacent face. This gives $R = \sum_p R_p$ exactly. With $\mu_{\lambda,U}$ from §13.2 define

$$\ell_p(U) = -\int_0^1 [\langle R_p\rangle_{\lambda,U} - \langle R_p\rangle_{\lambda,1}]\, d\lambda, \qquad |\ell_p(U)| \le 2A_u\sqrt{t} \le 2A_u t^\alpha. \tag{63}$$

Equation (55) proves this face by face with the same 16 cubic products as before; half-edge assignment adds $\beta(8/3)^{3/2}/3$ to the constant. Consequently $\sum_p \ell_p$ is precisely the identity-normalized $\log(Z_E/Z_{2,W})$. Assign amplitudes to faces in the same way and put $Q_p = \mathcal{J}_p(\xi) - \mathcal{J}_p(\xi_*)$. Integrating their expectations along the $s\mathcal{J}$ interpolation gives densities a_p for the centered amplitude log-ratio, with $|a_p| \le 12D_J(\sqrt{3}+3D_J)\sqrt{t}$. These bounds imply the requested value bound with $1 + |z_p|^2/t$ on the equal-layer slice, even at $z_p = 0$. Their dependence on U still comes through the full measure and saddle.

Here is the precise decay supplied by Theorem 5 **if its soft-barrier realization condition holds uniformly along these interpolations**. For a potential perturbation θk_T supported in T, hold the local Taylor coefficients and saddle fixed, and retain the convexity bounds and, for R, (55)'s fourth moments throughout the perturbation. Set $b_T = \sup \|\nabla k_T\|_{L^2}$ and $d = d(\partial p, T)$. The derivative Taylor bound and (55) with $k = 4$ give $\|\nabla R_p\|_{L^2} \le A_R = (320/3)(L_F + \beta)$; incidence counting gives $\|\nabla Q_p\|_{L^2} \le A_J = 6D_J$. Using $\kappa \le 1/2$ (including amplitudes), (21) proves

$$|\partial_\theta\langle f_p\rangle| = |\operatorname{Cov}(f_p, k_T)| \le \tfrac{2t}{7} A_f b_T e^{-\gamma d}, \qquad \gamma = \log(12/5), \quad (f, A_f) = (R, A_R) \text{ or } (Q, A_J). \tag{64}$$

The uniformity in λ, s permits integration; (25) sums remote scores. For actual boundary changes the exact response also contains $\langle\partial_\theta f_p\rangle$ and the score $\partial_\theta V$. Their quasi-local bounds require compatible boundary coordinates: $\partial_\theta\xi_* = -A^{-1}\partial_\theta\nabla E(\xi_*)$ has Lemma 1 decay for a local link score, whereas independent face-log variations require the additional transport control identified in §7.4. Finally (56) also contains $b(U) - b(1)$, where $b(U) = \log(Z_{2,W}/G_U)$. Its rectangle bound controls only the total cost. Completing the requested quasi-local remainder requires a uniform soft-barrier realization of (22), the boundary-score bounds just specified, and local densities for this Gaussian barrier cost with summable decay and size $O(t^\alpha)$. Equations (63)–(64) establish the local small densities and the conditional response estimate; the full quasi-local assertion retains these premises.

15. The barrier realization and local barrier cost

The following realization supplies §14.2's premises (i) and (iii). With (i), the localized-increment bounds (23)–(24) of §7.3 hold for the barriered measures themselves, which reduces the quasi-local remainder to one kernel comparison (§15.3). Notation and conditions are those of §§13–14; $\Lambda = L_F + \beta$.

Internal scope check. REFINE the extension: block-row counting gives the sufficient constant 14000 in (65), replacing 13000; the amplitude constant 770 survives. **REFINE** the approximation: mollification preserves convexity and convergence, with equality to the barrier required only in the limit. **ACCEPT** (66), (55) for partially barriered Gaussians, and (67) with $k_t \ge 2$. **REFINE** the decay claim after (67): it requires boundary-score control for the saddle Gaussian and a specified barrier profile. **REFINE** §15.3: use enlarged score supports and the Gaussian of (59), whose parameter scores can be quadratic. Corrections are made below; premise (iii)'s small densities are proved, their full spatial response remains conditional.

15.1 Premise (i): Theorem 5 for the barriered measures

Assumption (E1). §4's local bounds, with the constants L_F, β, D_J, hold on the enlarged chart $|\xi_e| \le 2c_0$. Since $c_0 \le 1/16$, this chart stays far from the cut locus.

Extension. Split the potential of §§13–14 into face densities, $S_{\lambda,s} = S_2 + \sum_p (\lambda R_p + sQ_p)$, with half of each edge term assigned to each adjacent face as in §14.2 and $Q_p = \mathcal{J}_p(\xi) - \mathcal{J}_p(\xi_*)$. Fix a smooth radial χ with $\chi = 1$ for $|\xi| \le c_0$, $\chi = 0$ for $|\xi| \ge 2c_0$, $|\chi'| \le 2/c_0$, $|\chi''| \le 8/c_0^2$, put $\chi_p = \prod_{e\in\partial p} \chi(|\xi_e|)$ and $\tilde S = S_2 + \sum_p \chi_p(\lambda R_p + sQ_p)$ on $\mathbb{R}^{6N}$. On the chart $D = \prod_e B(0, c_0)$, $\tilde S = S_{\lambda,s}$ up to a constant. On $\operatorname{supp}\chi_p$, $|y_e| \le 2c_0 + 6\varepsilon/7 \le 3c_0$, so $\sigma_p = \sum_{e\in\partial p} |y_e| \le 12c_0$ and the cubic remainder obeys $t|R_p| \le \Lambda\sigma_p^3/6 \le 288\Lambda c_0^3$, $t|\nabla_e R_p| \le 72\Lambda c_0^2$ and $t\sum_{e'} \|(\nabla^2 R_p)_{ee'}\| \le 48\Lambda c_0$. The cutoff has $\|\nabla\chi_p\| \le 8/c_0$ and block row sum $\|\nabla^2\chi_p\| \le 20/c_0^2$ (a radial Hessian is $\chi''\Pi_r + (\chi'/r)\Pi_\perp$ with $r \ge c_0$ where $\chi' \ne 0$). The product rule gives the block row sum $t\|\nabla^2(\chi_p R_p)\| \le (48 + 1152 + 5760)\Lambda c_0 \le 7000\Lambda c_0$. Here the two cross terms cost $2\cdot 288 + 8\cdot 72 = 1152$. For amplitudes, $|\nabla_e Q_p| \le 3D_J/2$, $|Q_p| \le 18 D_J c_0$ and Hessian row sum at most $4.5D_J$: one face plus half an edge amplitude per incident edge. Thus the product rule gives $(4.5c_0 + 24 + 360)tD_J/c_0 \le 385 tD_J/c_0$ for $c_0 \le 1/16$. Each edge meets two faces. Hence, if

$$K_u\varepsilon + 14000\,\Lambda c_0 + 770\,tD_J/c_0 \le \tfrac12. \tag{65}$$

then $\|t\nabla^2\tilde S - 6I\| \le \frac52$ on all of $\mathbb{R}^{6N}$, uniformly in $\lambda, s \in [0,1]$, with range one because S_2 and each $\chi_p R_p, \chi_p Q_p$ couple only edges of one face. This is (20) with $\kappa = \frac12$. Since $c_0 > 2\varepsilon = 2t^\alpha$, the last term of (65) is at most $385 D_J t^{1/2+\delta}$.

Approximation. For integers k with $c_0 - 1/k > \eta$, continue w_η from $r_k = c_0 - 1/k$ by its tangent plus $k(r - r_k)^2/2$. This is nonnegative, radial, convex and C^1, with bounded Hessian for each k. Convolve in $\mathbb{R}^3$ with a nonnegative radial smooth mollifier of radius $o(1/k)$, chosen to give local C^1 convergence inside D; call the result w_k. Convexity and nonnegativity survive. Then $\mu_k \propto e^{-\tilde S - \sum_e w_k(\xi_e)}$ satisfies Theorem 5's smooth hypotheses and (20) with $\kappa = \frac12$, for every k, λ, s. As $k \to \infty$, $e^{-w_k(\xi)} \to e^{-w_\eta(\xi)} 1_{|\xi|<c_0}$ off the sphere $|\xi| = c_0$ (where w_η diverges), with domination by $e^{-\tilde S}$, integrable by uniform convexity. So $\mu_k \to \mu$ in total variation (Scheffé). The functions needed here for (21), (23), (24), (64), namely $\chi_p R_p$, $\chi_p Q_p$, and cut-off local scores, are bounded with bounded gradients and agree with the original ones on D. Their covariances and gradient norms therefore converge: apply total variation to f, g, fg and $|\nabla f|^2$, then take square roots for the gradient norms. Finite sums at fixed N are allowed; general unbounded observables need uniform integrability. The bounds pass to the limit with the constants $t/(4-\kappa) = 2t/7$ and $q = 5/12$.

Consequence. Under (E1) and (65), (21), (23), (24) and (64) hold for every barriered interpolated measure of §§13–14. Premise (i) of §14.2 is proved.

15.2 Premise (iii): local densities of the barrier cost

Order the edges $e_1, \dots, e_{2N}$ and let $\mu_{<j}$ be the Gaussian of §13.2 at the saddle, restricted by the barriers of $e_1, \dots, e_{j-1}$. Then exactly

$$b(U) = \log\frac{Z_{2,W}}{G_U} = -\sum_{j=1}^{2N} \tau_j(U), \qquad \tau_j = -\log E_{\mu_{<j}}\big[e^{-w_\eta(\xi_{e_j})}\big] \ge 0. \tag{66}$$

The proof of (55) applies to each $\mu_{<j}$: S_2 has the diagonal dominance of A, and edges carrying a barrier contribute the nonnegative term $y_e \cdot \nabla w_\eta(\xi_e)$. Since $w_\eta = 0$ for $|\xi| \le \eta = 2\varepsilon$ and $|\xi_*| \le 6\varepsilon/7$, $E_{\mu_{<j}}[e^{-w_\eta}] \ge 1 - p_j$ with $p_j \le P(|y_{e_j}| > 8\varepsilon/7) \le M_k^{\mathrm{bd}} t^{k/2} (7/8\varepsilon)^k$. With $\varepsilon = t^{1/2-\delta}$ this is $M_k^{\mathrm{bd}}(7/8)^k t^{k\delta}$. Take $k+1 = \lfloor 3t^{-2\delta}/8\rfloor$; then $2(k+1)/3 \le t^{-2\delta}/4$, so $M_k^{\mathrm{bd}} \le 2^{-k} t^{-k\delta}$ and

$$0 \le \tau_j \le 2p_j \le 2(7/16)^{k_t}, \qquad k_t = \lfloor 3t^{-2\delta}/8\rfloor - 1, \tag{67}$$

provided $k_t \geq 2$ (equivalently $t^{-2\delta} \geq 8$), which also gives $p_j \leq (7/16)^2 < 1/2$. Thus the barrier cost splits into edge densities that are super-polynomially small, below t^α for small t, and the identity-normalized $b(U) - b(1)$ has densities $\tau_j(1) - \tau_j(U)$ of the same size (both costs lie in $[0, 2p_j^{\mathrm{bd}}]$). For a local potential perturbation h_R, (24) bounds the derivative of τ_j by $(4t/7)q^{d(R,e_j)}\omega_\eta p_j^{1/2}\|\nabla h_R\|_2$; the factor two uses $Ee^{-w_\eta} \geq 1/2$. Here $\omega_\eta = \sup_r w'_\eta(r)e^{-w_\eta(r)}$ must be finite with controlled t dependence; a fixed polynomial-divergence profile supplies this. For actual boundary changes, $A(U)$ and $\xi_*(U)$ in S_2 depend on the entire field. Their score needs a separate quasi-local bound. Equation (66) establishes small densities independently of that bound.

15.3 What remains for the quasi-local remainder

With §15.1, §7.3's (23) applies to the barriered, amplitude-weighted measure of $\mathcal{D}_s$ itself. In link coordinates a boundary change alters only incident bridge and face potentials, so its score is local (distance below is between these enlarged supports), and the mixed second difference of $\mathcal{D}_s$ for changes in regions R, T is at most $(2t/7)\, ab\, (5/12)^{d(R,T)}$. The Gaussian reference $\mathcal{D}^{\mathrm{cov}}_{xy}$ obeys (21) as a Gaussian measure with Hessian H/t; differentiating its $C(U)$ gives quadratic as well as linear scores. Lemma 1 alone handles the linear-score part. Each bound separately is of order $\varepsilon^2(5/12)^d/t = t^{-2\delta}(5/12)^d$ for changes of size ε, with no small factor. A quasi-local remainder needs the difference $\mathcal{D}_s - \mathcal{D}^{\mathrm{cov}}_{xy}$ to gain t^α there: this is the kernel estimate (27) of §7, now against the covariant reference. Premise (ii) of §14.2 is used in the interpolation form only; the direct route through (23) needs it for the actual measure, where scores are local.

16. SU(3) root constants and the conditional transfer

The project's gap goal is $SU(3)$, and the one-step bounds of §§12–15 use the group only through the items below. With the normalization of this note, $\langle X, Y\rangle = -2\,\mathrm{tr}\, XY$ (for $SU(2)$ the norm of X is its rotation angle, and $SU(2) \cong S^3$ has radius 2), every root obeys $|\alpha(X)| \leq |X|$ on $\mathfrak{su}(3)$: for $X = i\,\mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$, $(\lambda_i - \lambda_j)^2 \leq 2(\lambda_i^2 + \lambda_j^2) \leq |X|^2$. Hence $\|\mathrm{Ad}_{e^X} - I\| \leq \|\mathrm{ad}_X\| \leq |X|$, as for $SU(2)$, and the curvature-to-holonomy bounds of (44), the $d\log$ bound of §11.1 and Lemma 1 (orthogonal adjoint blocks) hold verbatim. Write $n = \dim G$: $n = 3$ for $SU(2)$, $n = 8$ for $SU(3)$.

Internal scope check. ACCEPT the root bound and the table's dimension, Sylvester, B, wrapping, C_D, moment, M_2, bridge-Hessian (with the spectral convention below), bridge-trace and k_t rows. **REFINE** the Šidák row: the exponent is 3/16 for eight components. **REFINE** the flux row by fixing the Fourier sign. **ACCEPT** the clock-and-shift spectrum and its lower bound, using the complexified adjoint basis. **REJECT** the identification of this two-dimensional magnetic twist with a four-dimensional fractional charge. **REFINE** the bridge-tail arithmetic to 43/15 and propagate dimension changes into the remainder constants. Corrections are made in place.

Item	$SU(2)$	$SU(3)$	Reason
mid-plane coordinates	$\mathbb{R}^{6N}$	$\mathbb{R}^{16N}$	$2N$ edges $\times\, n$
Sylvester factor in (48)	8^{3N}	8^{8N}	edge minus face dimension, nN
B in (48)	$\frac{3}{4}256^2 \cdot \frac{33}{8}$	$2 \cdot 256^2 \cdot \frac{33}{8} = 540\,672$	$n - \mathrm{tr}\, R = \frac{1}{2}\|R - I\|_F^2 \leq \frac{n}{2}\|R - I\|^2$ for $R \in SO(n)$
wrapping term in (48)	$\frac{9}{2}N3^{-n_{\min}}$	$12N3^{-n_{\min}}$	$\|\,\mathrm{tr}(R - I)\| \leq 2n$

Item	$SU(2)$	$SU(3)$	Reason
C_D in (54)	$\frac{46}{7}L_F$	$\frac{368}{21}L_F$	face rank $4n$; $C_D = \frac{4n}{6}\cdot\frac{23}{7}L_F$
moment bound (55)	$[2(k+1)/3]^{k/2}$	$[2(k+6)/3]^{k/2}$	$\operatorname{div}(\lvert y\rvert^{k-2}y) = (k+n-2)\lvert y\rvert^{k-2}$ in $\mathbb{R}^n$
Šidák term in (56)	$48Ne^{-t^{-2\delta}/2}$	$128Ne^{-3t^{-2\delta}/16}$	rectangle radius $\varepsilon/\sqrt{n}$, variance $\le t/3$
$k=2$ bound in §13.3	$M_2 \le (\sqrt{3}+3D_J)\sqrt{t}$	$M_2 \le (2\sqrt{2}+3D_J)\sqrt{t}$	$\langle y\cdot\nabla V\rangle = n$
bridge Hessian (§14)	$4\Pi_b + 4h_b\Pi_b^{\perp}$	$4f(\mathrm{ad}_{b/2})$	$f(\theta) = \frac{\theta}{2}\cot\frac{\theta}{2}$ on the pairs $\pm i\theta$ of $\mathrm{ad}_{b/2}$, $f=1$ on its kernel
bridge trace in (61)	$\frac{\beta}{2}T$	$\frac{4}{3}\beta T$	rank n per edge: $\frac{n}{6}\beta T$
k_t in (67)	$\lfloor 3t^{-2\delta}/8\rfloor - 1$	$\lfloor 3t^{-2\delta}/8\rfloor - 6$	same choice with $k+n-2$
centre and fluxes (§12.3)	$\mathbb{Z}_2$; $\frac{1}{4}\sum_k(-1)^{e\cdot k}Z_{k,m}$	$\mathbb{Z}_3$; $\frac{1}{9}\sum_k \omega^{-e\cdot k}Z_{k,m}$	convention $Z_{k,m} = \sum_e \omega^{e\cdot k}Z_{e,m}$

The constants L_F, β, M, D_J of §4 and §14 are finite on the $SU(3)$ chart by the same analyticity; their values change and are left symbolic. The bridge function means $f(i\theta) = (\theta/2)\cot(\theta/2)$. Use $k_t \ge 2$, $2e^{-3t^{-2\delta}/16} \le 1/2$ for $SU(3)$; replace $(8/3)^{3/2}$ by $6^{3/2}$ in A_L, A_u, and $\sqrt{3}$ by $\sqrt{8}$ in amplitude costs. In B_δ replace it by $[\alpha/(2\delta ce)]^{\alpha/(2\delta)}$, $c = 3/16$. These substitutions and the table apply when assembling C_0, C, C_u.

Twisted flat sector. For $m = 1 \in \mathbb{Z}_3$ take the clock and shift matrices $\Gamma_2 = P = \mathrm{diag}(1,\omega,\omega^2)$ and $\Gamma_3 = Q$, $Qe_j = e_{j+1}$. Both lie in $SU(3)$ (a 3-cycle is even), and $PQ = \omega QP$. The eight traceless matrices Q^aP^b, $(a,b) \in \mathbb{Z}_3^2 \setminus \{0\}$, span $\mathfrak{sl}_3$, and $\mathrm{Ad}_P(Q^aP^b) = \omega^a Q^aP^b$, $\mathrm{Ad}_Q(Q^aP^b) = \omega^{-b}Q^aP^b$. The joint phases in (51) are therefore $(\phi_2,\phi_3) = (2\pi a/3, -2\pi b/3)$ over all eight nonzero pairs: as for $SU(2)$ at $m=1$, the twisted sector has no adjoint zero mode, and $L_m \ge 2 - 2\cos(2\pi/(3n_{\max})) > 0$ there, since each mode has a phase $\pm 2\pi/3$ in at least one direction. For $m = 0$, commuting Cartan transitions give phases 0 (twice) and $\pm\alpha(\theta)$ for the three positive roots. The centre note separately identifies central images at trisections. A four-dimensional fractional topological charge requires four-dimensional twist data; this magnetic two-torus calculation supplies its adjoint spectrum only.

Lemma 2 for any compact group. Lemma 2's explicit tail, $k_t \le C_+ t^{-5/2}e^{-\theta^2/(2t)}$, comes from the $SU(2)$ image formula. §§13–15 do not use it; the large-field step and the bridge tail (7) do. A bi-invariant metric has $\mathrm{Ric}(X,X) = \frac{1}{4}\sum_i \lvert[X,e_i]\rvert^2 \ge 0$, so the Li–Yau bounds for $\mathrm{Ric} \ge 0$ apply (Li and Yau 1986, Acta Math. 156, 153–201; Crossref metadata verified 2026-09-29; **passage**, Theorems 3.2 and 4.2 in the original-paper transcription, checked 2026-09-29): for every $\epsilon' \in (0,1)$ and $0 < t \le 1$, in this note's normalization ($e^{t\Delta/2}$, small balls of volume comparable to $t^{n/2}$),

$$c_{\epsilon'}t^{-n/2}e^{-d^2/((2-\epsilon')t)} \le k_t(g) \le C_{\epsilon'}t^{-n/2}e^{-d^2/((2+\epsilon')t)}, \qquad d = d(g,1).$$

The proof of (7) then gives, for the bridge midpoint at distance $\rho \ge r$ from m_*, $\beta\{d(m,m_*) \ge r\} \le C'_{\epsilon'}t^{-n/2}\exp\{-[4(r-\lvert X\rvert/2)^2/(2+\epsilon') - \lvert X\rvert^2/(2-\epsilon')]/t\}$, which at $\epsilon' = 0$ is (7)'s exponent $2r(r-\lvert X\rvert)/t$. At $r = 2\varepsilon$, $\lvert X\rvert \le \varepsilon$, $\epsilon' = \frac{1}{2}$, the gain is $e^{-43\varepsilon^2/(15t)}$ since $9/(5/2) - 1/(3/2) = 43/15$, for each fixed compact simple group, with group-dependent non-explicit constants. The exact image formula of the centre note would give explicit constants and the sharper prefactor $t^{-n/2-\lvert\Delta_+\rvert}$ (t^{-7} for $SU(3)$, $t^{-5/2}$ for $SU(2)$ as in (5)); that refinement is conjectural here. The bulk order-t coefficients of the order-t note scale by the adjoint-Casimir ratio, $\frac{3}{2}$ for $SU(3)$ (§7b there).

Consequence for STATE. The one-step normalized small-field bounds of §§13–15 hold for $SU(3)$ in the same $1+2$ mid-plane setting with the constants tabulated above, and the bridge tail needed for the large-field step holds for $SU(3)$ by Li–Yau with non-explicit constants; the $\mathbb{Z}_3$ flux sectors follow 't Hooft's projection with the clock-and-shift representative.

17. Shrinking chart and the remaining resolvent term

The proposed chart $c_0 = 3\varepsilon$ makes the smooth Hessian error small.

Internal scope check. ACCEPT as a reduction. The identity (68)–(69) checks: with $f_a = \nabla\partial_a S$, $g_a = \nabla\partial_a V_2$ and (22), the covariance difference splits as the two score-difference terms, $t\langle g_a, (R_\mu - R_0)g_b\rangle_\mu$ with $R_\mu - R_0 = -R_\mu(\Delta + tW_k'')R_0$, and the change of Hilbert space $\mathcal{M}$; the smooth insertion carries $C_*\varepsilon$. I also accept the corrections to §§15–16 made in §17: the Šidák exponent $3t^{-2\delta}/16$ for eight components, the constant 14000, the gain $43/15$, and the rejection of my identification of the two-dimensional $\mathbb{Z}_3$ twist with a four-dimensional fractional charge. Two route tests for (69), developed in §§18–19. (a) For $\mathcal{B}$, use the barrier's own square root: since $tL_\mu + tS'' \geq 0$, $\|(tW_k'')^{1/2}R_\mu^{1/2}\| \leq 1$, so $|\mathcal{B}_{ab}| \leq t\,\|R_\mu^{1/2}g_a\|_\mu\,\|(tW_k'')^{1/2}R_0 g_b\|_\mu$. The last factor is an expectation of tW'' against $|R_0 g_b|^2$, supported where some $|\xi_e| > \eta = 2\varepsilon$, which (55) makes rare at rate $(7/16)^{k_t}$; integration by parts against e^{-W} trades w'' for w'^2 and boundary decay, and $R_0 g_b$ decays from b by Theorem 5's Neumann bound, so a weighted estimate of size $q^{d(a,b)}(7/16)^{k_t/2}$ times score norms looks within reach, uniformly in k. (b) For $\mathcal{M}$, interpolate $G \to \mu$ as in §13.2 (same barrier added at both ends through the approximants): the constant parts of the scores cancel exactly, as noted, and the linear parts give second moments whose change along the path is a third cumulant against $V - V_2$, bounded by (55) and the $C_*\varepsilon$ Hessian deviation. The direct resolvent proof still requires the barrier insertion (69) below. Write $V = S + W_\eta$, with S the extension in §15.1, and H from (59). For $6\varepsilon \leq 1/16$, (E1) and the earlier smallness conditions, (65) becomes $(K_u + 42000\Lambda)\varepsilon + (770/3)D_J t/\varepsilon \leq 1/2$. Since $t/\varepsilon \leq \varepsilon$, set $C_* = K_u + 42000\Lambda + (770/3)D_J$. Then globally $\|tS'' - H\|_{\infty,b} \leq C_*\varepsilon$; the norm is $\sup_e \sum_{e'} \|M_{ee'}\|$. This includes $A - H$ and the amplitudes. The full Hessian is $tV'' = tS'' + tW_\eta''$, whose last term is unbounded.

Normalization and identity. A unit parameter changes one boundary link at velocity εUT, $|T| = 1$ in the $-2\,\mathrm{tr}$ metric. Let $V_2 = E_w^{(2)}/t$, $G \propto e^{-V_2}$, and $f_a = \nabla_\xi\partial_a S$, $g_a = \nabla_\xi\partial_a V_2$. For $d(a,b) \geq 2$ in the link graph, the local mixed potential derivative vanishes; thus $K_{ab} = \partial_a\partial_b(\mathcal{D}_s - \mathcal{D}^{\mathrm{cov}})$ is minus the difference of score covariances. Scores have support radius one. The requested remote bound is $\sup_a \sum_{b:d(a,b)\geq 2} e^{\gamma d(a,b)}|K_{ab}| \leq C\varepsilon^3/t$; unscaled link derivatives divide this by ε^2.

Work first with the smooth barrier approximants, and write $R_\mu = [tL_\mu + tS'' + tW_k'']^{-1}$, $R_0 = [tL_\mu + H]^{-1}$, $R_G = [tL_G + H]^{-1}$ and $\Delta = tS'' - H$. Both R_μ, R_0 act in the **same** $L^2(\mu_k)$; Theorem 5's Neumann argument bounds each edge block by $(2/7)(5/12)^d$. With $\gamma = \frac{1}{2}\log(12/5)$ their weighted block-row norms are at most $R_\gamma = (2/7)C(5/12)$, with the explicit series (25). The resolvent identity, followed by (22), gives exactly

$$\begin{aligned} -K_{ab} = {} & t\langle f_a - g_a, R_\mu f_b\rangle_\mu + t\langle g_a, R_\mu(f_b - g_b)\rangle_\mu \\ & - t\langle g_a, R_\mu \Delta R_0 g_b\rangle_\mu + \mathcal{B}_{ab} + \mathcal{M}_{ab}. \end{aligned} \tag{68}$$

$$\begin{aligned} \mathcal{B}_{ab} &= -t^2\langle g_a, R_\mu W_k'' R_0 g_b\rangle_\mu, \\ \mathcal{M}_{ab} &= t\langle g_a, R_0 g_b\rangle_\mu - t\langle g_a, R_G g_b\rangle_G. \end{aligned} \tag{69}$$

Here at finite k, $-K$ denotes the score-covariance difference; bounded local score extensions recover the physical kernel in the limit. The smooth insertion has operator bound $\|R_\mu \Delta R_0\|_{\infty,b,\gamma} \leq R_\gamma^2 e^\gamma C_*\varepsilon$, since Δ has range one. This supplies the desired small factor in that term. Even granting the score-gradient estimates from Taylor expansion and (55), (69) remains. Theorem 5 controls R_μ by the **sign** of W_k''; it supplies no vanishing bound on its sandwiched insertion,

uniformly in k. The multiplication norm of tW_η'' is infinite at every $t > 0$. The term $\mathcal{M}$ retains the change of measure and diffusion: $L_\mu - L_G = \nabla(V - V_2) \cdot \nabla$. For constant g_a, g_b it vanishes because $R_0 g_b = H^{-1} g_b$; actual link scores also have linear gradients from $\partial_a H$, so this simplification covers only their constant parts. A uniform weighted bound of size ε^3/t on $\mathcal{B} + \mathcal{M}$ is the precise missing estimate in this route. Equation (68) records a proof obligation, rather than a counterexample to the requested kernel bound; changing Hilbert spaces silently would omit it.

Tail cost of the smaller chart. For the partially barriered saddle Gaussian with the tested edge still free, (55) and §15.2's $k_t \geq 2$ give $P(|\xi_e| \geq 2\varepsilon) \leq (7/16)^{k_t}$ and $P(|\xi_e| \geq 3\varepsilon) \leq (7/30)^{k_t}$, using $|\xi_*| \leq 6\varepsilon/7$. A union bound costs $2N$. Thus the annulus above 3ε joins the large-field problem; these reference tails leave comparison with the full group integral open.

18. The barrier insertion and the change of measure

The square-root route gives (70), with the remaining weighted moment specified in (72). The interpolation route is tested separately below.

Internal scope check. ACCEPT as route tests with correctly identified residuals. Checked: (70) from $T_k \leq R_\mu^{-1}$ and $R_\mu \leq \frac{2}{7}$ (the smooth part of tV'' is at least $\frac{7}{2}$); (74) for a constant score paired with an affine one; (75) from the Lyapunov equation $HD_b + D_b H = B_b$ and $tL_\nu(D_b y) = D_b H y + tD_b(\lambda \nabla T + \nabla W_k)$. Both residuals, the barrier-weighted fourth moment (72) and the moving-diffusion term J_{ab} of (73), are higher-moment and derivative bounds on Helffer–Sjöstrand resolvent solutions. That problem has a literature for uniformly convex gradient models: Naddaf and Spencer 1997 introduced the random-walk representation, and Delmotte and Deuschel 2005 estimate derivatives of the corresponding kernels in stationary random environments with applications to the $\nabla\phi$ interface model (Crossref metadata verified 2026-09-29; results recalled, not read here). Their hypotheses are uniform ellipticity of the edge couplings; here the couplings are uniformly elliptic ($\frac{7}{2} \leq$ smooth part $\leq \frac{17}{2}$ in block norm) and the barrier enters only on the diagonal of the edge index. (As made explicit in §19.1, in R_0 the barrier enters the drift, and it acts as killing only in R_μ and in the differentiated equation; positivity of the block Hessian alone supplies no scalar random-walk conductances.) Their annealed estimates do not by themselves establish (72) or uniform control of J_{ab}; §19 gives the applicability test. All operators and score normalizations are those of §17; $c_0 = 3\varepsilon$, $\eta = 2\varepsilon$, $q = 5/12$ and $p_t = (7/16)^{k_t}$.

18.1 Part 1: the barrier's square root

Put $u_a = R_\mu g_a$, $v_b = R_0 g_b$ and $T_k = tW_k'' \geq 0$. The form inequality $T_k \leq R_\mu^{-1}$ gives exactly

$$|\mathcal{B}_{ab}| \leq t\|R_\mu^{1/2} g_a\|_\mu Q_b^{1/2}, \qquad Q_b = t\sum_e E_\mu[v_{b,e}^T w_k'' v_{b,e}], \qquad \|R_\mu^{1/2} g_a\|_\mu \leq \sqrt{2/7}\|g_a\|_\mu. \tag{70}$$

This estimate retains the small-tail question in Q_b and loses the separation from a; recovering spatial decay requires an edge-local version before the final Cauchy–Schwarz step.

Here is the proposed integration by parts, including its derivative terms. Write $v = v_{b,e}$, $h = \nabla_e w_k$, $A = \nabla_e S$, and let $D_e v$ and $\operatorname{div}_e v$ differentiate only ξ_e. Integrating $\sum_{ij} \partial_j(v_i v_j \partial_i w_k) e^{-S - W_k}$ gives

$$E_\mu[v^T w_k'' v] = E_\mu[(v \cdot h)^2 + (v \cdot h)(v \cdot A) - h \cdot ((D_e v)v + v \operatorname{div}_e v)]. \tag{71}$$

One may first cut off v and then pass to the limit when these terms are integrable. Equation (71) exhibits the positive Fisher term $E[(v \cdot h)^2]$, together with barrier-weighted derivatives of v. Boundary decay removes the surface integral; bounding the surviving terms still requires weighted integrability of the resolvent solution.

For a concrete profile choose $w_\eta(r) = x^4$, $x = (r - \eta)_+/(c_0 - r)$ inside the ball. This is convex and C^2, and has polynomial divergence. With $a_j = (j/(2\mathrm{e}))^{j/4}$, direct differentiation

gives $\sup w_\eta' e^{-w_\eta/2} \le \Omega_0/\varepsilon$ and $\sup \|w_\eta''\| e^{-w_\eta/2} \le K_0/\varepsilon^2$, where $\Omega_0 = 4(a_3 + 2a_4 + a_5)$ and $K_0 = 12a_2 + 58a_3 + 100a_4 + 74a_5 + 20a_6$. In particular $\omega_\eta \le \Omega_0/\varepsilon$. For approximants the needed additional profile hypothesis is the same weighted-Hessian bound, uniformly in k, with a specified constant K_{app} replacing K_0; local C^1 convergence alone supplies less. Mollification enlarges the active set to $r > \eta - \rho_k$, where ρ_k is its radius. Tail estimates must use this threshold before taking limits.

The precise missing moment can also be seen without differentiating v. Let $\nu_{e,k}$ remove only the barrier on edge e from μ_k, and put $A_{e,k} = \{|\xi_e| > \eta - \rho_k\}$, $p_{e,k} = \nu_{e,k}(A_{e,k})$. Local reweighting and Cauchy–Schwarz give, under the profile hypothesis,

$$E_{\mu_k}[v_{b,e}^T w_k'' v_{b,e}] \le \frac{K_{\mathrm{app}}}{\varepsilon^2(1-p_{e,k})} p_{e,k}^{1/2} \big(E_{\nu_{e,k}} |v_{b,e}|^4\big)^{1/2}. \tag{72}$$

Even granting $p_{e,k} \le 2p_t$ and $1 - p_{e,k} \ge 1/2$, this needs an $L^4(\nu_{e,k})$ bound on $v_{b,e}$, uniform in e, k, N, with spatial decay. For the partially barriered saddle Gaussian the tail follows from (55), with sufficiently small ρ_k; transferring it to the extended, amplitude-weighted $\nu_{e,k}$ also needs its moment argument. Theorem 5 bounds $\|v_{b,e}\|_{L^2(\mu_k)} \le (2/7) q^{d(e,T_b)} \|g_b\|_\mu$, where T_b is the enlarged score support. Its norm and measure differ from the fourth moment in (72). The removal density is proportional to e^{w_k}, so this change of norm has no uniform bounded-density shortcut. Thus (71)–(72) identify the failing term of this route: the local barrier-weighted resolvent moment. Finite Fisher information for each fixed barrier and total-variation convergence in §15.1 leave its uniform $k \to \infty$ bound, and the weighted estimate for $\mathcal{B}$, open.

18.2 Part 2: interpolate the measure and the diffusion

Fix the boundary parameters and k. Set $T = S - V_2$, $\nu_\lambda \propto e^{-V_2 - \lambda T - W_k}$, $A_\lambda = tL_{\nu_\lambda} + H$, $u_{\lambda,b} = A_\lambda^{-1} g_b$ and $F_{ab}(\lambda) = tE_{\nu_\lambda}[g_a \cdot u_{\lambda,b}]$. Differentiating the density and $A_\lambda u_{\lambda,b} = g_b$, with $A_\lambda' = t\nabla T \cdot \nabla$, gives the exact finite-k identity

$$F_{ab}' = -t \operatorname{Cov}_{\nu_\lambda}(g_a \cdot u_{\lambda,b}, T) - t^2 E_{\nu_\lambda}[u_{\lambda,a} \cdot (\nabla T \cdot \nabla) u_{\lambda,b}]. \tag{73}$$

Smooth cutoffs justify differentiation first; passage to the polynomial scores requires the displayed integrability. The second term records how the inverse changes with the drift, even though H is fixed. Adding the same barrier at both ends leaves the endpoint correction $F_{ab}(0) - t\langle g_a, R_G g_b\rangle_G$ in $\mathcal{M}$.

Write $y = \xi - m_0$, where $m_0 = E_G \xi$, and $g_b = c_b + B_b y$; $B_b = (\partial_b H)/t$ for the quadratic parameter score. For two constant gradients, $u_b = H^{-1} c_b$ pointwise and both terms in (73) vanish. For a constant c_a paired with an affine g_b, integrating $A_\lambda u_b = g_b$ gives instead

$$F_{ab}(\lambda) = t c_a^T H^{-1}(c_b + B_b E_{\nu_\lambda} y), \qquad F_{ab}' = -t c_a^T H^{-1} B_b \operatorname{Cov}_{\nu_\lambda}(y, T). \tag{74}$$

Thus cancellation covers the constant–constant part exactly; the mixed parts require control of the moving mean. The centered second moment $\Sigma_{ij} = \operatorname{Cov}_{\nu_\lambda}(y_i, y_j)$ does satisfy $\Sigma_{ij}' = -\operatorname{cum}_{\nu_\lambda}(y_i, y_j, T)$, where the cumulant is $E[(y_i - Ey_i)(y_j - Ey_j)(T - ET)]$.

A direct affine calculation locates the additional resolvent term. Let $HD_b + D_b H = B_b$, so $D_b = \int_0^\infty e^{-rH} B_b e^{-rH}\, dr$ and $\|D_b\| \le \|B_b\|/7$ using $H \ge 7I/2$. Then $v_b = H^{-1} c_b + D_b y = R_G g_b$ and, exactly,

$$u_{\lambda,b} = v_b - t A_\lambda^{-1} D_b(\lambda \nabla T + \nabla W_k). \tag{75}$$

Indeed $tL_{\nu_\lambda}(D_b y) = D_b H y + tD_b(\lambda \nabla T + \nabla W_k)$, which proves (75). Consequently a moment-only interpolation would omit the last term of (75). At $\lambda = 0$ it already contains the barrier endpoint correction. The small edge costs (67) bound partition-function values; their second boundary derivatives require the response estimates sought here.

For the smooth path, the precise extra term in (73) is $J_{ab} = t^2 \sum_{ij} E[u_{\lambda,a,i}(\partial_j T)\partial_j u_{\lambda,b,i}]$, with scalar coordinate indices i, j. Hölder gives

$$|J_{ab}| \le t^2 \sum_{ij} \|u_{\lambda,a,i}\|_4 \|\partial_j T\|_4 \|\partial_j u_{\lambda,b,i}\|_2.$$

The bounds (55) control local polynomial moments, and $\|tT''\|_{\infty,b} \le C_*\varepsilon$ controls smooth local coefficients. Theorem 5 supplies edge-index L^2 decay for u; closing this estimate requires spatially summable bounds also in its differentiation index j, with the displayed higher norms uniform in k. The first term of (73) likewise involves u_b, rather than just a quadratic polynomial. Equations (73)–(75) identify the failing terms of the proposed third-cumulant shortcut and retain the barrier endpoint explicitly. A bound $C\varepsilon^3/t$ on the weighted norm of $\mathcal{M}$ therefore remains an obligation, alongside the weighted moment (72).

19. Semigroup representation and the applicability test

The finite-k representation gives uniform pointwise bounds for bounded local scores.

Internal scope check. ACCEPT (76)–(78) and the replacement lemma (80) as a conditional result. Checked: (76) because H is configuration-independent and commutes with the diffusion generator; the pointwise bound (77) from $\int_0^\infty e^{-5s}e^{sC}ds = (5-C)^{-1}$, which removes the fourth-moment question of (72) for bounded scores; (78) by differentiating $\nabla V \cdot \nabla u$, so only V'' enters, with the two-index base $tL + 10I$ plus a nonnegative right multiplication; (81) by removing one edge's barrier; the scale bookkeeping in (80). Most of (79) is already in hand. The deterministic majorant C^S with sums at most $\frac{7}{2}$ follows from §17's $\|tS'' - H\|_{\infty,b} \le C_*\varepsilon \le \frac{1}{2}$. The proof of (55) uses only diagonal dominance and the sign of the barrier terms on the edges that carry barriers, so it applies verbatim to the removed-edge measures $\nu_{e,k}$ and to the interpolated ν_λ; it gives $p_{e,k} \le 2(7/16)^{k_t}$ for small ρ_k, hence $p/(1-p) \le P\varepsilon^3/t$ for small t, and $E|\nabla_e T| \lesssim \Lambda(1 + \varepsilon/\sqrt{t})$, far inside $L\varepsilon/t$. The profile constants $K_{\mathrm{app}}, \Omega_{\mathrm{app}}$ are a choice of approximants (the x^4 profile of §18 and its mollifications). What remains is the score-extension step: extended scores with the stated marginal constants and a Gaussian covariance error $E_0\varepsilon^3/t$ against the original ones. With it, (80) gives the covariant kernel estimate (27) at decay exponent $\frac{1}{2}\log(10/7)$. A concrete replacement lemma below uses local drift moments and score extensions to close the comparison conditionally, including its barrier endpoint. These additional inputs remain to be established for §17.

19.1 Part 1: the representation and its test

Fix $k, N, t > 0$, write $V = S + W_k$, and let $dX_s = -t\nabla V(X_s)\,ds + \sqrt{2t}\,dB_s$. Its generator is $-tL_\mu$. Since H is configuration-independent,

$$R_0 g(x) = \int_0^\infty e^{-sH} E_x[g(X_s)]\,ds. \tag{76}$$

Indeed the two generators commute; integrating their semigroup gives $(tL_\mu + H)^{-1}$. For a variable matrix potential $K(x)$ replace e^{-sH} by the ordered multiplicative functional U_s, with $U_0 = I$, $\dot{U}_s = -U_s K(X_s)$, inside the expectation. This is the matrix Feynman–Kac representation; bounded-Hessian finite-k approximants justify it by the product formula and differentiation.

For plane side lengths at least three, (59) gives $H_{ii} = 5I$ and off-diagonal block-row sum at most 3 (six incidences of size 1/2). Thus $C_{ij} = \|(H - 5I)_{ij}\|$ has row and column sums at most 3. The spectral 5/2 bound about $6I$ in Theorem 5 concerns sums of signed products; absolute path counting requires this different bound. The series for e^{-sH} is dominated blockwise by $e^{-5s}e^{sC}$. It can be read as a continuous-time walk with nonnegative jump weights, including holding weights, and killing at least 2. The original colour matrices carry signs and rotations; their spectral bounds alone supply no positive random-walk transition rates. In particular the

barrier occurs in the **drift** of (76); the extra barrier killing belongs to R_μ and to the differentiated equation. The literature analogy in the §18 source comparison therefore requires this distinction.

For this section set $\gamma = \frac{1}{2}\log(10/7)$ and $m = 5 - (7/2)e^\gamma > 0$, a weaker weight than §17's. For $K^H = (5I - C)^{-1}$, range one and the geometric series give

$$\sup_i \sum_j e^{\gamma d(i,j)} K^H_{ij} \le m^{-1}, \qquad \|(R_0 g)_i\|_\infty \le \sum_j K^H_{ij} \|g_j\|_\infty. \tag{77}$$

The second inequality bounds every $L^4(\nu_{e,k})$ norm as well, independently of its density relative to μ_k. All t-dependence here is in the score size; the resolvent constants are dimensionless.

For $u = (tL_\nu + H)^{-1} g$, let $Z_{ij} = \partial_j u_i$ in edge blocks. Differentiation gives the closed equation

$$tL_\nu Z + HZ + Z(tV'') = Dg. \tag{78}$$

Thus first configuration derivatives require V'', rather than V'''. If $tS'' - 5I$ has a deterministic range-one block majorant C^S with row and column sums at most $7/2$, the two-index Neumann series has base $tL + 10I + tW_k''$ on the differentiation index, and perturbation weighted norm at most $7e^\gamma$. Its inverse has weighted row and column sums at most $(2m)^{-1}$ on the product graph. The positive on-site matrices tw_k'' are contractions in each colour block. This proves summable derivative bounds from bounded, summable Dg, uniformly in k, N. A spectral norm bound alone would lose this block majorant information. The affine scores of (74) have bounded Dg but unbounded g on the approximating whole space. Hence (77) directly settles the bounded-score case; its application to those affine scores needs a score extension or a moment argument. Part 2 states the former as a precise replacement, rather than assuming an annealed theorem.

19.2 Part 2: a replacement lemma with local inputs

Replacement lemma (conditional). Use smooth local score potentials with bounded gradients g_a, agreeing with the quadratic scores on D. For arrays $A_{ai} = \|g_{a,i}\|_\infty$ and $D_{aij} = \|\partial_j g_{a,i}\|_\infty$ (Euclidean and Frobenius norms), assume every one-index marginal sum is bounded by $G = G_0\varepsilon/t$ and $D_* = D_0\varepsilon/t$, respectively, after weighting A by $e^{\gamma d(a,i)}$ and D by $e^{\gamma[d(a,i)+d(a,j)]}$. Here a marginal fixes any one index and sums all the others. Assume the deterministic Hessian majorants of §19.1 for every $V_2 + \lambda T + W_k$, and the following local measure hypotheses:

$$\begin{gathered} \sup_{e,k,\lambda} E_{\nu_\lambda}|\nabla_e T| + \sup_{e,k} E_{\mu_k}|\nabla_e w_k| \le L\varepsilon/t, \\ \sup_{e,k} p_{e,k} \le p < 1, \qquad \sup_k \|w_k''\| e^{-w_k/2} \le K_{\mathrm{app}}/\varepsilon^2, \qquad \frac{p}{1-p} \le P\varepsilon^3/t. \end{gathered} \tag{79}$$

All constants $G_0, D_0, L, K_{\mathrm{app}}, P$ are fixed, independent of k, N, t; the last inequality specifies the allowed small-t range explicitly. For $\|Q\|_\gamma = \sup_a \sum_b e^{\gamma d(a,b)}|Q_{ab}|$, the residuals (69), formed with these scores, obey

$$\|\mathcal{B} + \mathcal{M}\|_\gamma \le \frac{K_{\mathrm{app}} G_0^2 P + 2G_0 D_0 L}{m^2} \frac{\varepsilon^3}{t}. \tag{80}$$

Proof. The Neumann bounds (77)–(78) propagate all the stated marginal norms: u has norm at most G/m, and Du at most $D_*/(2m)$. For $R_\mu g$ the same G/m bound uses C^S and positive on-site killing. In particular (77) supplies (72)'s fourth moment under the removed-edge measure, with spatially summable decay. Using the pointwise bound directly improves the tail power in (72):

$$E_\mu \|w_k''(\xi_e)\| \le \frac{K_{\mathrm{app}} p}{\varepsilon^2(1-p)}, \qquad \|\mathcal{B}\|_\gamma \le \frac{t^2 G^2 K_{\mathrm{app}} p}{m^2 \varepsilon^2 (1-p)}. \tag{81}$$

Both inequalities follow by removing that edge barrier; its normalizing denominator is at least $1 - p$. The weighted row bound then follows by inserting the two score envelopes on either side.

Also (78) inserted into (73) gives, uniformly along its smooth path,

$$\|J\|_\gamma \le \frac{t^2 G D_*}{2m^2} \sup_e E_{\nu_\lambda} |\nabla_e T|. \tag{82}$$

Thus its configuration-index sum is controlled before summing boundary labels. The block norms already include the three colour components.

To include the barrier endpoint in $\mathcal{M}$, couple a stationary X of potential $S + W_k$ to the stationary Gaussian diffusion Y of potential V_2 with the same Brownian noise. Strong convexity gives this joint stationary coupling by starting in the remote past. Their difference satisfies $\dot{Z} = -HZ - tF(X)$, where $F = \nabla(T + W_k)$. Variation of constants and the block majorant give $\sup_e E|Z_e| \le th/m$, with $h = \sup_e E_\mu |F_e| \le L\varepsilon/t$. By (76), each resolvent pairing is the time integral of $tE[g_a(X_0) \cdot e^{-sH} g_b(X_s)]$ (and likewise for Y). Subtract the two products, use the mean-value formula for each score, and integrate $e^{-5s} e^{sC}$. The two terms each cost $tGD_* \sup_e E|Z_e|/m$ in weighted row norm. Therefore $\|\mathcal{M}\|_\gamma \le 2t^2 G D_* h/m^2$. This compares directly to G, including the entire barrier endpoint. Combining with (81) proves (80). All steps hold at finite k, N with the displayed constants. □

Applicability and cost. This is outcome **(b)**: its inputs concern local scores, tails and first drift moments, rather than a resolvent residual. A local quadratic score can retain its linear part globally and smoothly truncate its quadratic part outside D; on the shrinking chart this plausibly gives G_0, D_0 independent of t. Applying (80) to §17 requires constructing these extensions as gradients, checking their marginal constants and bounding their Gaussian covariance error by $E_0 \varepsilon^3/t$ in weighted row norm. That explicit additional score-extension hypothesis adds E_0 to (80)'s constant when comparing to the original Gaussian scores; fixed-N barrier limits also require convergence of the extended score covariances. The Gaussian extension error is a local polynomial-tail calculation. For (79), the cubic remainder suggests $E|\nabla_e T| = O(\varepsilon^2/t)$; a uniform profile bound $|w_k'| e^{-w_k/2} \le \Omega_{\mathrm{app}}/\varepsilon$ gives $E_\mu |\nabla_e w_k| \le \Omega_{\mathrm{app}} p/[\varepsilon(1-p)]$. The work still owed is the interacting removed-edge tail, this profile bound, and deterministic Hessian envelopes. The Gaussian tail (55) makes the scales plausible, while its transfer to the extended measure is an additional proof. The stop rule is met by this conditional lemma; (80) becomes applicable only after those local hypotheses are proved.

20. The score extensions and the kernel estimate

Internal scope check. ACCEPT. §20.1: centring (85) at the saddle s_λ of the extended potential, displaced from ξ_* by at most h, keeps the barrier sign ($|s_{\lambda,e}| < \eta - \rho_k$); the profile family has the weighted bounds (84), with the convolution costing at most a factor e because the oscillation across a mollifier ball stays below 2; $p \le 2(7/16)^n$ and (83) give $P = 1$; L collects the drift terms. §20.2: the extended potentials (86) are gradients agreeing with the scores on D; the tail $G(A_a) \le 42 e^{-25\varepsilon^2/(6t)}$ from the Gaussian mean $\le \varepsilon/2$ and variance $\le t/4$ per component; $E_G|\xi_i|^4 \le 8\varepsilon^4$; Hölder gives the $p_G^{1/4}$ factor, and (88) follows from splitting the difference of covariances. The constants G_0, D_0, A_0 are loose and explicit.

20.1 Moment transfer and a fixed profile family

The transfer of (55) uses the saddle of the **extended smooth potential**. Put $B = 4c_M + 6K_u/7$, $h = 2B\varepsilon^2/3 + 2tD_J$ and $n = k_t = \lfloor 3t^{-2\delta}/8 \rfloor - 1$. In addition to §17 impose

$$C_*\varepsilon \le \tfrac{1}{2}, \quad n \ge 2, \quad h \le \frac{2\varepsilon \log 2}{7n}, \quad 4(7/16)^n \le \varepsilon^3/t. \tag{83}$$

These hold for sufficiently small t at each fixed $0 < \delta < 1/6$. Indeed $nh/\varepsilon = O(t^{1/2-3\delta} + t^{1/2-\delta})$. For $S_\lambda = V_2 + \lambda(S - V_2)$, let s_λ be its unique whole-space saddle. At ξ_*, the energy gradient vanishes, $t|\nabla_e V_2(\xi_*)| \le B\varepsilon^2$ and $t|\nabla_e S(\xi_*)| \le 3tD_J$: use $|Y - w| \le 4c_M\varepsilon^2$, $\|C^*\|_{\infty,b} \le 2$,

and the radius-ε Hessian error $K_u\varepsilon$. Global diagonal dominance at least $3/2$ therefore gives $\|s_\lambda - \xi_*\|_\infty \le h$, by the maximum-block argument on $\int_0^1 tS''_\lambda(\xi_* + u(s_\lambda - \xi_*))\,du$. The majorants here are deterministic: apply §15.1's product-rule bounds block by block, add $\|(H-5I)_{ij}\|$, and use their row and column sums $3 + C_*\varepsilon \le 7/2$.

Here is one concrete family, indexed by integers $k \ge 1$. At $r_k = 3\varepsilon - \varepsilon/(k+1)$ continue $w = x^4$ by $w(r_k) + w'(r_k)(r-r_k) + w''(r_k)(r-r_k)^2/2$; denote the radial whole-space continuation by v_k. Convolve it in three dimensions with the probability density proportional to $\exp[-1/(1-|z|^2)]1_{|z|<1}$, rescaled to radius

$$\rho_k = \min\left\{\frac{\varepsilon}{1000(k+1)^6}, \frac{2\varepsilon\log 2}{7n}\right\}.$$

The resulting w_k is smooth, nonnegative and convex, vanishes on $r \le \eta - \rho_k$, and has bounded Hessian for each k. It converges locally in C^1 to w inside D and diverges outside. With §18's constants one may take

$$K_{\mathrm{app}} = \mathrm{e}K_0, \qquad \Omega_{\mathrm{app}} = \mathrm{e}(\Omega_0 + \sqrt{2K_0/\mathrm{e}}). \tag{84}$$

For verification, beyond r_k the radial Hessian is at most $w''(r_k)$, and $w'\exp(-w/2) \le w'(r_k)e^{-k^4/2} + \sqrt{2w''(r_k)/\mathrm{e}}\,e^{-k^4/2}$. Inside $r_k + 2\rho_k$, $\sup|v'_k| \le 32(k+1)^5/\varepsilon$, so the oscillation across a mollifier ball is less than 2. Convolving the weighted bounds thus costs at most e. For $r \ge r_k + \rho_k$, use $v_k \ge k^4$ on the convolution ball; the same quadratic bound controls the gradient there. This proves (84) for both $\|w''_k\|e^{-w_k/2}$ and $|w'_k|e^{-w_k/2}$.

For any subset of the edge barriers, integration by parts about s_λ gives, for every real $q \ge 2$,

$$\sup_e E|\xi_e - s_{\lambda,e}|^q \le [2(q+1)t/3]^{q/2}. \tag{85}$$

The barrier term has the required sign since $|s_{\lambda,e}| \le 6\varepsilon/7 + h < \eta - \rho_k$. Whole-space Gaussian domination and bounded finite-k Hessians justify integration by parts, including on the removed edge. Thus (85) applies to $\nu_{e,k}$, ν_λ and μ_k. The active threshold is at distance at least $8\varepsilon/7 - h - \rho_k$ from this saddle. Markov with $q = n$, and $-\log(1-u) \le 2u$ for $u \le 1/2$, gives $p_{e,k} \le 2(7/16)^n =: p < 1/2$. Equations (83) give $p/(1-p) \le \varepsilon^3/t$, so choose $P = 1$. Finally the Hessian bound on $T = S - V_2$ and (85) with $q = 2$ give $E|\nabla_e T| \le B\varepsilon^2/t + 3D_J + C_*\varepsilon(h + \sqrt{2t})/t$. Removing one barrier gives $E_{\mu_k}|\nabla_e w_k| \le \Omega_{\mathrm{app}}p/[\varepsilon(1-p)]$. Consequently (79) holds with $L = B + 3D_J + 3C_* + \Omega_{\mathrm{app}}$, uniformly in N, k, λ.

20.2 Gradient extensions and their Gaussian error

Use one scalar direction per boundary link (three directions and two layers cost a factor six in the incidence counts). A score has insertion support T_a of at most seven edges, each at distance at most one from a; each insertion edge belongs to at most 42 such supports. Write $\sigma_a = \partial_a V_2 = \mathrm{const} + l_a \cdot \xi + q_a(\xi)$, $q_a = \xi^T Q_a \xi/2$. Fix $b \ge 1$ as the supremum of $(t/\varepsilon)|l_{a,i}|$ and $(t/\varepsilon)\|(Q_a)_{ij}\|_F$ over these local coefficients, all allowed backgrounds and t; equivalently take the corresponding unit-link derivatives of $E^{(2)}_w$. This is a fixed finite-dimensional compact-chart supremum, independent of N, t. With §15.1's cutoff set

$$\widehat{\sigma}_a = l_a \cdot \xi + \chi_a q_a, \qquad \chi_a = \prod_{i \in T_a} \chi(|\xi_i|), \qquad \widehat{g}_a = \nabla\widehat{\sigma}_a. \tag{86}$$

These smooth potentials retain the entire linear part globally and agree with σ_a up to a constant on D. On the cutoff support, $|q_a| \le 98b\varepsilon c_0^2/t$ and $|\nabla_i q_a| \le 14b\varepsilon c_0/t$. The product rule, including the radial Hessian's Frobenius norm, gives $|\widehat{g}_{a,i}| \le 1000b\varepsilon/t$ and $\|\partial_j \widehat{g}_{a,i}\|_F \le 2000b\varepsilon/t$. Counting

at most 7^2 pairs per score and $42 \cdot 7$ per fixed insertion index proves every marginal in §19.2 with the explicit choices

$$G_0 = 50000be^{\gamma}, \qquad D_0 = 600000be^{2\gamma}. \tag{87}$$

Here is the local polynomial-tail check against the Gaussian (59). Its mean has block norm at most $\varepsilon/2$, and each scalar variance is at most $t/4$. Thus $E_G|\xi_i|^4 \le 8\varepsilon^4$. For $A_a = \{\max_{i\in T_a} |\xi_i| > 3\varepsilon\}$, a union over seven edges and three components gives $G(A_a) \le p_G := \min\{1, 42\exp[-25\varepsilon^2/(6t)]\}$. The error gradient vanishes off A_a; on it the product rule gives $|\nabla_i(\widehat{\sigma}_a - \sigma_a)| \le 2000b\varepsilon t^{-1}\sum_{j\in T_a}|\xi_j|$. Hölder and the fourth moment give its $L^2(G)$ bound $24000b\varepsilon^2 p_G^{1/4}/t$. Its weighted marginal sums are therefore at most $A_0\varepsilon^2 p_G^{1/4}/t$ with $A_0 = 1100000be^{\gamma}$. The original gradient's L^2 marginals are at most $(G_0 + A_0)\varepsilon/t$. Apply the Gaussian covariance representation to the two differences of products, using the weighted inverse bound m^{-1}. This proves

$$\|\operatorname{Cov}_G(\widehat{\sigma},\widehat{\sigma}) - \operatorname{Cov}_G(\sigma,\sigma)\|_{\gamma} \le E_0 p_G^{1/4}\varepsilon^3/t, \qquad E_0 = A_0(2G_0 + A_0)/m. \tag{88}$$

At fixed N, t, these extended potentials grow at most linearly. The densities are bounded by a fixed multiple of e^{-S} for all sufficiently large k, since their partition functions converge to a positive limit. Strong convexity of S supplies an integrable Gaussian dominator for their products. Dominated convergence therefore proves convergence of their means and covariances to the barriered covariances; on D these are exactly the original quadratic-score covariances.

20.3 Assembly of the kernel estimate

Use §19's weight, $\gamma = \frac{1}{2}\log(10/7)$, $m = 5 - \frac{7}{2}e^{\gamma}$, the unit-link normalization of §17, and the extended scores of (86) on the Gaussian side; on the barriered measure μ the extended quadratic scores agree with σ_a up to constants, and their insertion gradients agree exactly, since μ lives on D. The actual scores $\partial_a S$ remain distinct from σ_a. Define $h_{ai} = \sup_D |\nabla_i \partial_a (S - V_2)|$ and

$$C_T = \sup_{N,t,U} \frac{t}{\varepsilon^2} \max\left\{ \sup_a \sum_i e^{\gamma d(a,i)} h_{ai}, \sup_i \sum_a e^{\gamma d(a,i)} h_{ai} \right\}.$$

Uniform finiteness of this supremum is an additional hypothesis. For radius-one supports with the counts of (87), the sufficient local estimate $h_{ai} \le M_T\varepsilon^2/t$ gives $C_T \le 42e^{\gamma}M_T$. Establishing M_T requires mixed boundary–insertion Taylor bounds for the energy and amplitudes, including derivatives of the moving saddle and Hessian used in the extension. The insertion-only bounds Λ, c_M, K_u, B supply no such mixed-jet estimate by themselves. Compactness at fixed t, N leaves the scaled supremum as $t \downarrow 0$ unsettled; the shift estimate in §20.1 bounds a value, rather than its boundary derivative. Amplitude mixed derivatives also need a specified bound extending D_J.

Internal scope check. REFINE the first term of C_{27}: (68) pairs $f_a - g_a$ with f_b, adding $C_T^2\varepsilon/m$ to its coefficient. **ACCEPT**, conditionally on the displayed score hypothesis, the smooth-insertion, barrier/measure, and Gaussian-extension terms. **REFINE** score coincidence as above: it holds modulo constants for quadratic potentials, exactly for their gradients, and after the barrier limit. **ACCEPT** the conversion from ε-velocity to unscaled link derivatives and its relative t^{α} gain. **REJECT** the asserted derivation of uniform C_T from the listed constants; the explicit missing estimate is above. **REFINE** the identification with (27): (89) is a link-coordinate analogue; §7.4's curvature-coordinate conversion remains an obligation.

Conditional covariant link-kernel estimate. Under (83), $C_T < \infty$ and the hypotheses of §§17–20, for boundary links at link distance $d(a,b) \ge 2$,

$$\sup_a \sum_{b:\, d(a,b)\ge 2} e^{\gamma d(a,b)} \big|\partial_a\partial_b(\mathcal{D}_s - \mathcal{D}^{\mathrm{cov}})\big| \le C_{27}\,\frac{\varepsilon^3}{t},$$

$$C_{27} = \frac{2C_T G_0 + C_T^2}{m} + \frac{e^\gamma C_* G_0^2}{m^2} + \frac{\mathrm{e} K_0 G_0^2 + 2 G_0 D_0 L}{m^2} + E_0 p_G^{1/4}. \tag{89}$$

Proof. Use (68) on the limiting barriered measure, with the extended quadratic scores. Their covariance comparisons follow by §20.2's dominated limit. Since $f = g + (f - g)$, the two score-difference terms cost $(2C_T G_0 + C_T^2 \varepsilon)\varepsilon^3/(mt)$ by (77) and its R_μ version; use $\varepsilon \le 1$ for the constant in (89). The smooth insertion costs $t(G_0\varepsilon/t)^2 e^\gamma C_* \varepsilon / m^2$, since Δ has range one and block row sum at most $C_*\varepsilon$, and each resolvent has weighted norm at most m^{-1}. The residuals (69) obey (80) with $P = 1$ and $K_{\mathrm{app}} = \mathrm{e}K_0$ from §20.1. Returning from extended to original scores on the Gaussian side costs (88). Collect the four terms. □

In unscaled link derivatives the bound reads $C_{27}\varepsilon/t$, so the difference kernel is smaller than the Gaussian kernel scale $1/t$ of (29) by the factor $\varepsilon = t^\alpha$. This proves the conditional link-coordinate estimate with decay rate $\frac{1}{2}\log(10/7)$, uniformly in plane size. Finite-k quadratic-score comparisons converge to this estimate; equality with the original scores is asserted on the limiting measure.

Pair response and the next estimate. Together with the face-local sizes (63), (89) controls the pair dependence of the remainder $r = \mathcal{D}_s - \mathcal{D}^{\mathrm{cov}}$ on boundary data: a change at distance d from a region moves the link derivative of r by at most $C_{27}\varepsilon^3 e^{-\gamma d}/t$ per unit change. Polymer activities require all-order connected response estimates, an admissible family of link interpolation paths and control of their order-dependent constants. The scale $\varepsilon^3/t = t^{\alpha - 2\delta}$ also retains the energy budget of (62); bare $O(t^\alpha)$ activities need a stronger estimate or a specified normalization.

21. Third response and an all-order polymer criterion

The third response reduces to a differentiated vector resolvent. Its additional source is the third insertion derivative of the potential, including the barrier. This specifies the next estimate.

Internal scope check. I accept the corrections to my assembly: the added C_T^2/m, score coincidence modulo constants, the rejection of my claim that C_T follows from the insertion-only constants (the mixed boundary–insertion jets need their own bound), and (89) as the link-coordinate analogue of (27). §21: ACCEPT (90); (91) by applying (22) twice to $\mathrm{Cov}(A_a^\circ A_b^\circ, A_c)$ with $\nabla(A_a^\circ A_b^\circ) = A_a^\circ f_b + A_b^\circ f_a$; (92) by differentiating $(tL_\mu + tV'')u_c = f_c$, whose drift term contributes $Z_c t V''$ and whose matrix potential contributes $tV''Z_c + Y_c$; (93) as a conditional bound; and (94)–(95) as the standard anchored Möbius conversion. The source Y looks closable with the tools at hand: u_c is pointwise bounded by (77), so Y needs only $E_\mu \|w_k'''\|^4$, which the edge-removal argument of (81) controls once the profile gives $\|w_k'''\| e^{-w_k/4} \le K_3/\varepsilon^3$; the x^4 profile has this property with an explicit K_3.

21.1 Third mixed derivative

Work at finite k, N, in fixed insertion coordinates, with $V = S + W_k$, $A_a = \partial_a S$, $f_a = \nabla A_a$, $R = [tL_\mu + tV'']^{-1}$ and $u_a = R f_a$. Take three distinct boundary directions whose local potential supports are pairwise separated, so $S_{ab} = S_{ac} = S_{bc} = 0$ identically; use the same condition for V_2. Separation here means disjoint interaction terms, as in §17's link-graph convention. Differentiating EF gives $E\partial_c F - \mathrm{Cov}(F, A_c)$, hence

$$\partial_a \partial_b \partial_c r = \mathrm{cum}_\mu(A_a, A_b, A_c) - \mathrm{cum}_G(\sigma_a, \sigma_b, \sigma_c). \tag{90}$$

For overlapping supports the additional terms are $ES_{abc} - \mathrm{Cov}(S_{ab}, A_c) - \mathrm{Cov}(S_{ac}, A_b) - \mathrm{Cov}(S_{bc}, A_a)$, and their Gaussian counterparts. These terms matter at higher orders.

The covariance formula (22), applied twice, gives the exact identity

$$\mathrm{cum}_\mu(A_a, A_b, A_c) = t^2 E_\mu[f_a \cdot R\nabla(f_b \cdot u_c) + f_b \cdot R\nabla(f_a \cdot u_c)]. \tag{91}$$

Indeed apply it first to $\operatorname{Cov}(A_a^\circ A_b^\circ, A_c)$; the product rule leaves $t\operatorname{Cov}(A_a, f_b\cdot u_c)$ and $t\operatorname{Cov}(A_b, f_a\cdot u_c)$. This proves (91) without differentiating a parameter-dependent Hilbert-space pairing. Differentiating the equation for u_c, with $Z_{c,ij}=\partial_j u_{c,i}$, gives, in colour components,

$$tL_\mu Z_c + (tV'')Z_c + Z_c(tV'') = Df_c - Y_c, \qquad (Y_c)_{ij} = t\sum_l (\partial_j V''_{il})u_{c,l}. \tag{92}$$

Thus the second derivative of the scalar Poisson solution introduces W'''_k as a source; the positive W''_k still supplies killing.

Here are sufficient quantitative inputs for a weighted third bound. Let $\ell(I)$ be the minimum number of graph edges in a tree joining the labels I. For each tensor take the maximum of its one-index marginal sums of $e^{\gamma\ell(I)}$ times its block $L^4(\mu_k)$ norm. Assume these norms for f, Df, Y are at most $F\varepsilon/t, D_F\varepsilon/t, Y_0\varepsilon/t$, respectively, uniformly in k, N, t, U. Smooth approximation and integrability of (91) are included; the resulting cumulants must converge to the barrier limit. The block semigroup bounds give $\|u\| \le F\varepsilon/(mt)$ and $\|Z\| \le (D_F + Y_0)\varepsilon/(2mt)$: (92) has two Hessian indices, base $10I$ and perturbation norm at most $7e^\gamma$. Minkowski and stationarity give the same bounds in L^4 as in L^∞. Tree weights multiply under contraction since joining the constituent trees joins their external labels. Hölder in (91) therefore proves

$$\sup_a \sum_{b,c \text{ separated}} e^{\gamma\ell(a,b,c)}|\partial_a\partial_b\partial_c r| \le C_3\varepsilon^3/t.$$

$$C_3 = \frac{F^2(3D_F + Y_0) + 3(G_0 + A_0)^2 D_G}{m^2}, \quad D_G = 294be^{2\gamma}. \tag{93}$$

For the Gaussian term, g is affine, its L^2 marginal bound is $(G_0 + A_0)\varepsilon/t$ by §20.2, and Dg is constant with marginal bound $D_G\varepsilon/t$ (at most $42\cdot 7$ blocks). Here $Y = 0$, Du is constant with bound $D_G\varepsilon/(2mt)$, so Cauchy–Schwarz alone gives the second summand in C_3. For the interacting term the two product-rule contributions in (91) cost $2F^2D_F/m^2 + F^2(D_F + Y_0)/m^2$.

Equation (93) is a precise reduction. The new work is the uniform Df and especially Y bounds in (92), including the barrier limit. The profile estimates (84) control w'_k and w''_k with a density weight; a bound on $w'''_k u_c$ in the stated L^4 tensor norm remains to be proved. Differentiating (78) as if its matrix potential stayed constant would omit this term. The scale in unscaled derivatives is C_3/t; the scaled third response retains the cubic size of (89).

21.2 All orders and anchored Möbius inversion

Fix admissible independent link paths $U_a(z_a)$, $0 \le z_a \le 1$, of speed at most ε, with every mixed corner and intermediate point in the chart. Assume $r(0) = 0$. A sufficient all-order estimate is, for every $n \ge 1$, with fixed $A, q, \eta > 0$,

$$\sup_a \sum_{\substack{X \ni a \\ |X|=n}} e^{\eta\ell(X)} \sup_{z\in[0,1]^{\mathcal{E}}} |\partial_X r(U(z))| \le At^\alpha q^n. \tag{94}$$

It includes adjacent links and all mixed potential derivatives. Repeated differentiation uses $\partial_a EF = E\partial_a F - \operatorname{Cov}(F, A_a)$; each new score adds a cumulant, with contact terms from differentiated scores. Resolvent differentiation adds the higher jets of (92). Uniform control of their growth in n is part of (94).

Define $r_X = \sum_{Y\subseteq X}(-1)^{|X|-|Y|} r(U_Y, 0_{Y^c})$. Finite Boolean inversion gives $r(U) = \sum_X r_X$; repeated fundamental theorems of calculus give $|r_X| \le At^\alpha q^{|X|}e^{-\eta\ell(X)}$. Assign each X to a deterministic minimum connecting tree with vertex set $H(X)$, and collect $\rho_H = \sum_{H(X)=H} r_X$. Since $X \subseteq H$ and $\ell(X) = |H| - 1$, summing over its subsets proves

$$|\rho_H| \le Ae^\eta t^\alpha e^{-[\eta - \log(1+q)]|H|}. \tag{95}$$

Thus (94) with $\eta > \log(1+q)$ gives the requested connected activities, $C = Ae^{\eta}$, $\gamma' = \eta - \log(1+q)$, at finite volume. An infinite-volume polymer norm additionally needs summability over connected sets. Pairwise-separated third derivatives alone leave the contact terms, admissible paths and order growth in (94) open. Also (93) supplies t^{α} for $(t/\varepsilon^2)r$; bare r needs the stronger scale in (94), or activities carrying (62)'s energy budget.

22. The mixed score jets and the barrier source

The barrier source closes with an explicit third-profile constant and a strengthened small-time condition. All tensor norms below include colour Frobenius norms.

Internal scope check. ACCEPT. Checked: for the x^4 profile, $3\varepsilon - r = \varepsilon/(1+x)$ and $dx/dr = (1+x)^2/\varepsilon$, whence $\varepsilon w' = 4x^3(1+x)^2$; the weighted third-derivative bound feeds the edge-removal argument as $\|D^3 w_k\|_{L^4(\mu_k)} \le (K_3/\varepsilon^3)(p/(1-p))^{1/4}$, and (98) turns the last factor into ε^3/t, so the barrier part of Y is $K_3 F\varepsilon/(mt)$ as in (99); (100) from differentiating the second-order remainder at $\pm b/2$; the face, bridge and amplitude contributions to M_T in (101). The local suprema live on the fixed enlarged chart, so they are uniform as $t \downarrow 0$; the amplitude constants rest on §4's analytic zero-image factors, as stated. With (89) and (93) now on the chart, the pair and separated-triple responses of the remainder are controlled; the all-order growth in (94) is the next step.

22.1 The source Y

Use scalar score extensions before taking the barrier limit. For a local score A_c set $a_c = A_c(0)$, $l_c = \nabla A_c(0)$ and $\widehat{A}_c = a_c + l_c\cdot\xi + \chi_c(A_c - a_c - l_c\cdot\xi)$, with the product cutoff of (86). This agrees with A_c on D and has bounded gradient; the original finite-k score can have an unbounded affine gradient outside D. Equations (91)–(92) hold for these scalar observables, with $f_c = \nabla\widehat{A}_c$. Write $F\varepsilon/t$ for their pointwise weighted gradient marginals; §22.2 supplies F. The R_μ version of (77) gives pointwise marginals $F\varepsilon/(mt)$.

Put $b_j = (j/\mathrm{e})^{j/4}$ and

$$K_3 = \mathrm{e}(24b_1 + 324b_2 + 1257b_3 + 2178b_4 + 1881b_5 + 780b_6 + 120b_7). \tag{96}$$

Then the precise family of §20.1 satisfies $\|D^3 w_k\|_F e^{-w_k/4} \le K_3/\varepsilon^3$ uniformly in k. Here is a direct check. On the active interval, $dx/dr = (1+x)^2/\varepsilon$, $\varepsilon w' = 4x^3(1+x)^2$ and $\varepsilon^2 w'' = 12x^2 + 56x^3 + 96x^4 + 72x^5 + 20x^6$. A further derivative gives coefficients $24, 216, 744, 1296, 1224, 600, 120$ in degrees 1 through 7. The radial tensor formula bounds its Frobenius norm by $|w'''| + 9(w''/r + w'/r^2)$. Since $r \ge 2\varepsilon$, the larger polynomial $\varepsilon^3 w''' + 9\varepsilon^2 w'' + (9/4)\varepsilon w'$ has exactly the coefficients in (96). Beyond r_k, write $A = w''(r_k)$ and $B = w'(r_k)$: the quadratic continuation has $D_r^3 v_k = 0$ and $v_k'/r^2 \le B/(4\varepsilon^2) + A/(2\varepsilon)$. Its radial tensor is bounded by $9A/\varepsilon + 9B/(4\varepsilon^2)$, while $v_k \ge k^4$. The bound follows from $\sup_{x\ge 0} x^j e^{-x^4/4} = b_j$. The continuation is C^2, so its weak third derivative has no interface measure. Near the interface §20.1's oscillation bound costs at most $\mathrm{e}^{1/2}$ under convolution; beyond $r_k + \rho_k$ use $v_k \ge k^4$ on the entire ball. The factor e in (96) covers both regions.

For completeness fix the smooth cutoff once, and let $K_\chi \ge 1$ bound c_0^r times the sum of all colour-block norms of $D^r\chi_p$ for $0 \le r \le 3$. Enlarge $\Lambda_3 \ge \Lambda$ to bound the local third insertion jets in these norms, and enlarge D_J to bound all first through third insertion derivatives of each local amplitude. These constants are finite on the enlarged compact chart. The product rule in §15.1 gives the global third-jet marginal bound

$$\|tS'''\|_{\gamma,\mathrm{marg},\infty} \le J_S := e^{2\gamma} K_\chi(4000\Lambda_3 + 100 D_J). \tag{97}$$

Indeed the four differentiated cubic-remainder terms per face cost $(64 + 576 + 864 + 288)K_\chi\Lambda_3 = 1792 K_\chi\Lambda_3$; each edge meets two faces. The amplitude terms cost at most $19 K_\chi D_J t/c_0^2$ per face, and $t/c_0^2 \le 1/9$. The quadratic extension has zero third insertion derivative.

Removing edge i exactly as in (81) gives

$$\|D^3 w_k(\xi_i)\|_{L^4(\mu_k)} \le \frac{K_3}{\varepsilon^3}\left(\frac{p}{1-p}\right)^{1/4}.$$

In addition to (83), impose

$$4(7/16)^n \le (\varepsilon^3/t)^4. \tag{98}$$

This holds for sufficiently small t at each fixed $0 < \delta < 1/6$. The barrier tensor is diagonal in its three edge indices. Contracting its bound and (97) with the pointwise envelope of u_c, using the joining-tree inequality of §21, proves

$$\|Y\|_{\gamma,\mathrm{marg},4} \le Y_0\varepsilon/t, \qquad Y_0 = \frac{F}{m}(J_S + K_3). \tag{99}$$

22.2 Mixed boundary–insertion jets

Write $\partial_a = \varepsilon D_a$, where D_a changes one boundary link at unit speed and holds the insertion coordinates fixed. Fix $J_1 \ge 1$ bounding the local first derivatives of b, Y, the four rotations O, and the midpoint-coordinate and endpoint-displacement maps, using sum norms. These maps are analytic on the compact chart; compact group-valued midpoint links include arbitrary free-cycle holonomies. Let $L_{F,3}$ bound all third derivatives of F in $(Y, z_1, \dots, z_4)$, and $L_{B,3}$ all third derivatives of B in (b, ξ). Let M_3 bound the mixed midpoint derivative of the second insertion derivative of the face-log map. Enlarge M to bound that second insertion derivative throughout the same chart. All are fixed local suprema, independent of t, N. Enlarge D_J also to bound the composed amplitudes' $D_a\nabla_i$ and $D_a\nabla_i\nabla_j$ derivatives. Their uniformity as $t \downarrow 0$ follows from §4's analytic zero-image factors and polynomial-times-$e^{-c/t}$ bounds for differentiated images.

The second-difference formula for the endpoint face logs gives

$$|Y - w| \le 4c_M\varepsilon^2, \qquad |\partial_a(Y - w)| \le J_\Delta\varepsilon^2, \quad J_\Delta = 8J_1(M + M_3). \tag{100}$$

Indeed differentiate the integral second-order remainder at displacements $\pm b/2$: their sum norm is at most 2ε, their unscaled first derivative is bounded by J_1, and the differentiated Hessian costs M_3J_1. Increasing the former M preserves $c_M = M/2$ and all previous inequalities, with the correspondingly stronger smallness conditions.

Put $F_0 = |Y + \sum z_i|^2/4$. The function $F - F_0$ has zero two-jet at the origin. On D, $|Y| + \sum|z_i| \le 15\varepsilon$; Taylor's formula bounds its Hessian by $15L_{F,3}\varepsilon$ and its gradient by $225L_{F,3}\varepsilon^2/2$. Differentiating $z_i = O_i\xi_i$, including $\partial_a O_i$, therefore costs at most $1000J_1L_{F,3}\varepsilon^2$ per incident face in $|\nabla_i\partial_a(E - E_w^{(2)})|$. The bridge costs at most $4J_1L_{B,3}\varepsilon^2$, since $D_bD_\xi(B - 2|\xi|^2)$ vanishes at the origin. The difference between $F_0(Y)$ and $F_0(w)$ has insertion gradient $C^*(Y - w)/2$; differentiating both factors and using (100) costs at most $(4c_MJ_1 + J_\Delta)\varepsilon^2$ over two faces. This accounts for the background dependence of both w and C_{m_*}. The old-face term $-(\|x\|^2+\|y\|^2)/8$ has zero insertion derivative. On D the extension equals the original energy plus amplitudes, so moving-saddle and moving-Hessian contributions cancel in that equality. The three incident amplitude terms cost $3D_J\varepsilon$. Since $t \le \varepsilon$, we have proved

$$h_{ai} \le M_T\varepsilon^2/t, \qquad M_T = J_1(2000L_{F,3} + 4L_{B,3} + 4c_M) + J_\Delta + 3D_J, \quad C_T \le 42e^\gamma M_T. \tag{101}$$

Here are also explicit inputs for the extended scores of §22.1. Form A_c there from the original local action on the enlarged chart; its scalar extension serves as an observable for the measure with potential $S + W_k$. Let $J_2, J_3 \ge 1$ bound, per local energy or amplitude term, respectively $D_a\nabla_i$ and $D_a\nabla_i\nabla_j$ in colour-block norm on the enlarged chart; include the amplitude contribution with $t \le 1/2$. These are finite by the same local analytic argument. Before cutoff, $|l_{c,i}| \le 3J_2\varepsilon/t$ and $|D_j\nabla_i A_c| \le 3J_3\varepsilon/t$. Taylor's formula on seven blocks, with $\sum|\xi_i| \le 42\varepsilon$, and the cutoff product rule give gradient and Hessian bounds $3000(J_2 + J_3)\varepsilon/t$ and $10000J_3\varepsilon/t$. Counting the seven-edge supports gives

$$F = 126000e^\gamma(J_2 + J_3), \qquad D_F = 2940000e^{2\gamma}J_3. \tag{102}$$

22.3 The chart bounds and their scope

Put $C_T^* = 42e^{\gamma}M_T$. Equations (89) and (93) now hold on the barriered chart, with explicit constants

$$C_{27} = \frac{2C_T^* G_0 + (C_T^*)^2}{m} + \frac{e^{\gamma}C_* G_0^2 + \mathrm{e}K_0 G_0^2 + 2G_0 D_0 L}{m^2} + E_0 p_G^{1/4},$$

$$C_3 = \frac{F^2[3D_F + F(J_S + K_3)/m] + 3(G_0 + A_0)^2 D_G}{m^2}.$$

Thus the respective weighted pair and separated-triple sums are bounded by $C_{27}\varepsilon^3/t$ and $C_3\varepsilon^3/t$, uniformly in plane size. The remaining hypotheses are side lengths at least three, the fixed link-coordinate convention, (E1), the chart and numerical smallness conditions of §§14 and 17–20 with the enlarged constants, (83), (98), and the separation conditions of (89) and (90). The local suprema above are finite by analyticity; both former open jet hypotheses are discharged. At fixed N, t the scalar score extensions grow at most linearly, so Gaussian domination gives convergence of all products through degree three. Apply (91)–(99) to those observables and then pass to the barrier limit, where they equal the original scores. This proves (93) for the actual response.

23. The n-th response and its growth

The exact response has an iterated covariance representation with $(n-1)!$ ordered histories. Its unordered marginal bound below separates this count from higher-jet growth. Uniform higher barrier jets are additional hypotheses.

Internal scope check. ACCEPT (103)–(107) as exact identities and conditional bounds, and the growth verdict. Checked: the cumulant recursion behind (104) by the generating function $E[e^{\sum z_i h_i} g]$ and (22); the partition formula (107) by the rule that differentiating a block inserts a label and differentiating the law opens a singleton; the $(n-1)!$ of (106) as the ordered-versus-unordered count; and the fifth-jet obstruction, since the x^4 onset is C^3 with a jump of $24/\varepsilon^4$ in the fourth derivative. One remark on the remedies. Any C^∞ onset that is flat at $r = \eta$ is non-analytic there, so its jets $K_{j,P}$ grow faster than any C^j; a smoother profile trades the divergence for super-geometric growth in j, which (105) would pass on to the order growth. The geometric bound in (94) is therefore more likely to come from the second remedy: keep the barrier undifferentiated and treat the region where it is active as a large-field polymer, whose activity per edge carries the super-polynomially small $p \le 2(7/16)^{k_t}$ of (67) and §20.1. That is the standard small-field/large-field split, and it merges this step with the large-field comparison already on the list.

23.1 Separated responses and an explicit majorant (Part 1)

Use §22's scaled directions and scalar score extensions, fixed insertion coordinates and fixed barrier. For $n \ge 2$ mutually separated labels, $d\mu \propto e^{-S-W}d\xi$ and $dG \propto e^{-V_2}d\xi$ give

$$\partial_{a_1} \cdots \partial_{a_n} r = (-1)^{n-1}\{\kappa_\mu(A_{a_1}, \dots, A_{a_n}) - \kappa_G(\sigma_{a_1}, \dots, \sigma_{a_n})\}. \tag{103}$$

Here r has the free-energy sign of (90); deterministic local terms vanish for separated labels. Define $\Gamma(h, g) = \nabla h \cdot R\nabla g$. At finite k, (22) applied to a product, followed by the moment–cumulant identity, gives

$$\kappa(h_1, \dots, h_s, g) = t\sum_{i=1}^{s} \kappa(h_1, \dots, \widehat{h}_i, \dots, h_s, \Gamma(h_i, g)).$$

One verification is to apply (22) to $e^{\sum z_i h_i}$ and g, divide by its expectation, and compare the coefficient of $z_1 \cdots z_s$; finite-order truncation and integrable cutoffs suffice for this identity.

Iterating with the composite observable last proves the exact formula

$$\kappa(A_1,\dots,A_n)=t^{n-1}\sum_{\pi\in\mathfrak{S}_{n-1}}E\Gamma(A_{\pi(n-1)},\Gamma(A_{\pi(n-2)},\dots,\Gamma(A_{\pi(1)},A_n)\dots)). \tag{104}$$

Each contraction attaches a fresh score to the existing connected expression. These are rooted contraction histories; expansion by the product rule gives decorated trees, with differentiated resolvents at internal vertices. Their multiplicities are retained in (105), rather than identifying the histories with the n^{n-2} labelled simple trees.

Here is a finite-order bound with every combinatorial coefficient explicit. Use §21's tree-weighted one-index marginal norm, now with block L^{4n} norms for elementary tensors. Set $v_0=F$, $v_1=D_F$, and $v_r=J^{\mathrm{sc}}_{r+1}$ for $2\le r\le n-2$, where $\|D^j_\xi\widehat{A}\|_{\gamma,\mathrm{marg},4n}\le J^{\mathrm{sc}}_j\varepsilon/t$ defines the higher insertion score jets. Thus $J^{\mathrm{sc}}_1=F$, $J^{\mathrm{sc}}_2=D_F$; these symbols are separate from §22's local map constants. For $3\le j\le n$ put

$$B_j=t\|D^jS\|_{\gamma,\mathrm{marg},4n}+t\varepsilon^{-j}K_{j,4n}\left(\frac{p}{1-p}\right)^{1/(4n)},\qquad K_{j,P}=\sup_{k,\xi}\varepsilon^j\|D^jw_k(\xi)\|_Fe^{-w_k(\xi)/P}.$$

The edge-removal argument proves this majorant for tD^jV whenever $K_{j,4n}<\infty$; finite-k versions use the same supremum at that k. Both the score jets and B_j may depend on t and order. Uniformity is an explicit premise wherever these constants are used uniformly.

For a nonnegative jet vector h, define recursively

$$\begin{aligned}U_0(h)&=h_0/m,\\ U_r(h)&=\frac{h_r+\sum_{j=2}^{r+1}\binom{r+1}{j}B_{j+1}U_{r+1-j}(h)}{(r+1)m},\\ \mathcal{B}(v,h)_r&=\sum_{l=0}^{r+1}\binom{r+1}{l}v_lU_{r+1-l}(h).\end{aligned} \tag{105}$$

Start $h^{(0)}=v$ and take $h^{(s+1)}=\mathcal{B}(v,h^{(s)})$, losing one available jet each time. Write $H_n=(F/m)h_0^{(n-2)}$. The Gaussian quantity H^G_n uses $B_j=0$, $v^G_0=\sqrt{4n}(G_0+A_0)$, $v^G_1=D_G$ and $v^G_r=0$ for $r\ge2$; Gaussian moment recursion bounds its affine gradient in L^{4n} by this $v^G_0\varepsilon/t$. Then the ordered separated marginal is bounded by

$$\sup_{a_1}\sum_{a_2,\dots,a_n\ \mathrm{separated}}e^{\gamma\ell(a_1,\dots,a_n)}|\partial_{a_1}\cdots\partial_{a_n}r|\le(n-1)!\,\frac{\varepsilon^n}{t}(H_n+H^G_n). \tag{106}$$

To check (105), differentiate the scalar Poisson equation $r+1$ times. The rank-$(r+1)$ resolvent has norm at most $1/((r+1)m)$; the remaining commutators are $\binom{r+1}{j}tD^{j+1}V\,D^{r+1-j}u$, $j\ge2$. Leibniz gives $\mathcal{B}$. Tree weights multiply under contraction. Every expanded term contains n score factors and at most $n-2$ higher-potential factors, so Hölder with elementary L^{4n} bounds and the L^p semigroup estimates justifies each intervening product. Finally (104) supplies $(n-1)!$. Passage to the barrier limit requires these bounds uniformly in k and convergence of the products in (104). This is a conditional all-order majorant; §22 already proves the sharper third-order estimate with its lower moment requirements.

23.2 Contacts, paths and the growth verdict (Part 2)

For arbitrary distinct links let $S_B=\partial_BS$, $\sigma_B=\partial_BV_2$. The exact contact formula, with $\kappa(h)=Eh$, is

$$\partial_Xr=\sum_{\pi\in\mathcal{P}(X)}(-1)^{|\pi|-1}\{\kappa_\mu(S_B:B\in\pi)-\kappa_G(\sigma_B:B\in\pi)\}. \tag{107}$$

Thus order two adds ES_{ab}, and order three adds ES_{abc} and the three negative covariances displayed after (90). Induction uses $\partial_a EF = E\partial_a F - \mathrm{Cov}(F, A_a)$: differentiating a block inserts a into it, while differentiating the law creates a new singleton block. This generates every partition exactly once.

For contact blocks of size b, include the local constants $M_{b,j} = (t/\varepsilon^b)\|D^j_\xi S_B\|_{\gamma,\mathrm{marg},4n}$ for $1 \le j \le n$, and the corresponding L^{4n} norm $M_{b,0}$ for $j = 0$, including Gaussian counterparts. The marginal includes all boundary labels of B. These are mixed jets of the same local action, cutoff and coordinate maps used in §22; locality bounds their support, and compact-chart smoothness makes each fixed-order constant finite. Their order growth and powers of ε remain in $M_{b,j}$. Apply (105) with these jets at each vertex of (104). A partition with s blocks contributes at most $(s-1)!\varepsilon^n/t$ times its contraction majorant; for $s = 1$ use $M_{n,0}$. Summing these bounds over partitions proves a contact bound with explicit multiplicities: there are $\left\{{n \atop s}\right\}$ partitions with s blocks, each carrying $(s-1)!$ histories. This bounds every contact by local mixed jets and the same B_j, m; a cubic remainder scale needs the cancellations between interacting and Gaussian contact terms, especially at orders one and two. Separate absolute bounds can retain the larger ε^2/t.

Admissibility in (94) means one fixed product of link paths, each depending only on its own parameter, of speed $\le \varepsilon$, such that the entire parameter cube, including all corners, stays inside the same chart with all §§14, 17–22 inequalities uniform and insertion coordinates fixed. The barrier and its domain stay independent of these parameters. A sufficient local construction is a product coordinate box contained in the open chart, with straight coordinate segments rescaled to this speed. Its existence near an interior background follows by continuity at finite volume; a box of the required size for every target configuration is an additional geometric requirement. The based-cycle data remain included.

Growth verdict. For distinct labels the ordered marginal in (106) is exactly $(n-1)!$ times the unordered marginal in (94). Accordingly its separated bound is $\varepsilon^n(H_n + H^G_n)/t$: permutation counting alone supplies a factorial upper bound in the ordered convention, and cancels in the required convention. Remaining jet and contact growth decides whether a fixed q works. The present profile already obstructs this particular all-order majorant: near $r = 2\varepsilon$, $w = (r-2\varepsilon)^4_+/\varepsilon^4 + O((r-2\varepsilon)^5_+/\varepsilon^5)$, so its fourth radial derivative jumps by $24/\varepsilon^4$. Mollification makes the fifth jet of size proportional to $\varepsilon^{-4}\rho_k^{-1}$ on a shell of width ρ_k. Hence $K_{5,P} = \infty$ for every finite P, and its L^P norm diverges for $P > 1$ at fixed t as $k \to \infty$. The density is positive at this inner shell; the barrier weight tends to one there. This is an obstruction to differentiated-resolvent estimates, while finite-order boundary cumulants themselves remain well defined.

A constructive remedy is a smoother flat onset with quantified jets, or a connected expansion controlling the barrier without differentiating it. Such a route must bound the *unordered* contact majorant geometrically, retain the small remainder activity (including orders one and two), and prove $q < e^\eta - 1$ for some $0 < \eta \le \gamma$. A residual $n!c^n$ upper bound cannot be absorbed by fixed small t: $(n!c^n)^{1/n} \sim cn/e$. Connected expansion weights with compensating factorials or stronger cancellations are then required. Equations (103)–(107) therefore leave (94), its bare activity scale and geometric order growth open on this chart.

24. Small-field and large-field polymers

Uniform convexity upgrades the one-edge tail to a joint bound with $C = 1$ and $c = 1/16$.

Internal scope check. ACCEPT (108)–(111), the counting behind (112), and (113) as a sufficient target. Checked: (110) from the Hessian bound $3/(2t)$, which survives linear tilts, so (13) bounds every tilted variance by $2t/3$ and the log moment-generating function by $t|v|^2/3$; the Chernoff step with 6^M sign choices; $d \ge \varepsilon$ from (83); and the conversion to $p^{|H|/16}$. An independent route gives the same joint bound with a slightly better constant: uniform log-concavity yields a logarithmic Sobolev inequality (Bakry–Émery), and Gaussian concentration

of the 1-Lipschitz function $(\sum_{e\in H}|\xi_e - s_e|^2)^{1/2}$, whose mean is at most $\sqrt{2|H|t}$ by (85), gives $\exp\{-\frac{3}{4t}(d-\sqrt{2t})^2|H|\}$ with no factor $6^{|H|}$. The scale remark in §24.2 is right and worth keeping in view: the pair and triple constants certify $t^{\alpha-2\delta}$ for the bare remainder, and the missing $t^{2\delta}$ must come from cancellation or the energy-budget form of (62). The resulting connected-set weights retain correlations between components; their conversion into a convergent polymer expansion requires the additional estimates in Part 2.

24.1 Exact split and a joint bad-set estimate (Part 1)

Work with §20's extended S_λ and finite-k profile. For any $J \subseteq \mathcal{E}$, write $\nu_J \propto e^{-S_\lambda - \sum_{e\in J} w_k}$, Z_J for its integral, and $\nu_\varnothing = \nu_0$. Fix insertion coordinates and set $b_e = 1_{\{|\xi_e|>\eta-\rho_k\}}$, $g_e = 1 - b_e$. Then the exact indicator decomposition, with every barrier undifferentiated, is

$$Z_{\mathcal{E}} = \sum_{B\subseteq\mathcal{E}} Z(B), \qquad Z(B) = \int e^{-S_\lambda - \sum_e w_k} \prod_{e\in B} b_e \prod_{e\notin B} g_e \, d\xi. \tag{108}$$

Join two edges when they border the same mid-face. This graph has degree at most $\Delta = 6$. A bad polymer is a nonempty connected component H of B. Compatible polymers are disjoint and have no joining graph edge. Define the exact sector activity of a compatible family $\mathcal{H}$ by $a_{\mathrm{sec}}(\mathcal{H}) = Z(\bigcup\mathcal{H})/Z_{\mathcal{E}}$, including the empty family; (108) says their sum is one.

A useful insertion version puts $f_e = e^{-w_k} - 1$, so $|f_e| \le b_e$, and

$$\frac{Z_{\mathcal{E}}}{Z_0} = \sum_{\mathcal{H}\ \text{compatible}} a_{\mathrm{ins}}(\mathcal{H}), \qquad a_{\mathrm{ins}}(\mathcal{H}) = E_{\nu_0} \prod_{e\in\bigcup\mathcal{H}} f_e, \quad a_{\mathrm{ins}}(\varnothing) = 1. \tag{109}$$

In particular $a_{\mathrm{ins}}(\{H\})$ defines a connected insertion activity. Both family activities generally differ from products of single-polymer activities: the background measure couples the components. Equation (109) is an exact expansion with correlated weights, prior to any factorization.

Here is a joint estimate uniform in J, k, λ, N and in the admissible boundary chart. Retain §§20–22's conditions and impose, with $x = \varepsilon^2/t = t^{-2\delta}$, the explicit additional smallness condition $x \ge 64\log 6$. Then $n = \lfloor 3x/8\rfloor - 1 \ge 4$, and (83) gives $h + \rho_k \le 4\varepsilon\log 2/(7n) \le \varepsilon/7$. Thus every bad edge satisfies $|\xi_e - s_{\lambda,e}| > d$, where $d = 8\varepsilon/7 - h - \rho_k \ge \varepsilon$.

Proof by linear exponential moments. Diagonal dominance in §20 gives $\nabla^2(S_\lambda + \sum_{e\in J} w_k) \ge 3I/(2t)$. A linear tilt preserves this bound. Applying (13) to that tilted measure and integrating the second derivative of its log moment-generating function twice yields

$$E_{\nu_J} e^{v\cdot(\xi - E\xi)} \le e^{t|v|^2/3}. \tag{110}$$

The finite-k convex Gaussian tails justify the tilts. Equation (85) at $q = 2$ gives $|E\xi_e - s_{\lambda,e}| \le \sqrt{2t}$. For each bad edge one of its six signed coordinate projections exceeds $d/\sqrt{3}$. For a fixed choice on each of M edges, their sum exceeds $Md/\sqrt{3}$, its mean is at most $M\sqrt{2t}$, and its coefficient vector has squared norm M. Chernoff optimization of (110), followed by the 6^M choices, proves

$$\nu_J(b_e = 1\ \forall e\in H) \le \left[6\exp\left\{-\frac{3}{4t}(d/\sqrt{3} - \sqrt{2t})^2\right\}\right]^{|H|} \le e^{-x|H|/32} \le p^{|H|/16}, \quad p = 2(7/16)^n. \tag{111}$$

Indeed $x \ge 24$ gives $d/\sqrt{3} - \sqrt{2t} \ge \varepsilon/(2\sqrt{3})$; $x \ge 32\log 6$ absorbs the factor 6. Finally $\log(1/p) \le n\log(16/7) \le 3x/8$, so the last inequality follows. This proof holds for every set H, including connected sets, and passes to the barrier limit by dominated convergence. Consequently $|a_{\mathrm{ins}}(\mathcal{H})|, a_{\mathrm{sec}}(\mathcal{H}) \le p^{|\bigcup\mathcal{H}|/16}$ for nonempty families.

Successive edge removal using (85) alone supplies marginal probabilities; conditioning on earlier bad events changes that class of measures. The linear-tilt argument supplies the missing joint step using the already proved convexity constant, with no barrier derivatives. Its scope is the extended chart model; comparison to the original group integral outside that chart still requires a separate estimate.

24.2 Conditions for geometric responses and the bare scale (Part 2)

The joint estimate permits continuing past the proposed stop rule. Three additional conditions would complete this route to (94).

Small-field expansion and component correlations. On the sector $g_e = 1$ every barrier vanishes. A connected expansion of its interacting integral relative to the covariant Gaussian must control its boundary scores, its truncation surface, and the correlations in (109), uniformly in volume. Require dressed connected activities z_H (after these correlations have been expanded), with $|z_H| \le b_*^{|H|}$ and the same bound in the source domain used below. A useful explicit summability budget on the degree-six graph is

$$\sup_e \sum_{H \ni e} |z_H| e^{(1+\eta)|H|} \le \frac{1}{7}, \qquad e^{1+\eta} b_* \le \frac{1}{43}. \tag{112}$$

Indeed a canonical spanning-tree traversal bounds the number of size-j sets through an edge by $6^{2(j-1)} = 36^{j-1}$. The resulting geometric sum is $e^{1+\eta} b_* / (1 - 36 e^{1+\eta} b_*)$. Each polymer H has at most $7|H|$ sites in its closed graph neighbourhood, so (112) bounds its incompatibility sum by $|H|$, with an exponential spatial reserve η. The raw activities satisfy this numerical budget if $x \ge 32(1 + \eta + \log 43)$, by (111). Passing that bound to the dressed activities, including clusters linking distant components, is additional work.

Contacts and scale. Combine the interacting, Gaussian and deterministic local subtractions before taking absolute values in (107). Require this cancellation for singleton and adjacent-link contacts as well as separated responses, preserving t^α for the bare remainder r. The proved $C_{27}\varepsilon^3/t$ and $C_3\varepsilon^3/t$ equal $C_{27} t^\alpha t^{-2\delta}$ and $C_3 t^\alpha t^{-2\delta}$. Thus those constants certify a weaker bound; recovering the missing $t^{2\delta}$ requires further cancellation or the energy-budget variant of (62). Exponential bad-set rarity can absorb any fixed power of t^{-1}; its derivative costs at unbounded order still need uniform control.

All-order source control. One precise sufficient target is an absolutely convergent subtracted connected expansion $r = \sum_C R_C$, with boundary support A_C, analytic on complex polydiscs of radius $R > 0$ about every point of the admissible real path cube. Require constants $A, u, R > 0$, independent of t, N, k, such that

$$\sup_a \sum_{C : a \in A_C} e^{\eta \ell(A_C)} \|R_C\|_R (1 + u/R)^{|A_C|} \le A t^\alpha, \qquad q = u^{-1} < e^\eta - 1. \tag{113}$$

Cauchy's bound for distinct labels is $\|R_C\|_R R^{-|X|}$, with no factorial. Expanding the binomial in (113) and using $\ell(X) \le \ell(A_C)$ gives (94) at each order with precisely A, q, η. This also specifies the source margin required in (112). The paths must satisfy §23.2's full product-cube condition and cover the target backgrounds.

Chart verdict. Equations (108)–(111) and the raw summability budget hold with the stated extra smallness conditions. Sections 20–22 give real convexity and the pair and separated-triple constants; they leave (112) for dressed activities, (113), low-order contact cancellation at the bare scale, and target-covering paths unproved. The split avoids the fifth-barrier-jet obstruction, while geometric growth in (94) remains open.

25. The scale choice and truncated bad-set correlations

The weaker error exponent absorbs the proved bare pair and triple scales while preserving the small-field domain.

Internal scope check. ACCEPT. §25.1: with $\varepsilon = t^\beta$, $\varepsilon^3/t = t^{3\beta-1} = t^{1/2-3\delta} = t^{\alpha'}$, and $\alpha' > 2\delta$ iff $\delta < 1/10$; the inventory is consistent, so Hypothesis P holds with the error exponent α' on the domain fixed by β, and the bare-scale request is met for the separated pair and triple bounds, with the singleton and adjacent contacts still owed. (115): checked $|\nabla f_e| \le$

$e^{-w_k/2}|\nabla w_k|b_e \le (\Omega_{\mathrm{app}}/\varepsilon)b_e$, the orthogonal-block gradient norm with (111), the covariance bound (21) at $\kappa = \frac{1}{2}$, and $\sup_M (M/2)e^{-LM/64} = 32/(\mathrm{e}L)$. One remark on the dressing step. The objection that distance decay $(5/12)^d$ cannot pay for good edges along connecting paths applies to supports enlarged by those paths, whose number grows like 36^n. The tree-graph form (116) sums instead over the positions of the components: on the planar edge graph only $O(d)$ edges lie at graph distance d from a given one, so $\sum_d O(d)e^{-\gamma_* d} < \infty$ for every $\gamma_* > 0$, and the rarity $p^{c_*|H|}$ of the components themselves pays for their own entropy. Once (116) holds, a standard cluster expansion for polymers with tree-decaying correlations should give the dressed bound (112) without enlarging supports. Throughout this section retain all chart hypotheses of §22.3 and §24.1 and take $0 < t \le 1$.

25.1 Scale choice (Part 1)

Distinguish the chart exponent $\beta = 1/2 - \delta$ from the error exponent:

$$\varepsilon = t^\beta, \qquad \alpha' = 1/2 - 3\delta = \beta - 2\delta, \qquad \varepsilon^3/t = t^{\alpha'}, \qquad \alpha' > 2\delta \iff 0 < \delta < 1/10. \tag{114}$$

Here the equivalence is within the standing range $\delta > 0$. Hypothesis P, §4 (**passage**) fixes the domain using β and permits any error exponent exceeding 2δ. Thus fix $\delta < 1/10$; all earlier $\delta < 1/6$ conditions survive. For $t \le 1$, $t^\beta \le t^{\alpha'}$. The complete exponent inventory in §§13–24 is as follows; earlier occurrences of $\alpha = \beta$ keep that meaning.

Location	Meaning and consistent weaker statement
§13, (53)–(58)	The chart uses $\varepsilon = t^\beta$. Determinant Young bounds require only an error exponent $\le 1/2$; classical, amplitude and Laplace bounds weaken to $t^{\alpha'}$ with the same constants. If reoptimizing the exponential tail, use $B_{\delta,\alpha'} = [\alpha'/(\delta \mathrm{e})]^{\alpha'/(2\delta)}$.
§14, (62)–(64)	The energy-budget bound weakens with unchanged C_u; (63) gives $2A_u\sqrt{t} \le 2A_u t^{\alpha'}$ per face. Conditional spatial decay keeps its original hypotheses.
§15, (65)–(67) and §15.3	The identity $2\varepsilon = 2t^\alpha$ and the tail optimization use β; $t/c_0 \le t^{1/2+\delta}/2$ stays unchanged. Small edge costs admit $t^{\alpha'}$; the requested relative kernel gain still has the stronger factor t^β.
§16	The tail constant becomes $[\alpha'/(2\delta c\mathrm{e})]^{\alpha'/(2\delta)}$, $c = 3/16$, if reoptimized; root notation $\alpha(X), \alpha(\theta)$ denotes a Lie root.
§§17–19	The shrinking chart, path velocity, scores and resolvents are expressed in ε, t; keep them fixed. Their implicit inherited exponent is β.
§20, (83)–(89)	The condition $nh/\varepsilon = O(t^{1/2-3\delta} + t^{1/2-\delta})$ still tends to zero. The relative unscaled gain is t^β; the bare scaled pair bound is $C_{27}t^{\alpha'}$.
§21, (93)–(95)	The separated triple bound is $C_3 t^{\alpha'}$. State the sufficient all-order target (94) with α'; its implication (95) then carries $Ae^\eta t^{\alpha'}$, with unchanged q, η.

Location	Meaning and consistent weaker statement
§22	The constants closing (89), (93) are unchanged; both bare response bounds use (114).
§23, (103)–(107)	For orders $j \geq 3$, $\varepsilon^j/t = t^{\alpha'}\varepsilon^{j-3} \leq t^{\alpha'}$; higher-jet growth and contacts remain separate obligations.
§24, (108)–(113)	Joint rarity and (112) depend on $x = t^{-2\delta}$, unchanged. Replace the target prefactor in (113) by $At^{\alpha'}$, consistently with (94)–(95).

The exact chart identities and the stronger relative kernel statement genuinely use β; the error budget itself admits α' throughout. This resolves the extra $t^{2\delta}$ request for the proved separated pair and triple bounds. Singleton and adjacent-link cancellations still need proof at the cubic scale: separate terms in (107) can cost $\varepsilon^2/t = t^{-2\delta}$, larger than $t^{\alpha'}$ by $1/\varepsilon$. The revised (94) and (113) remain sufficient targets, with their original path and all-order hypotheses; the scale choice alone proves neither.

25.2 Truncated correlations and the remaining dressing step (Part 2)

Let H_1, H_2 be disjoint nonempty edge sets, $h_i = |H_i|$, $M = h_1 + h_2$, $d = d(H_1, H_2)$ and $F_H = \prod_{e\in H} f_e$. Use §24.1's finite-k measures ν_J, including $J = \varnothing$ as required in (109). Under (83) and $x \geq 64\log 6$, uniformly in J, k, N, λ and the admissible real background, one has

$$\begin{aligned} |E_{\nu_J} F_{H_1\cup H_2} - E_{\nu_J} F_{H_1} E_{\nu_J} F_{H_2}| &\leq C_{\mathrm{bad}} p^{M/64} e^{-\gamma_{\mathrm{bad}} d}, \\ C_{\mathrm{bad}} = \frac{64\Omega_{\mathrm{app}}^2}{7\mathrm{e}\log 2}, &\qquad \gamma_{\mathrm{bad}} = \log(12/5). \end{aligned} \tag{115}$$

The profile constant Ω_{app} is exactly (84).

Proof. Since w_k vanishes on the good ball, (84) gives $|\nabla f_e| = e^{-w_k}|\nabla w_k| \leq (\Omega_{\mathrm{app}}/\varepsilon) b_e$, while $|f_e| \leq b_e$. Different edge derivatives occupy orthogonal coordinate blocks. Thus

$$|\nabla F_H|^2 \leq \frac{\Omega_{\mathrm{app}}^2}{\varepsilon^2}|H| \prod_{e\in H} b_e, \qquad \|\nabla F_H\|_{L^2(\nu_J)} \leq \frac{\Omega_{\mathrm{app}}}{\varepsilon}\sqrt{|H|}\, p^{|H|/32}.$$

The last step is (111), applied to the whole set, including the differentiated edge. The smooth part satisfies (20) with $\kappa = 1/2$; each retained barrier is on-site convex. Hence (21), on gradient supports H_1, H_2, bounds the covariance by $(2\Omega_{\mathrm{app}}^2/7)(t/\varepsilon^2)\sqrt{h_1 h_2}\, p^{M/32}(5/12)^d$. Put $L = \log(1/p) \geq \log 2$. The inequalities $\sqrt{h_1 h_2} \leq M/2$ and $\sup_{M>0}(M/2)e^{-LM/64} = 32/(\mathrm{e}L)$ absorb the cardinality factor into $p^{M/64}$; $t/\varepsilon^2 = t^{2\delta} \leq 1$ proves (115). Bounded products and Gaussian domination give convergence of their expectations at fixed volume as $k \to \infty$, so the same uniform bound holds in the barrier limit. □

Equation (115) controls the second truncated correlation of bad components, with both rarity and distance decay. The exact moment–cumulant identity for $E\prod_i F_{H_i}$ also contains every $\kappa_{\nu_0}(F_{H_1}, \ldots, F_{H_r})$, $r \geq 3$. A concrete remaining estimate is the all-order tree bound

$$|\kappa_{\nu_0}(F_{H_1}, \ldots, F_{H_r})| \leq B_*^r p^{c_* \sum_i |H_i|} \sum_{T\in\mathcal{T}_r} \prod_{ij\in T} e^{-\gamma_* d(H_i, H_j)}. \tag{116}$$

Here $\mathcal{T}_r$ is the set of labelled spanning trees, and fixed $B_* \geq \max(1, \sqrt{C_{\mathrm{bad}}})$, $0 < c_* \leq 1/64$, $0 < \gamma_* \leq \gamma_{\mathrm{bad}}$ must work uniformly in r, k, N, t for compatible families. Equation (115) proves its $r = 2$ case only. Expanding the correlated family weights into such connected blocks, controlling

the hard-core incompatibilities, and proving their summed norm (112) are the missing dressing step. A shortest connecting path of length d receives only $e^{-\gamma_{\mathrm{bad}} d}$ from (115); rarity pays for M bad edges, rather than the intervening good edges. Consequently the raw $p^{|H|/16}$ budget cannot simply be assigned to the enlarged connected supports. The source-domain version and the subtracted contact bounds in (113), now with $t^{\alpha'}$, remain additional obligations. Real convex covariance decay controls (115); it supplies neither complex-source control nor the higher connected bounds asserted as targets in (116). This is the stopping point.

26. Low-order contact cancellation

On the barriered chart, the complete responses of orders one through three obey (94) with exponent α' and explicit constants below.

Internal scope check. ACCEPT. The key step checks: per face, $F - F_0$ and $B(b,\xi) - 2|\xi|^2$ have zero two-jets (B is even in b with $B(0,\xi) = 2|\xi|^2$), and $\frac{1}{4}[|Y + C\xi|^2 - |w + C\xi|^2] = \frac{1}{4}\langle Y - w, Y + w + 2C\xi\rangle$ is cubic because $Y - w = O(b^2)$; so $T = S - V_2$ is cubic in the small variables, each boundary derivative at velocity ε costs one factor ε, and every $\partial_B T$ with $|B| \le 3$ is $O(\varepsilon^3/t) = O(t^{\alpha'})$ as in (117). The change of measure in (118) costs $\|\nabla\sigma_B\|\, E|X - Y| \le (G_c\varepsilon^{|B|}/t)(L_2\varepsilon^2/m)$ by the stationary coupling, and amplitudes cost $\varepsilon^{|B|} \le \varepsilon^3/t$. With (89) and (93) valid for all label pairs and triples, (119) gives (94) through order three, contacts included, on admissible chart boxes. Retain §22.3, (114), and $20\varepsilon \le r_0 = 1/16$. Derivatives use fixed insertion coordinates and straight boundary-coordinate segments with velocity ε; repeated labels mean commuting coordinate derivatives on one link. The assertion is uniform on any admissible product box of §23.2.

26.1 Local blocks, including their constant parts

Use the physical local potential on D, restoring all boundary-only constants to S and V_2 before applying (107). Such constants leave all insertion estimates unchanged. Write $T_B = \partial_B(S - V_2)$. The common old-face energy in (59) cancels identically, including every boundary derivative. The remaining local energy difference is cubic in small variables: per face use $z = (Y, b_1, \dots, b_4, \xi_1, \dots, \xi_4)$, $|z|_1 \le 19\varepsilon$, and compact transport parameters v. Indeed $F - F_0$ has zero two-jet, $B(b,\xi) - 2|\xi|^2$ has zero two-jet, and $w - Y = O(b^2)$ makes $[|Y + C\xi|^2 - |w + C\xi|^2]/4$ cubic. These statements hold for every v, so transport derivatives preserve their vanishing jets. Assign half of each bridge to each incident face as in §14.2.

Here are fixed local supremum definitions for the additional constants. Let $J \ge 1$ bound the sum norms of boundary-coordinate derivatives through order three of (v, z), with insertion variables held fixed. Let $K_E \ge 1$ bound the sum of all joint derivative norms through order six of these local energy differences and of the original and Gaussian local energies, on the fixed enlarged chart. Let $K_J \ge 1$ bound the corresponding local log-amplitude jets through order six, including boundary-only amplitudes. Their uniformity follows from the analytic zero-image factors and differentiated image bounds used in §22.2. These are unscaled, fixed-dimensional suprema, rather than suprema of the desired response. Compact transport charts retain free-cycle holonomies. Set $D_c = 100^4 e^{6\gamma}$, $\gamma = \frac{1}{2}\log(10/7)$, and

$$C_L = D_c J^3(2000K_E + 5K_J), \qquad H_2 = 3000 D_c J^3(K_E + K_J).$$

Taylor's formula gives $\|D^j(T_{\mathrm{energy}})\| \le K_E(20\varepsilon)^{3-j}$ for $j \le 3$, before division by t. The chain rule then bounds each $\partial_B T_{\mathrm{energy}}$ by $2000 J^3 K_E \varepsilon^3/t$ for $1 \le |B| \le 3$. Amplitude derivatives cost at most $5J^3 K_J \varepsilon^{|B|}$; $\varepsilon \le \varepsilon^3/t$ absorbs them. Therefore

$$\|E_\mu T_B\|_{\gamma,\mathrm{marg}} \le C_L \varepsilon^3/t, \qquad 1 \le |B| \le 3. \tag{117}$$

Here and below the marginal fixes any one boundary label and sums the others, allowing repetitions. A local term has at most 48 directional boundary labels and seven insertion blocks;

a fixed label meets fewer than 100 terms. Through order three their incidence sums are below 100^4, and the joining trees have at most six edges. This proves the count used in D_c. The scalar extensions of §22.1 and their product rule also give gradient marginals $H_2\varepsilon^2/t$ for both S_{ab} and σ_{ab}. The six-jet suprema cover the needed mixed boundary jets, extending the one-boundary jets J_2, J_3, M_T of §22.

26.2 Change of measure and assembly

Equation (85) and the drift estimate after it sharpen (79) to

$$\sup_e E_\mu|\nabla_e(T+W_k)| \le L_2\varepsilon^2/t, \qquad L_2 = B + 3D_J + (1+\sqrt{2})C_* + \Omega_{\mathrm{app}}.$$

Use $h \le \varepsilon$, $\sqrt{t} \le \varepsilon$, $t \le \varepsilon^2$ and $p/(1-p) \le \varepsilon^3/t$. In particular the saddle displacement $B\varepsilon^2$ is retained. The stationary coupling in §19.2, which includes the barrier endpoint of §18.2, now gives $\sup_e E|X_e - Y_e| \le L_2\varepsilon^2/m$. Let $b_c \ge b$ bound the linear and quadratic coefficients of every $\varepsilon^{-|B|}t\sigma_B$, $1 \le |B| \le 3$, on the fixed chart. Put $G_c = 1000D_cb_c$ and $E_c = 1000D_cb_c$. The extensions (86) have gradient marginals $G_c\varepsilon^{|B|}/t$. Their Gaussian mean errors are at most $E_c\varepsilon^{|B|+2}p_G^{1/2}/t$: apply Cauchy–Schwarz to the quadratic cutoff error and $E_G|\xi_i|^4 \le 8\varepsilon^4$. Constants in σ_B agree exactly at both ends. Consequently

$$\|E_\mu S_B - E_G\sigma_B\|_{\gamma,\mathrm{marg}} \le C_M\varepsilon^3/t, \qquad C_M = C_L + G_cL_2/m + E_c. \tag{118}$$

This proves the expectation-block cancellation before absolute values. All comparisons can first use finite k and the scalar extensions; (85) supplies domination, and the barrier limit restores the physical blocks.

The proof of (89) bounds the covariance difference for **all** label pairs; its separation restriction only removed the expectation block from the derivative of r. Likewise (91)–(99) bound the third cumulant difference for all triples, including repeated labels. Use C_{27} with $p_G^{1/4}$ replaced by 1, and C_3 of §22.3. Each order-three contact covariance costs at most $H_2[F+G_0+A_0]\varepsilon^3/(mt)$ by (22), for the interacting and Gaussian terms together. The three partitions of type $(2,1)$ thus give

$$\|\partial^n r\|_{\gamma,\mathrm{marg}} \le C_n^* t^{\alpha'}, \quad C_1^* = C_M, \quad C_2^* = C_M + C_{27}, \quad C_3^* = C_M + C_3 + 3H_2(F+G_0+A_0)/m. \tag{119}$$

The one-block, two-block and three-block terms in (107) exhaust these orders. Taking $\eta = \gamma$, any fixed $q > 0$, and $A = \max_{1\le n\le 3} C_n^*/q^n$ proves (94) through order three: its unordered distinct-link sums are bounded by the ordered marginals.

27. Three bad components: a tree bound from the two-set estimate

The third truncated correlation satisfies (116) with the same rarity exponent $c_* = 1/64$ and decay rate $\gamma_* = \frac{1}{2}\log(12/5)$. This uses (115) for disjoint unions of components and needs no additional chart hypothesis or barrier derivative. It is a finite-order result; the constants for arbitrary order and the dressed expansion (112) remain open.

27.1 The three-component estimate

Retain all hypotheses of §25.2, including (83), $x \ge 64\log 6$ and $0 < t \le 1$. Let H_1, H_2, H_3 be nonempty disjoint edge sets, put $h_i = |H_i|$, $M = h_1+h_2+h_3$, $d_{ij} = d(H_i, H_j)$, and use the same $F_H = \prod_{e\in H} f_e$. Compatibility and connectivity may be imposed as in (109), but are unnecessary for this estimate. For every ν_J, uniformly in J, k, N, λ, t and the admissible real background,

$$\begin{gathered}|\kappa_{\nu_J}(F_{H_1}, F_{H_2}, F_{H_3})| \le 3C_{\mathrm{bad}}p^{M/64}\sum_{T\in\mathcal{T}_3}\prod_{ij\in T} e^{-\gamma_* d_{ij}},\\ \gamma_* = \tfrac{1}{2}\gamma_{\mathrm{bad}} = \tfrac{1}{2}\log(12/5).\end{gathered} \tag{120}$$

In particular the $r = 2, 3$ cases of (116) hold simultaneously with

$$B_* = \max\{1, \sqrt{C_{\mathrm{bad}}}, (3C_{\mathrm{bad}})^{1/3}\}, \qquad c_* = 1/64, \qquad \gamma_* = \tfrac{1}{2}\gamma_{\mathrm{bad}}. \tag{121}$$

Here $C_{\mathrm{bad}} = 64\Omega_{\mathrm{app}}^2/(7\mathrm{e}\log 2)$ and Ω_{app} is the fixed profile constant (84).

Proof. Write $F_i = F_{H_i}$ and $m_i = EF_i$. For any permutation (i, j, k) the exact identity is

$$\kappa(F_1, F_2, F_3) = \mathrm{Cov}(F_i, F_jF_k) - m_j \,\mathrm{Cov}(F_i, F_k) - m_k \,\mathrm{Cov}(F_i, F_j). \tag{122}$$

Equation (111) gives $|m_j| \le p^{h_j/16}$. Apply (115) to H_i and $H_j \cup H_k$ in the first term; its proof and statement allow any disjoint edge sets. In particular $F_jF_k = F_{H_j \cup H_k}$ and $d(H_i, H_j \cup H_k) = \min(d_{ij}, d_{ik})$. The other two terms have rarity at most $p^{h_j/16+(h_i+h_k)/64} \le p^{M/64}$, and similarly with j, k exchanged. Thus every choice of i gives

$$|\kappa(F_1, F_2, F_3)| \le 3C_{\mathrm{bad}} p^{M/64} e^{-\gamma_{\mathrm{bad}} \min(d_{ij}, d_{ik})}.$$

Order the three pair distances as $a \le b \le c$. The largest of the three nearest-neighbour distances $\min(d_{ij}, d_{ik})$ is exactly b. The two pairs realizing a, b form a spanning tree on three labels, even if the distances tie. Since $b \ge (a + b)/2$,

$$e^{-\gamma_{\mathrm{bad}} b} \le e^{-\gamma_*(a+b)} \le \sum_{T \in \mathcal{T}_3} \prod_{ij \in T} e^{-\gamma_* d_{ij}}.$$

This proves (120), and weakening (115)'s decay to γ_* proves (121). No triangle inequality between distances of sets is used. Bounded F_i and Gaussian domination permit passage to the barrier limit at fixed volume, exactly as in §25.2. □

The empty-component case is consistent: $F_\varnothing = 1$ makes the third cumulant zero, and a nontrivial cumulant with a constant argument always vanishes. At diverging component separations (120) tends to zero; at fixed separations and $t \downarrow 0$ its rarity tends to zero with constants independent of plane size. All quantities in (120) are dimensionless chart quantities. This $SU(2)$ statement transfers to another group only after its analogues of (111) and (115) are verified; §16 alone does not verify those later estimates for $SU(3)$.

27.2 What the arbitrary-order step must add

For a finite label set I, write $\kappa_B = \kappa(F_i : i \in B)$, with $\kappa_{\{i\}} = EF_i$. Moment–cumulant inversion across any nontrivial cut $I = A \sqcup B$ gives the exact recurrence

$$\kappa_I = \mathrm{Cov}\left(\prod_{i \in A} F_i, \prod_{j \in B} F_j\right) - \sum_{\substack{\pi \in \mathcal{P}(I),\ \pi \ne \{I\} \\ \exists C \in \pi:\ C \cap A \ne \varnothing,\ C \cap B \ne \varnothing}} \prod_{C \in \pi} \kappa_C. \tag{123}$$

Partitions with no crossing block are exactly the product of the two moment expansions and cancel in the covariance. Formula (122) is its three-label, singleton-cut case.

Equation (115) bounds the first term by $C_{\mathrm{bad}} p^{\sum_i h_i/64} e^{-\gamma_{\mathrm{bad}} d(H_A, H_B)}$, where $H_A = \bigcup_{i \in A} H_i$. It supplies one crossing factor. The remaining partition products in (123) retain connected blocks, but taking their absolute values does not generate a fresh factor for each of the $|I| - 1$ tree edges. The three-component proof uses the special fact that its two shortest edges already span all labels. For r equally distant components a single-cut factor would instead require weakening the rate to $\gamma_{\mathrm{bad}}/(r - 1)$ to pay for a tree. This describes the limitation of this cut estimate, rather than a counterexample to (116).

The next lemma must construct an undifferentiated-barrier expansion that attaches one summable spatial connector per new component, with a fixed positive decay rate and a geometric constant per attachment. Intercomponent interaction interpolation is a candidate: its derivatives

act on smooth coupling terms, while each on-site w_k stays inside its measure. It must preserve convexity and (111) for every interpolated measure and handle unbounded insertion coordinates. Until that lemma is proved, (120) gives neither arbitrary-order (116), dressing (112), nor the complex-source estimate (113). Abandon a cut-only induction if its decay rate tends to zero with order; abandon the coupling interpolation if convexity or joint rarity fails along its decoupling path.

28. A decoupling family with the barrier retained

Freezing other blocks at the common smooth saddle gives a factorizing reference and a real interpolation family preserving the convexity, local moments, joint rarity (111) and pair decay (115). A single smooth connector has the explicit response bound (128), with gain t^{δ}. The barrier stays inside every interpolated measure. This supplies inputs for a connector expansion; the arbitrary-order tree estimate remains open.

28.1 Freeze at the saddle before removing interactions

Fix the boundary background and λ and put $s = s_\lambda$ from §20.1. Let $\mathcal{P}$ be any partition of all insertion edges into nonempty blocks and let P_A denote coordinate projection onto block A. Define

$$S_{\mathcal{P}}(\xi) = S_\lambda(s) + \sum_{A\in\mathcal{P}} \left[S_\lambda(s + P_A(\xi - s)) - S_\lambda(s)\right]. \tag{124}$$

The one-block partition recovers S_λ. Every $S_{\mathcal{P}}$ has the same value and zero gradient at s. Its Hessian is the direct sum of principal blocks of S''_λ, each evaluated at its own partially frozen configuration. With the on-site barriers added, $\nu_{\mathcal{P},J} \propto e^{-S_{\mathcal{P}} - \sum_{e\in J} w_k(\xi_e)}$ factorizes over $A \in \mathcal{P}$. This exact factorization applies even when the blocks are disconnected. It is an auxiliary fixed-chart construction; it does not assert a new gauge-invariant group action.

Take any convex combination $S_\theta = \sum_{\mathcal{P}} \theta_{\mathcal{P}} S_{\mathcal{P}}$, with $\theta_{\mathcal{P}} \ge 0$ and $\sum \theta_{\mathcal{P}} = 1$. Under §25.2's conditions, uniformly in the partitions and coefficients,

$$\begin{aligned} \|tS''_\theta - 6I\| \le 5/2, \qquad (S''_\theta)_{ee'} = 0 \text{ if } d(e,e') > 1, \\ \nabla^2\left(S_\theta + \sum_{e\in J} w_k\right) \succeq 3I/(2t), \\ \sup_e E_{\theta,J}|\xi_e - s_e|^q \le [2(q+1)t/3]^{q/2} \quad (q \ge 2), \\ \nu_{\theta,J}(b_e = 1 \ \forall e \in H) \le p^{|H|/16}. \end{aligned} \tag{125}$$

Consequently (115) and (120), with unchanged constants, hold throughout this family, including when $J = \varnothing$.

Proof. A principal submatrix of $tS''_\lambda - 6I$ has operator norm at most $5/2$ by (20). A direct sum of such matrices, even evaluated at different points, has the same bound; convex combinations preserve it. Freezing deletes Hessian entries between blocks and creates none, so range one remains. This proves the first line and an even stronger spectral lower bound than the second line.

For the local moment estimate retain §20.1's deterministic entrywise envelope about $5I$: fixed nonnegative $K_{ee'}$ majorize $\|(tS''_\lambda - 5I)_{ee'}\|$, with row and column sums at most $3 + C_*\varepsilon \le 7/2$. Within a block the envelope is restricted to its principal indices and outside it the entries vanish. The same envelope therefore applies to every S_θ, regardless of the frozen configurations. The block maximum argument and integration by parts proving (85) apply with exactly $m = 3/2$ and centre s. The on-site barrier has nonnegative radial contribution about s_e because it is convex and vanishes at s_e; the choice of s leaves its distance to the bad threshold unchanged. Linear tilts preserve the spectral bound, so (110) and the Chernoff proof of (111) apply without

alteration. The gradient estimate for F_H and Theorem 5 then give (115); §27 gives (120). The common saddle and Hessian lower bound give Gaussian domination, allowing the same barrier limit at fixed volume. □

Freezing at arbitrary points would add gradients at the reference centre and require a new saddle displacement budget. Freezing at s_λ is the step that avoids that extra hypothesis. The real family (125) holds at each fixed background; differentiating its moving saddle in boundary-source directions still needs separate control.

28.2 An explicit local connector and its rarity-weighted cost

For a two-block cut $\mathcal{E} = A \sqcup B$, write the smooth potential as the local face sum $S_\lambda = \sum_p \phi_p$, assigning half of every one-edge term to each incident face, as before. Put $y = \xi - s$. The interaction removed at a face meeting both blocks is

$$\begin{aligned} K_p(\xi) &= \phi_p(s+y) - \phi_p(s+P_A y) - \phi_p(s+P_B y) + \phi_p(s), \\ S_\lambda - S_{\{A,B\}} &= \sum_{p \text{ crossing}} K_p. \end{aligned} \tag{126}$$

Each K_p is supported on the four edges of that face. One-edge terms cancel identically. Let $K_{\mathrm{loc}} \geq 1$ bound its parent face's unscaled Hessian row, $\sup_\xi \max_{e\in\partial p} \sum_{e'\in\partial p} t\|D_e D_{e'} \phi_p(\xi)\|$. This is a fixed-dimensional local derivative bound. The Gaussian face row is at most four (two from its half-bridges and two from its curl quadratic form); allowing its local saddle-Hessian perturbation by two, §15.1's product rule permits

$$K_{\mathrm{loc}} = 6 + K_u \varepsilon + 7000\Lambda c_0 + 385 D_J t/c_0 \leq 7 \tag{127}$$

under (65). Convex interpolation with V_2 preserves this choice. The background parameter $\Lambda = L_F + \beta$ here is §15's local constant, not a momentum cutoff.

Group the coordinates of each side of the cut. Differentiating (126) gives

$$D_A K_p = \int_0^1 H_{AB}(s + y_A + v y_B) y_B \, dv,$$

where H_{AB} is the mixed block of the parent face Hessian; the $D_B K_p$ formula integrates along $s + u y_A + y_B$. At each common configuration, symmetry and (127) bound both block row and column sums of H_{AB} by K_{loc}/t, so $\|H_{AB}\|_{\mathrm{op}} \leq K_{\mathrm{loc}}/t$. Consequently

$$|\nabla K_p| \leq \frac{K_{\mathrm{loc}}}{t} \left(\sum_{e\in\partial p} |\xi_e - s_e|^2 \right)^{1/2}, \qquad \|\nabla K_p\|_q \leq \frac{G_q}{\sqrt{t}}, \quad G_q = 2K_{\mathrm{loc}} \sqrt{2(q+1)/3} \quad (q \geq 2).$$

The last step uses (125) and Minkowski on the sum of squared block norms. In particular $G_2 = 2\sqrt{2} K_{\mathrm{loc}}$, preserving the constant used below. This corrects the earlier individual-coordinate argument: a configuration-dependent row bound cannot be pulled outside an L^2 sum as fixed coefficients. The grouped operator estimate requires no new deterministic-envelope hypothesis.

Take the convex path $S_u = S_{\{A,B\}} + u \sum_{p \text{ crossing}} K_p$, $0 \leq u \leq 1$, leaving every w_k unchanged. For a nonempty bad set H, $h = |H|$, one local connector obeys

$$\begin{aligned} |\operatorname{Cov}_{u,J}(F_H, K_p)| &\leq C_{\mathrm{conn}} t^\delta p^{h/64} e^{-\gamma_{\mathrm{bad}} d(H,\partial p)}, \\ C_{\mathrm{conn}} &= \frac{32 \Omega_{\mathrm{app}} K_{\mathrm{loc}}}{7\sqrt{\mathrm{e} \log 2}}. \end{aligned} \tag{128}$$

Indeed Theorem 5 and §25.2's gradient estimate first give

$$\frac{4\sqrt{2}\Omega_{\mathrm{app}} K_{\mathrm{loc}}}{7} \frac{\sqrt{t}}{\varepsilon} \sqrt{h}\, p^{h/32} e^{-\gamma_{\mathrm{bad}} d(H,\partial p)}.$$

Use $\sqrt{t}/\varepsilon = t^{\delta}$ and $\sup_{h>0} \sqrt{h}\, e^{-\log(1/p)h/64} = \sqrt{32/(\mathrm{e}\log(1/p))}$, with $\log(1/p) \geq \log 2$, to obtain (128). The connector grows at most quadratically in $\xi - s$; its gradient grows at most linearly. Thus Gaussian domination justifies the covariance identity and its barrier limit without differentiating the barrier.

Differentiating an expectation along this real path yields

$$\left| \frac{d}{du} E_{u,J} F_H \right| \leq C_{\mathrm{conn}} t^{\delta} p^{h/64} \sum_{p \text{ crossing}} e^{-\gamma_{\mathrm{bad}} d(H,\partial p)}. \tag{129}$$

There is no plane-size cost hidden in this sum. With $q = 5/12$ and $C_{\mathrm{edge}} = 2\sum_{d\geq 0}(4d+3)^2 q^d$ from §7.3's graph counting, the sum over even all faces is at most $2hC_{\mathrm{edge}}$: each edge lies in two faces and $e^{-\gamma_{\mathrm{bad}} \min_{e\in H, f\in\partial p} d(e,f)} \leq \sum_{e\in H, f\in\partial p} q^{d(e,f)}$. For explicit constants,

$$C_{\mathrm{edge}} = 2\left[\frac{16q(1+q)}{(1-q)^3} + \frac{24q}{(1-q)^2} + \frac{9}{1-q}\right].$$

This is a real, one-connector locality estimate with rarity retained. It controls the change of a component's bare activity, rather than a fully dressed activity in (112).

28.3 The surviving arbitrary-order obligation

The tested interaction route preserves convexity and joint rarity, so those two conditions no longer obstruct this particular decoupling. It also prices one attachment by (128). The missing lemma must control **mixed connected responses to several** K_p, with one fixed-rate connector per tree edge and a geometric constant per attachment. Repeated differentiation of an expectation involves cumulants of F_{H_i} and the unbounded quadratic K_p; (128) alone controls neither their growth with order nor the sum over decoupling choices. The real convex simplex of (124) is also smaller than the independent complex-source domain demanded by (113).

Abandon a proposed expansion if it differentiates w_k, requires a vanishing decay rate as the number of components grows, or leaves uncontrolled products of the K_p. A useful next proof must supply a summable multiple-connector norm on the real decoupling family, before dressing (112) and boundary sources (113) are claimed. The group-independent freezing algebra transfers immediately, but the $SU(2)$ tail and profile constants in (125)–(129) still require their explicit $SU(3)$ verification.

29. All-order connector moments and a uniform analytic source ball

The quadratic connectors of §28 have exponential moments uniform over the real decoupling family. Their connected responses with one bad set satisfy (132) at every order, with the rarity exponent independent of that order. The normalized expectation is analytic on the complex ℓ^1 source ball (134). This controls the previously unbounded connector products; a summable spatial tree and the boundary-source polydisc of (113) remain separate obligations.

29.1 The quadratic connector norm

Fix any real measure $\nu_{\theta,J}$ in (125), and take a finite collection of §28.2 connectors with the common saddle s and parent face Hessian row bound $K_{\mathrm{loc}} \leq 7$. Repetitions of a connector label are allowed. No derivative acts on a barrier. Mixed-Hessian rectangle integration in (126) gives

$$|K_p(\xi)| \leq \frac{K_{\mathrm{loc}}}{2t} \sum_{e\in\partial p} |\xi_e - s_e|^2. \tag{130}$$

Indeed integrate the parent face's mixed Hessian at $s + uP_A(\xi - s) + vP_B(\xi - s)$, $0 \leq u, v \leq 1$. At this common configuration its symmetric block matrix has row and column majorants bounded

by K_{loc}/t. Apply $2|y_e||y_f| \le |y_e|^2 + |y_f|^2$ to the cross-block sum and integrate. This proves the factor $1/2$ in (130), also when one side has no edge of the face and $K_p = 0$.

Put $C_K = (8\mathrm{e}/3)K_{\mathrm{loc}} \le 56\mathrm{e}/3$. For every integer $n \ge 1$, (125) and Minkowski give

$$\|K_p\|_{L^n(\nu_{\theta,J})} \le \frac{8K_{\mathrm{loc}}}{3}(n+\tfrac12), \qquad E|K_p|^n \le n!C_K^n. \tag{131}$$

For the first inequality, each of the four squared coordinate blocks has L^n norm at most $2(2n+1)t/3$. For the second use

$$\log(n!) \ge \int_{1/2}^{n+1/2} \log x\,dx = n\log(n+\tfrac12) - n + \tfrac12\log(2n+1).$$

The inequality follows by applying concavity of log on each unit interval centred at $1,\dots,n$. It implies $(n+\frac12)^n \le \mathrm{e}^n n!$. All constants are dimensionless, independent of plane size, barrier approximation, small heat time and real decoupling parameters.

29.2 Connected responses at arbitrary order

For F_H of §25.2 put $h = |H| \ge 1$ and $a_H = p^{h/32}$. Joint rarity (111), $|F_H| \le 1_{B_H}$ and Cauchy–Schwarz give

$$E\left|F_H \prod_{j=1}^{n} K_{p_j}\right| \le a_H C_K^n \sqrt{(2n)!} \le a_H (2C_K)^n n!.$$

Hölder and (131) supply the product's second moment. The last inequality uses $\binom{2n}{n} \le 4^n$. With no F_H insertion, the corresponding moment is at most $C_K^n n!$. For $n = 0$, $|EF_H| \le p^{h/16} \le a_H$.

Moment–cumulant inversion now proves, at every $n \ge 1$,

$$\begin{aligned} |\kappa(F_H, K_{p_1}, \dots, K_{p_n})| &\le a_H C_K^n n!(n+2)2^{n-1} \\ &\le a_H (4C_K)^n n!. \end{aligned} \tag{132}$$

For completeness, distinguish the partition block containing F_H. If it contains k connector labels, its moment costs $a_H(2C_K)^k k!$. The other blocks of sizes m cost $C_K^m m!$. The cumulant coefficient is the factorial of the number of those other blocks. After selecting the k labels, ordering the remaining blocks and choosing their positive sizes, the total coefficient of $a_H C_K^n n!$ is

$$2^n + \sum_{k=0}^{n-1} 2^k 2^{n-k-1} = (n+2)2^{n-1}.$$

Here 2^{n-k-1} counts the compositions of $n-k$; the $k = n$ term is the single block. Thus rarity is spent once, without an exponent that deteriorates with response order. This is an all-order growth estimate, rather than a spatial tree estimate.

29.3 A complex source domain with no partition-function zero

For finitely supported complex $z = (z_p)$ write

$$r = \sum_p |z_p|, \qquad Z(z) = Ee^{-\sum_p z_p K_p}, \qquad G_H(z) = \frac{E[F_H e^{-\sum_p z_p K_p}]}{Z(z)}. \tag{133}$$

Minkowski and (131) give $E(\sum_p |z_p||K_p|)^n \le n!(C_K r)^n$. Hence $Ee^{\sum_p |z_p||K_p|} \le (1 - C_K r)^{-1}$ when $C_K r < 1$. In particular, throughout $r < 1/(2C_K)$,

$$\begin{aligned} |Z(z) - 1| &\le \frac{C_K r}{1 - C_K r}, \qquad |Z(z)| \ge \frac{1 - 2C_K r}{1 - C_K r} > 0, \\ |G_H(z)| &\le a_H \frac{1 - C_K r}{(1 - 2C_K r)^{3/2}}. \end{aligned} \tag{134}$$

The numerator uses Cauchy–Schwarz against 1_{B_H} and the exponential bound at $2r$. Uniform domination on smaller closed balls proves analyticity in each finite collection of sources. On $r \le 1/(4C_K)$ the last bound is $|G_H(z)| \le 3a_H/\sqrt{2}$. Its derivatives at zero are $(-1)^n\kappa(F_H, K_{p_1}, \dots, K_{p_n})$. Higher moments and exponential integrability on a slightly larger ball give uniform integrability for passage to the existing barrier limit at fixed volume. Complex convexity and barrier jets are unused.

This source ball has radius uniform in heat time and plane size. It concerns connector sources at a fixed real background and saddle, rather than the boundary-source variations in (113).

29.4 The spatial estimate still needed

There is no connector-location dependence in (132). Summing it over face labels incurs a volume cost at each order. With m active coordinates a common polydisc fits in (134) only with radius of order m^{-1}, so it does not prove the volume-uniform polydisc in (113). Nor does it supply the dressed per-site budget (112) or a new tree factor for each bad component in (116). These are precise limitations of this norm calculation, not counterexamples to spatial clustering.

The anchored two-connector estimate, proved in §30, is

$$|\kappa(F_H, K_p, K_q)| \le D_2 p^{h/64} \sum_{T \in \mathcal{T}_{\{H,p,q\}}} \prod_{ij \in T} e^{-\gamma_2 d(S_i, S_j)}, \qquad S_H = H, \quad S_p = \partial p, \quad S_q = \partial q. \tag{135}$$

Its constants and the sum over both face labels are explicit there. The moments (131) control the quadratic insertions; Theorem 5 supplies the locality input. This finite-order result still needs an order-uniform attachment construction before dressing is claimed.

30. An anchored two-connector tree and its spatial sum

Equation (135) holds throughout (125), with $\gamma_2 = \gamma_{\mathrm{bad}}/2$ and the explicit D_2 in (139). The sum over both connector locations retains rarity and a positive spatial reserve, as (140) states. No barrier is differentiated and no additional chart or physical hypothesis is introduced. The result is finite-order; arbitrary-order spatial clustering remains open.

30.1 Three cuts with enough rarity left

Fix $0 < t \le 1$ under the existing small-time conditions, a nonempty bad set H, $h = |H|$, and any two §28.2 connectors, including repeated or overlapping labels. Put $K = K_{\mathrm{loc}}$, $\Omega = \Omega_{\mathrm{app}}$, $P = p^{h/16}$ and $\gamma = \log(12/5)$. All expectations below use the same real measure from (125). For $\bar{K}_p = K_p - EK_p$, (131) and $|EK_p| \le 4K$ give

$$\|\bar{K}_p\|_4 \le M_4 = 16K, \qquad \|\bar{K}_p\|_6 \le M_6 = 64K/3. \tag{136}$$

Section 28.2 gives $\|\nabla K_p\|_q \le G_q/\sqrt{t}$, with $G_q = 2K\sqrt{2(q+1)/3}$. Section 25.2 gives the pointwise estimate $|\nabla F_H| \le (\Omega\sqrt{h}/\varepsilon)1_{B_H}$, and $|F_H| \le 1_{B_H}$, $\Pr(B_H) \le P$.

The exact centred identities are

$$\kappa(F_H, K_p, K_q) = \mathrm{Cov}(F_H, \bar{K}_p\bar{K}_q) = \mathrm{Cov}(K_p, (F_H - EF_H)\bar{K}_q),$$

and the identity with p, q interchanged. For the first cut, $\|\nabla(\bar{K}_p\bar{K}_q)\|_2 \le 2M_4G_4/\sqrt{t}$ by Hölder. Theorem 5, with $\kappa = 1/2$, therefore gives

$$|\kappa(F_H, K_p, K_q)| \le \frac{4}{7}\Omega M_4 G_4 t^{\delta}\sqrt{h}\, p^{h/32} e^{-\gamma \min(d(H,\partial p), d(H,\partial q))}. \tag{137}$$

For the connector singleton cut, use sixth moments to obtain

$$\|\nabla[(F_H - EF_H)\bar{K}_q]\|_2 \le P^{1/3}\left[\frac{\Omega M_6\sqrt{h}}{\varepsilon} + \frac{2G_6}{\sqrt{t}}\right].$$

Indeed $\|1_{B_H}\bar{K}_q\|_2 \le P^{1/3}M_6$ and $\|1_{B_H}\nabla K_q\|_2 \le P^{1/3}G_6/\sqrt{t}$; the centring term costs at most $PG_2/\sqrt{t}$, bounded by the same $P^{1/3}G_6/\sqrt{t}$. Theorem 5 now yields

$$|\kappa(F_H, K_p, K_q)| \le \frac{2}{7}G_2[\Omega M_6 t^\delta\sqrt{h} + 2G_6]p^{h/48}e^{-\gamma\min(d(\partial p,H),d(\partial p,\partial q))}. \tag{138}$$

The third cut has the same constant. Fourth moments in (138) would leave only $p^{h/64}$ before absorbing $\sqrt{h}$. Sixth moments provide exactly the extra rarity this absorption needs.

Put

$$\begin{aligned} D_H &= \frac{4}{7}\Omega M_4 G_4\sqrt{32/(\mathrm{e}\log 2)},\\ D_K &= \frac{2}{7}G_2\left[\Omega M_6\sqrt{96/(\mathrm{e}\log 2)} + 2G_6\right],\\ D_2 &= \max(D_H, D_K). \end{aligned} \tag{139}$$

Since $t^\delta \le 1$ and $L = \log(1/p) \ge \log 2$, $\sup_{h>0}\sqrt{h}\,e^{-Lh/64} = \sqrt{32/(\mathrm{e}L)}$ absorbs (137)'s cardinality factor. For (138), the reserve is $1/48 - 1/64 = 1/192$, giving $\sup_{h>0}\sqrt{h}\,e^{-Lh/192} = \sqrt{96/(\mathrm{e}L)}$. Every singleton cut thus costs at most $D_2 p^{h/64}$ times its distance factor.

30.2 Tree conversion and the sum over locations

Order the three pair distances as $a \le b \le c$. The strongest singleton cut has nearest-support distance b. The pairs realizing a, b form a spanning tree on the three labels, and $b \ge (a+b)/2$. Therefore (137)–(139) prove (135), with $\gamma_2 = \gamma/2$. This uses no triangle inequality for distances between sets and permits repeated face labels.

There is also a useful weighted spatial sum. Let $\ell(H, p, q)$ be the minimum spanning-tree length on the three supports. For $0 \le \eta < \gamma/2$ put $r = e^{-(\gamma/2-\eta)}$ and define

$$C_E(r) = 2\left[\frac{16r(1+r)}{(1-r)^3} + \frac{24r}{(1-r)^2} + \frac{9}{1-r}\right].$$

The graph count in §28.2 gives $\sum_p r^{d(H,\partial p)} \le 2hC_E(r)$ and $\sup_p \sum_q r^{d(\partial p,\partial q)} \le 8C_E(r)$. For any tree T, $\ell \le \sum_{ij\in T} d(S_i, S_j)$; multiplying its term in (135) by $e^{\eta\ell}$ therefore leaves rate $\gamma/2 - \eta$ on each edge. The root-star tree costs $4h^2C_E^2$, and the two root-path trees together cost $32hC_E^2$. Hence

$$\begin{aligned} \sum_{p,q} e^{\eta\ell(H,p,q)}|\kappa(F_H, K_p, K_q)| &\le 4D_2C_E(r)^2(h^2 + 8h)p^{h/64}\\ &\le 4D_2C_E(r)^2\left[\frac{128}{\mathrm{e}\log 2} + 8\right]h\,p^{h/128}. \end{aligned} \tag{140}$$

The second line spends additional rarity using $\sup_{h>0} he^{-Lh/128} = 128/(\mathrm{e}L)$. All faces of a finite plane may be summed; restricting to crossing faces improves the bound. The choice $\eta = \gamma/4$ leaves a fixed positive spatial reserve. Constants are independent of plane size, barrier approximation, real decoupling parameters and small heat time.

The observables have polynomial growth, so (125)'s higher moments justify covariance truncation and uniform integrability in the existing fixed-volume barrier limit. These arguments use the convex barriers only inside the measures. All constants are dimensionless: $t\|\nabla K\|\|\nabla F_H\|$ scales as $\sqrt{t}/\varepsilon = t^\delta$, while two connector gradients have scale t^{-1}, cancelling the covariance factor t. At fixed h, rarity still vanishes as $t \downarrow 0$.

30.3 The next attachment must retain a fixed spatial reserve

The sum (140) closes the anchored second connected response and allows the second-order term of a real response expansion to be summed without plane-size cost. It does not give a radius for a uniform source polydisc or a fully dressed activity.

The next lemma must control the anchored sum of $\kappa(F_H, K_{p_1}, \dots, K_{p_n})$ for arbitrary n, with one fixed positive spatial reserve and a geometric constant per attachment after the response factorial is divided out. A sufficient target is a fixed $\eta_*, a_*, \rho_* > 0$ and $A_* < \infty$ such that

$$\sum_{n\ge 1} \frac{\rho_*^n}{n!} \sum_{p_1,\dots,p_n} e^{\eta_* \ell(H,p_1,\dots,p_n)} |\kappa(F_H, K_{p_1}, \dots, K_{p_n})| \le A_* |H| p^{a_* |H|}.$$

This is a target, not an inference from (132) and (140). An independent-attachment forest interpolation retaining the measures in (125) is one possible construction. Stop a cut-only recurrence when its decay rate decreases with order; reject a forest interpolation if its differentiation reaches a barrier, leaves the admissible real family or cannot price its unbounded insertions. Test at most two mathematically distinct constructions before changing this research decision. Bad-component dressing and boundary-source variation still require their own bounds after this anchored lemma.

31. All-order anchored attachments at the exact Gaussian endpoint

Under the additional exact-quadratic and unbarriered-measure assumptions below, the order-uniform anchored target of §30.3 holds with explicit constants (141) and the stronger sum (142). A volume-uniform complex source polydisc is also zero-free. This is a conditional reference construction, rather than a proof for all measures (125). Its mechanism is normalization of the marginal on H before taking absolute coefficients: outside contributions disconnected from H cancel exactly.

31.1 Assumptions and constants

Take $J = \varnothing$ and assume the smooth potential is exactly $S(y) = y^{\mathsf{T}} A y/(2t)$, $y = \xi - s$, up to an irrelevant constant. The real symmetric matrix A has three-dimensional coordinate blocks, range one, $\|A - 6I\| \le 5/2$, and fixed block row and column sums at most $17/2$. These hold at the covariant Gaussian endpoint and its quadratic real decouplings: the last bound follows from §20.1's majorant about $5I$. Retain the existing small-time assumptions giving $\Pr(B_H) \le p^{h/16}$, $p \le 1/2$, for $h = |H| \ge 1$. The observable F_H is measurable on y_H and obeys $|F_H| \le 1_{B_H}$, as in §25.2. No derivative of it is needed.

There is one fixed connector matrix per face: $K_p = y^{\mathsf{T}} B_p y/(2t)$, where B_p is real symmetric, supported on ∂p and has block row and column sums at most $K_{\mathrm{loc}} \le 7$. Equation (126) gives precisely this form when the parent potential is quadratic; constant and linear terms cancel. Additional independently sourced connector types would change the incidence constant below. Fix $0 < \eta < \log(12/5)$, put $q = 5/12$, and use C_E from §30.2 to set

$$\begin{aligned} g &= \frac{2}{7} C_E(q e^{\eta}), \qquad a = \frac{17}{2} e^{\eta}, \qquad b = 2K_{\mathrm{loc}} e^{\eta}, \\ D &= 1 + 6ag + 2a^2 g^2, \\ d_0 &= \min\{(256g)^{-1}, 7/1024\}, \qquad \rho_* = \frac{d_0}{bD} > 0. \end{aligned} \tag{141}$$

These constants are independent of h, plane size, heat time and the profile approximation. The face-edge graph of §2 joins two edges when they share a face, so the diameter of a face is one.

Proposition 8 (Gaussian anchored attachment norm). Define $\kappa(F_H) = EF_H$ and the empty-label tree length as zero. With $\ell(H, p_1, \dots, p_n)$ the minimum spanning-tree length on these supports, repetitions allowed,

$$\sum_{n\ge 0} \frac{\rho_*^n}{n!} \sum_{p_1,\dots,p_n} e^{\eta \ell(H,p_1,\dots,p_n)} |\kappa(F_H, K_{p_1}, \dots, K_{p_n})| \le p^{h/64}. \tag{142}$$

In particular this proves the restricted version of §30.3's target with $a_* = 1/64$, $A_* = 1$ and $\eta_* = \eta$.

31.2 A Schur marginal with only rooted contributions

For every principal submatrix A_{OO}, compression preserves the norm bound about $6I$. The Neumann operator estimate gives $\|(A_{OO}^{-1})_{ef}\| \le (2/7)q^{d(e,f)}$, using ambient graph distance. The weighted block row and column sums of this inverse are at most g. Treat each inverse block as a single hop charged its endpoint distance; do not expand it into bare lattice paths and assign this operator bound to individual paths.

For $E(z) = \sum_p z_p B_p$, the summed absolute source coefficients at radius ρ have weighted block row and column sums at most $b\rho$: each edge is in two faces and a face has diameter one. The same weighted sums for A are at most a. Let O be the complement of H. The normalized marginal precision on H is

$$M_z = (A+E)_{HH} - (A+E)_{HO}(A+E)_{OO}^{-1}(A+E)_{OH}.$$

Empty complementary blocks contribute zero. Put $C_H = (A^{-1})_{HH} = M_0^{-1}$ and $\Delta = M_z - M_0$. Expand the outside inverse in E_{OO} and cancel the source-free term **before** taking absolute coefficients. The remaining weighted row and column majorant is

$$d(\rho) = b\rho + g\left[\frac{(a+b\rho)^2}{1-gb\rho} - a^2\right]. \tag{143}$$

The first term is E_{HH}. Every other term contains at least one source insertion on a chain with both ends in H. For $v = b\rho \le a$ and $gv \le 1/2$,

$$\frac{d(\rho)}{v} = 1 + g\frac{2a + v + a^2 g}{1-gv} \le D.$$

The choice (141) satisfies both inequalities and gives $d(\rho_*) \le d_0$.

Each such chain visits every source face appearing in its coefficient. For geometric accounting, join the endpoints of an inverse hop by a shortest path. Collapsing each visited face to its support and the whole set H to the root gives a connecting tour; a spanning tree has no greater total length. Its $e^{\eta\ell}$ weight is therefore bounded by the product of the hop weights already counted in (143). This use of paths concerns support geometry, not a pathwise bound on the inverse operator.

31.3 The normalized density and its absolute coefficients

The exact Gaussian marginal density ratio, with its branch equal to one at zero, is

$$R_z(y_H) = \det(I + C_H\Delta)^{1/2} e^{-y_H^{\mathsf{T}}\Delta y_H/(2t)}.$$

Its logarithm is

$$\log R_z = \frac{1}{2}\sum_{j\ge 1}\frac{(-1)^{j+1}}{j}\operatorname{Tr}(C_H\Delta)^j - \frac{1}{2t}y_H^{\mathsf{T}}\Delta y_H.$$

The trace costs $3h$, the number of scalar coordinates in H; each C_H contraction joins root coordinates. The symmetric quadratic majorant uses $2|y_e||y_f| \le |y_e|^2 + |y_f|^2$. Consequently its absolute diagram sum, including source powers at radius ρ_* and all spatial weights, is at most

$$L(y_H) = \frac{3h}{2}\log\frac{1}{1-gd_0} + \frac{d_0}{2t}\sum_{e\in H}|y_e|^2. \tag{144}$$

Every trace or quadratic term is attached to H. Products in the exponential stay attached to that same root; the length of a tree for their union is at most the sum of their individual connecting tour lengths. Thus the weighted absolute Taylor sum of R_z is bounded pointwise by $e^{L(y_H)}$.

Integrate against F_H in the original Gaussian marginal. Its covariance satisfies $tC_H \preceq (2t/7)I$. Cauchy–Schwarz against 1_{B_H} and the Gaussian quadratic exponential formula give

$$E[1_{B_H} e^{L(y_H)}] \le p^{h/32} \left[(1-gd_0)^{-3/2}(1-4d_0/7)^{-3/4}\right]^h. \tag{145}$$

For the exponential formula, diagonalize C_H and integrate each scalar Gaussian:

$$E \exp[(d_0/t)|y_H|^2] \le (1-4d_0/7)^{-3h/2}.$$

Both quantities subtracted from one in (145) are at most $1/256$. Since $-\log(1-x) \le x/(1-x)$,

$$\log\left[(1-gd_0)^{-3/2}(1-4d_0/7)^{-3/4}\right] \le \frac{3}{340} < \frac{\log 2}{64} \le \frac{\log(1/p)}{64}.$$

The strict comparison follows, for example, from $\log 2 \ge 2/3$. Equation (145) is therefore at most $p^{h/64}$.

Finally $G_H(z) = E[F_H R_z]$ has derivatives $\partial_{p_1} \cdots \partial_{p_n} G_H(0) = (-1)^n \kappa(F_H, K_{p_1}, \dots, K_{p_n})$. For multiplicities α_p, the ordered-label sum divided by $n!$ gives exactly $1/\prod_p \alpha_p!$, its Taylor denominator. Repeated supports have zero connecting distance. The absolute majorant just proved therefore yields (142), including the zero-order term. Gaussian integrability in (145) justifies termwise integration. No factor depending on response order or plane size has been hidden.

Throughout $\sup_p |z_p| \le \rho_*$, $\operatorname{Re} A_z \succeq (7/2 - 2K_{\mathrm{loc}}\rho_*)I \succ 0$. The Gaussian partition function has its analytic nonvanishing determinant branch on this volume-uniform complex polydisc. Equation (142) supplies more than zero-freeness: it retains an anchored spatial reserve and bad-set rarity at every order. The matrices, source radius and weighted norm are dimensionless after the stated t scaling.

31.4 The next nonlinear test

The result uses the exact Gaussian marginal formula. A retained barrier or nonlinear smooth interaction changes that marginal and its conditional outside partition function; real convexity and moments alone do not supply (144). The next bounded gauge lemma is an analogue of the **rooted marginal logarithm estimate** (144) with those terms retained inside the measure. It must cancel all source-free outside contributions and attach every surviving source to H, before absolute summation. This identifies a concrete estimate to derive, rather than a premise that can be silently added. Test conditional block integration first, then an independent-attachment forest construction if that fails; stop after two distinct failures to improve this marginal norm. Reject barrier differentiation or an attachment rate tending to zero.

Equation (142) proves neither the full-family version of §30.3 nor the multi-bad-component estimate (116). Bad-component dressing, boundary-source variations and comparison to the original group integral remain subsequent steps. A Gaussian endpoint polydisc is not the boundary-source polydisc of (113).

32. A retained barrier: pointwise logarithms and a direct density norm

A compact radial outside barrier can make the conditional marginal vanish at complex sources arbitrarily near zero somewhere on an unbounded root. The pointwise analytic logarithm used in (144) therefore need not survive a retained barrier. The direct density norm (148) does survive, with fixed source radius, bad-set rarity and no barrier derivatives. This conditional integration test changes the next forest's target; the full nonlinear plane still requires rooted normalization.

32.1 Conditional integration exposes the numerator zero

Take two three-dimensional blocks x, y, with y the unbarriered root, and set

$$S_0(x,y) = 3(|x|^2 + |y|^2)/t, \qquad K = -x \cdot y/t, \qquad t > 0.$$

Put the hard radial barrier $|x| \le R$, $0 < R < \infty$, on the outside block. Its extended-valued penalty is convex. The parent $S_0 + K$ has scaled Hessian $\left(\begin{smallmatrix} 6I & -I \\ -I & 6I \end{smallmatrix}\right)$, with eigenvalues five and seven. Its norm difference from $6I$ is one, and its block row bound is seven. Freezing removes precisely K, as in (126); the real segment $S_0 + uK$, $0 \le u \le 1$, preserves these convexity bounds. This model tests the analytical Hessian, moment and barrier inputs. It is an additional test profile, rather than the entire $SU(2)$ plane or its particular w_η.

Let E_x denote the truncated density proportional to $e^{-3|x|^2/t} 1_{|x| \le R}$, independently of $y \sim N(0, tI/6)$. With the source convention e^{-zK}, the exact conditional integral and its normalization are

$$\begin{aligned} M_z(y) &= E_x e^{zx \cdot y/t}, \\ Z(z) = E_y M_z(y) &= E_x e^{z^2|x|^2/(12t)}, \qquad R_z(y) = M_z(y)/Z(z). \end{aligned} \tag{146}$$

Write the unnormalized radial Fourier transform as

$$\phi(k) = \int_{|x| \le R} e^{-3|x|^2/t} e^{ikx_1}\, dx = \frac{4\pi}{k} \int_0^R r e^{-3r^2/t} \sin(kr)\, dr.$$

Twice integrating by parts gives, for fixed R, t,

$$\phi(k) = -\frac{4\pi R e^{-3R^2/t}}{k^2} \cos(kR) + O_{R,t}(k^{-3}).$$

The first integration gives the endpoint cosine; integrating the remaining smooth derivative once more bounds its contribution by $O(k^{-3})$. At $k_n = n\pi/R$ the signs alternate for all large n, because the endpoint coefficient is positive. Continuity gives a finite positive zero k_*. For any $a > 0$, choose $z = ia$ and $y = (tk_*/a, 0, 0)$. Then

$$M_{ia}(y) = \phi(k_*)/\phi(0) = 0, \qquad Z(ia) = E_x e^{-a^2|x|^2/(12t)} > 0. \tag{147}$$

Every positive source disc thus contains a zero of the normalized marginal at some root coordinate. Each fixed root has a neighborhood of zero where its logarithm exists; (147) excludes a common neighborhood over the unbounded root. It defeats a general pointwise analogue of (144), while leaving the integrated anchored target (142) open.

The endpoint term is nonzero at every finite R, even when $R/\sqrt{t}$ is large. At $R = \infty$ the Gaussian Fourier transform is positive and this obstruction disappears. At fixed R, t, it also persists for sufficiently close radial smooth convex barrier approximations. For example, radially mollify $k(|x| - R)_+^2/2$ and let $k \to \infty$. Normalized densities converge in L^1 to the hard-ball density, so their Fourier transforms converge uniformly. Radial symmetry keeps them real: two fixed opposite-sign frequencies, and hence an intervening zero, survive. This proves persistence for that approximation, rather than for every soft barrier or the fixed profile w_η.

32.2 An all-order density norm with the barrier retained

For $f(z) = \sum_n f_n z^n$ set $\|f\|_\rho = \sum_n |f_n| \rho^n$; this coefficient norm is submultiplicative. Assume a measurable root observable obeys $|F_H(y)| \le 1_B(y)$ and $\Pr_y(B) \le p^{1/16}$, $0 < p \le 1/2$. This is the additional one-root rarity hypothesis; no observable derivative is used.

Proposition 9 (retained-barrier reference density norm). At $\rho = 1/16$, uniformly in R and $t > 0$,

$$\sum_{n \ge 0} \frac{\rho^n}{n!} |\kappa(F_H, \underbrace{K, \dots, K}_{n \text{ labels}})| \le p^{1/64}. \tag{148}$$

The cumulants use the barriered product endpoint, and the zero-label term is $|EF_H|$. The same bound holds for a soft radial barrier with nondecreasing radial penalty. The partition function is zero-free on $|z| \le \rho$.

Proof. The radius and uniform direction of $G \sim N(0, tI/6)$ are independent. Truncation decreases every increasing radial moment, so $E_x(x \cdot y)^{2n} \le E(G \cdot y)^{2n}$. Odd moments vanish by symmetry. Consequently, without a factor two,

$$\|M_\bullet(y)\|_\rho = M_\rho(y) \le e^{\rho^2|y|^2/(12t)}.$$

For a soft penalty, the reweighting factor $e^{-w(r)}$ is nonincreasing. Its covariance with an increasing function of the Gaussian radius is nonpositive: use two independent radii and expand the product of their differences. Division by the mean weight proves the same radial moment domination.

Z in (146) has nonnegative even Taylor coefficients. Radial domination and the Gaussian quadratic integral give

$$\|Z - 1\|_\rho \le q(\rho) = (1 - \rho^2/36)^{-3/2} - 1.$$

For $q < 1$, the geometric inverse series yields

$$\|Z^{-1}\|_\rho \le (1-q)^{-1}, \qquad \|R_\bullet(y)\|_\rho \le (1-q)^{-1} e^{\rho^2|y|^2/(12t)}.$$

Also $|Z(z)| \ge 1 - q > 0$ on the disc. Coefficientwise integration and Cauchy–Schwarz against 1_B give

$$\|E_y[F_H R_\bullet]\|_\rho \le p^{1/32} \frac{(1 - \rho^2/18)^{-3/4}}{1 - q(\rho)}. \tag{149}$$

The finite dominating quadratic exponential justifies the interchange of coefficient summation and integration.

At $\rho = 1/16$, put $u = 1/9216$. The derivative of $(1-u)^{-3/2}$ is at most two for $0 \le u \le 1/16$; hence $q \le 2u = 1/4608$. The prefactor in (149) is at most $(1 - 1/4608)^{-7/4} \le (1 - 1/4608)^{-2}$. Its logarithm is at most $2/4607 < 1/96 \le \log 2/64$, using $-\log(1-v) \le v/(1-v)$ and $\log 2 \ge 2/3$. Thus that prefactor is at most $2^{1/64} \le p^{-1/64}$. The derivatives of $E_y[F_H R_z]$ are $(-1)^n \kappa(F_H, K, \dots, K)$, which proves (148). □

This tensors to any finite number of independent pairs with independent connector sources. For an observable on h root blocks bounded by a joint bad event of probability at most $p^{h/16}$, the product density majorant and one Cauchy–Schwarz step give $p^{h/32}$ times the prefactor in (149) to power h, hence $p^{h/64}$. Sources on pairs disjoint from the root cancel exactly in the normalized expectation. Ordered-label sums divided by $n!$ give the multivariate Taylor denominators, as in §31.3. All surviving connectors already touch the root, so their tree length is zero. The tensor reference therefore has an all-order anchored density norm and a zero-free source polydisc independent of its number of pairs; this construction prices no chains between pairs.

The variables are dimensionless chart coordinates; t is heat time, and K, z, ρ and the constants are dimensionless. The density bound survives both radial barrier approximation and $R \to \infty$. The numerator-zero statement has the more restricted limit order specified in §32.1.

32.3 The next forest must normalize before taking absolute coefficients

Conditional integration has given a precise failure of the pointwise logarithm route and a useful retained-barrier density construction. The next distinct attempt is an interaction forest on (125), targeting the normalized marginal density after cancellation of every outside component disconnected from H. Available inputs are (148), the full-family moments (131), and convexity along frozen partitions.

Taking a separate absolute norm of the global inverse partition function introduces a factor for each outside component, including disconnected ones. The needed lemma instead attaches

every surviving source support to H before summation, with a fixed spatial reserve and a geometric cost per attachment. Marginal numerator zeros are compatible with this density norm. Stop this second construction if it retains a volume cost, loses the fixed reserve with order, or differentiates a barrier. Dressing (112), boundary sources (113) and the multi-component target (116) continue to depend on that rooted forest lemma.

33. Retained barriers and nonlinear connectors in a weak recoupling strip

The interaction expansion converges in a volume-uniform source polydisc around the singleton product endpoint of (125). It retains every on-site barrier and the nonlinear face interactions. Connected support has a fixed exponential reserve. This closes a weak recoupling case; the rarity-weighted marginal density estimate (142) and recoupling to the physical value one remain separate obligations.

33.1 The singleton subtraction and its quadratic majorant

Use the fixed-chart family of §28.1, with smooth saddle s and $y = \xi - s$. The plane has periods at least three, so each face has four distinct edges, each edge belongs to two faces, and the face graph under shared-edge adjacency has degree at most four. Take the singleton edge partition, keep all the on-site barriers, and denote its product probability by E_0. Assume the local moment bound (125) and the parent face Hessian row bound (127), $K = K_{\mathrm{loc}} \le 7$, throughout the fixed-chart extension. These are the inputs from the real family; no complex convexity or barrier derivative is assumed.

For each face define the singleton connector

$$K_p(y) = \phi_p(s+y) - \phi_p(s) - \sum_{e \in \partial p} [\phi_p(s + P_e y) - \phi_p(s)]. \tag{150}$$

Here every argument varies only the coordinates on that face. Summing (150) gives exactly $S_{\mathrm{parent}} - S_{\mathrm{singleton}} = \sum_p K_p$. Every one-edge term cancels. The constant and linear Taylor terms also cancel face by face; the individual faces need not be critical at s. Taylor's integral remainder and the symmetric Hessian row and column bound give

$$|K_p(y)| \le K \sum_{e \in \partial p} \zeta_e, \qquad \zeta_e = |y_e|^2/t. \tag{151}$$

Indeed the full-face remainder is at most $(K/2t)\sum_e |y_e|^2$, and the sum of the singleton remainders has the same bound. Their difference costs both. The sharper two-block rectangle coefficient in (130) cannot be used for this simultaneous singleton subtraction.

Put $C = 4\mathrm{e}/3$. Equation (125) and the factorial estimate used in (131) imply, for integers $n \ge 0$,

$$E_0 \zeta_e^n \le n! C^n, \qquad E_0[\zeta_e^d e^{\beta \zeta_e}] \le C^d d! (1 - C\beta)^{-d-1} \quad (C\beta < 1). \tag{152}$$

The second estimate follows by expanding the exponential and using $\sum_{j \ge 0} (d+j)! x^j / j! = d!(1-x)^{-d-1}$. Independence in the following step is supplied by the singleton reference, rather than assumed for the interacting family.

33.2 An exact polymer gas with an absolute coefficient bound

Give each face an independent complex source u_p and put

$$f_p = e^{-u_p K_p} - 1, \qquad Z(u) = E_0 \prod_p (1 + f_p), \qquad w_\Gamma(u) = E_0 \prod_{p \in \Gamma} f_p.$$

For a multivariate series use the sum of absolute Taylor coefficients weighted by $\rho^{|\alpha|}$. This norm is submultiplicative and $\|f_p\|_\rho \le \rho|K_p|e^{\rho|K_p|}$. Let Γ be a set of distinct faces, $m = |\Gamma|$, and set

$$\beta = 2\rho K, \qquad D = (1 - 2C\rho K)^{-1}.$$

The factor two counts the faces incident to an edge. Equations (151)–(152) give

$$\|w_\Gamma\|_\rho \le (8\rho KCD^5)^m \le (256KC\rho)^m \quad \text{if } \rho \le (4KC)^{-1}. \tag{153}$$

To see every factor, before integration the majorant is

$$(\rho K)^m \prod_{p\in\Gamma} \left(\sum_{e\in\partial p} \zeta_e \right) \exp\left[2\rho K \sum_{e\in \operatorname{supp}\Gamma} \zeta_e \right].$$

Expanding the product gives at most 4^m assignments. Their selected edge multiplicities satisfy $d_e \le 2$, $\sum_e d_e = m$, $|\operatorname{supp}\Gamma| \le 4m$, and $\prod_e d_e! \le 2^m$. Product integration and (152) therefore cost at most $C^m 2^m D^{m+4m}$. If $\rho \le (4KC)^{-1}$, then $D \le 2$, which proves the last bound in (153). Repeated powers of a source are already included in its Mayer factor; no order-dependent response factorial is suppressed.

The connected components of a face set have disjoint edge supports. Their expectations factor exactly under E_0. Thus Z is the abstract polymer partition function whose polymers are nonempty connected face sets, whose activities are w_Γ, and whose incompatibility means overlapping edge supports. This representation keeps the nonlinear connectors and barriers in their original positions.

33.3 Uniform convergence with a fixed support reserve

Fix $\eta > 0$ and define

$$\rho_0 = [256KC\, e^{1+\eta}(10 + 8\mathrm{e})]^{-1} > 0. \tag{154}$$

For a face graph of degree at most four, the usual rooted connected-set count is at most $(4\mathrm{e})^{n-1}$ for size n. The closed face neighborhood of an m-face polymer has at most $5m$ members. Consequently there are at most $5m(4\mathrm{e})^{n-1}$ incompatible n-face polymers. Writing $s_\rho = 256KC\rho e^{1+\eta}$, (153) gives

$$\sum_{\Gamma' \not\sim \Gamma} \|w_{\Gamma'}\|_\rho e^{(1+\eta)|\Gamma'|} \le \frac{5ms_\rho}{1 - 4\mathrm{e}s_\rho} \le \frac{5m}{10 + 4\mathrm{e}} < \frac{m}{2} \quad (\rho \le \rho_0). \tag{155}$$

Also $\rho_0 < (4KC)^{-1}$, as required by (153).

Apply the Kotecký–Preiss theorem with size function $a(\Gamma) = |\Gamma|$ to the absolute coefficient majorants, retaining the weight $e^{\eta|\Gamma|}$ in the activities. The required hypothesis is the sum in (155) bounded by $a(\Gamma)$. The theorem supplies absolute convergence of the connected expansion, local cluster bounds uniform in plane size, and $Z(u) \neq 0$ throughout $\sup_p |u_p| \le \rho_0$. The reference is Kotecký–Preiss 1986 [@KoteckyPreiss1986] (publisher metadata and abstract read; the convergence criterion is recalled explicitly here and in the Wilson note, §2). The theorem's scalar positive majorants dominate the Taylor series coefficientwise, so the same convergence applies to the stated coefficient norm. Its proof is borrowed, rather than re-audited.

A connected cluster's face union is connected. Its geometric distance is at most its total face count, so η supplies a fixed geometric reserve as well as a support-size reserve. The radius and rooted local sums are uniform; the total logarithm of the partition function remains extensive. Disconnected outside components cancel in the connected expansion before absolute summation. A marked bad observable still requires its own integrated marginal estimate.

For real $0 \le u \le \rho_0/2$ and $u_p = u + z_p$ with $\sup_p |z_p| \le \rho_0/2$, the same polydisc supplies a complex neighborhood of the real weak recoupling strip. Its real base action is the convex combination of the parent and singleton actions in (125). All constants are independent of heat time and barrier approximation whenever (125), (127) hold uniformly. They are dimensionless after the t scaling. No $SU(3)$ group estimate has been transferred from this $SU(2)$ chart construction.

34. A marked bad observable in the retained-barrier weak strip

The normalized expansion of §33 admits one marked bad observable with the all-order rarity and spatial norm (157). The root is normalized before absolute summation: outside components disconnected from it cancel. Every barrier remains undifferentiated. This proves the weak-strip version of (142), rather than recoupling to one or the connected bound for several distinct bad components.

34.1 Assumptions and the marked norm

Retain exactly the singleton reference, nonlinear connectors, incidence counts and uniform moment assumptions of §33. Let H be a nonempty edge set, $h = |H|$, and let F_H be measurable on y_H, with $|F_H| \le 1_{B_H}$. Equation (125) supplies $P_0(B_H) \le p^{h/16}$, $0 < p \le 1/2$. This joint rarity is an input already established in the real frozen-block family; it is not an independence assumption about the individual bad edges.

With $K = K_{\mathrm{loc}} > 0$, $C = 4\mathrm{e}/3$, $\eta > 0$ and ρ_0 from (154), set

$$\rho_1 = \rho_0/16, \qquad r_* = \rho_1/2, \qquad \eta_* = \eta/2. \tag{156}$$

Let E_u be the normalized measure proportional to $e^{-u\sum_p K_p} dP_0$, for a **common scalar real** $0 \le u \le r_*$. This base action is in (125). Independent complex sources below are used for analyticity; no convexity claim is made for arbitrary nonuniform real face coefficients.

The tree length $\ell(H, p_1, \dots, p_n)$ uses the face-edge metric of §31.1 and treats the entire support H as **one root vertex**. If H is disconnected, all its pieces are contracted to that root. Different marked bad components are not contracted to a common root in their future connected-correlation problem. Repeated face labels are allowed, and the empty-label length is zero.

Proposition 10 (marked weak-strip attachment norm). Uniformly in plane size, barrier approximation and the admitted small heat times,

$$\sum_{n\ge 0} \frac{r_*^n}{n!} \sum_{p_1,\dots,p_n} e^{\eta_* \ell(H,p_1,\dots,p_n)} |\kappa_u(F_H, K_{p_1}, \dots, K_{p_n})| \le p^{h/64}. \tag{157}$$

The zero-order term is $|E_u F_H|$. The source polydisc is zero-free by §33. No derivative of F_H or of a barrier occurs in this result.

34.2 One Cauchy step prices the marked integral

For any set Γ of distinct faces, connected or not, put $m = |\Gamma|$ and $r_{H,\Gamma} = E_0[F_H \prod_{p\in\Gamma} f_p]$. Use the absolute coefficient norm of §33, now at $\rho \le (8KC)^{-1}$. The pointwise Mayer majorant, (151), and Cauchy–Schwarz give the rarity factor $P_0(B_H)^{1/2} \le p^{h/32}$.

Expand the product of the four-term edge sums as in §33.2. There are at most 4^m assignments, with $d_e \le 2$, $\sum_e d_e = m$ and at most $4m$ edges. Squaring the integrand gives an exponential $e^{4\rho K \sum_e \zeta_e}$. Write $D_2 = (1 - 4C\rho K)^{-1} \le 2$. Equation (152) and product integration give

$$\begin{aligned} \|r_{H,\Gamma}\|_\rho &\le p^{h/32} (4\rho KC\, 24^{1/4} D_2^3)^m \\ &\le p^{h/32} a_\rho^m, \qquad a_\rho = 256 KC\rho. \end{aligned} \tag{158}$$

Indeed $\prod_e (2d_e)! \le 24^{m/2}$, and the square root of the moment bound costs $C^m 24^{m/4} D_2^{(2m+4m)/2}$. The last inequality uses $4 \cdot 24^{1/4} \cdot 8 < 256$. For $\Gamma = \varnothing$, $|E_0 F_H| \le p^{h/16} \le p^{h/32}$ gives the same bound. This step spends rarity only once, independently of connector order.

34.3 The exact root quotient cancels the outside volume

Call a face set Γ a root configuration if every one of its face-connected components has edge support meeting H; the empty set is allowed. Set $S_\Gamma = H \cup \operatorname{supp}\Gamma$. Let $Z_{\mathrm{avoid}}(S)$ be the ordinary polymer partition function with only polymers whose edge supports avoid S. Expansion of the finite numerator gives exactly

$$\begin{aligned} G_H(v) &= \frac{E_0[F_H e^{-\sum_p v_p K_p}]}{Z(v)} \\ &= \sum_{\Gamma \text{ root}} r_{H,\Gamma}(v) \frac{Z_{\mathrm{avoid}}(S_\Gamma; v)}{Z(v)}. \end{aligned} \tag{159}$$

To check the identity, collect all selected face components touching H into Γ. Every remaining component avoids both H and $\operatorname{supp}\Gamma$, and is independent of the root integral under E_0. Conversely each allowed outside configuration reconstructs one term of the numerator. This uses no pointwise logarithm of a conditional marginal, so §32's numerator zeros present no obstruction.

In the logarithm of the quotient in (159), clusters whose polymers all avoid S_Γ cancel **before** absolute coefficients are taken. Use a diagram norm that weights every polymer face occurrence by e^η, as in §33.3. Occurrences count with multiplicity, including all polymers in all quotient clusters; root faces will also be counted. Set

$$s_\rho = a_\rho e^{1+\eta}, \qquad \sigma_\rho = \frac{s_\rho}{1 - 4\mathrm{e} s_\rho}.$$

The Kotecký–Preiss pointed-cluster bound implies

$$\left\| \log \frac{Z_{\mathrm{avoid}}(S_\Gamma)}{Z} \right\|_{\rho,\eta}^{\mathrm{diagram}} \le (2h + 5m)\sigma_\rho. \tag{160}$$

Here at most $2h$ faces touch H, and the closed face neighborhood of Γ has at most $5m$ faces. Any forbidden polymer contains one of these faces. For each specified face the rooted animal count and the pointed bound give $\sum_{n\ge1}(4\mathrm{e})^{n-1}s_\rho^n = \sigma_\rho$. Counting a cluster at each forbidden polymer overcounts it and is therefore a valid absolute upper bound.

The borrowed pointed result is the absolute cluster sum containing a specified polymer γ bounded by $\|w_\gamma\|_\rho e^{(1+\eta)|\gamma|}$ under (155). Its precise source is Fernández–Procacci, §2, (2.5)–(2.15) [@FernandezProcacci2007] (passage read, arXiv v2, pages 3–4). Their pinned log-ratio series and positive derivative series give this bound with size function $a(\gamma) = |\gamma|$. We borrow that theorem; the application and incidence constants in (160) are derived here.

Exponentiating (160) costs at most $e^{(2h+5m)\sigma_\rho}$. Every surviving quotient cluster touches S_Γ and is connected in the incompatibility graph, hence is geometrically attached to the marked root configuration. No outside volume factor remains.

34.4 Root summation leaves both rarity and spatial reserve

Multiplying (158), the quotient bound and the root face weight $e^{\eta m}$ gives $p^{h/32}e^{2h\sigma_\rho}b_\rho^m$, where

$$b_\rho = a_\rho e^{\eta+5\sigma_\rho} = s_\rho e^{-1+5\sigma_\rho}.$$

Choose a fixed ordering of the at most $2h$ faces incident to H. Encode each component of Γ by its least such face and its rooted connected animal. Distinct components have distinct chosen faces. Allowing an optional arbitrary animal at each incident face overcounts these encodings, so

$$\sum_{\Gamma \text{ root}} b_\rho^{|\Gamma|} \le \left(1 + \frac{b_\rho}{1 - 4\mathrm{e} b_\rho}\right)^{2h}.$$

At $\rho = \rho_1$, the constants in (154), (156) give

$$4\mathrm{e}s_{\rho_1} < \frac{1}{32}, \qquad \sigma_{\rho_1} \le \frac{2}{31(10+8\mathrm{e})} < \frac{1}{403}.$$

Thus $5\sigma_{\rho_1} < 1$, $b_{\rho_1} \le s_{\rho_1}$, and the complete marked diagram sum obeys

$$\begin{aligned} \|G_H\|^{\mathrm{diagram}}_{\rho_1,\eta} &\le p^{h/32} e^{4h\sigma_{\rho_1}} \\ &\le p^{h/64}. \end{aligned} \tag{161}$$

The last step uses $4\sigma_{\rho_1} < 4/403 < 1/96 \le \log(1/p)/64$; $\log 2 \ge 2/3$ supplies the final comparison. All bounds apply also to the empty root configuration.

To obtain (157) about the real base, translate every source as $v_p = u + z_p$, with $u \le r_*$ and $|z_p| \le r_*$. Absolute Taylor coefficients after translation are dominated by the original radius $u + r_* \le \rho_1$. Keep the hidden u-faces in their diagrams. After contracting H, the union of root components and quotient clusters is a connected network attaching every retained z-face support to H. If N is its total number of face **occurrences**, a closed traversal has metric length at most $2N$: each face has diameter one, and shared-edge incidences join its pieces. Consequently $\ell \le 2N$ and $e^{\eta_*\ell} \le e^{\eta N}$. The fixed reserve in (161) therefore pays the spatial weight in (157) at every response order.

Finally $\partial_{p_1} \cdots \partial_{p_n} G_H(u) = (-1)^n \kappa_u(F_H, K_{p_1}, \dots, K_{p_n})$. For a multiplicity multi-index α, the ordered-label sum divided by $n!$ is exactly the coefficient denominator $\prod_p \alpha_p!$. This converts (161) into (157), including repeated labels and the zero-order term. Moment domination (152) justifies the expansions and the admitted barrier-approximation limits at fixed volume. Constants are uniform in volume and dimensionless after the t scaling. □

Internal scope check. The root quotient, pointed cluster sum, moment constants and rarity absorption were checked. Contracting H as one support and counting polymer occurrences are essential conventions in the stated spatial bound.

35. Separately marked bad components at arbitrary order

Composite polymers made of ordinary faces and separately marked bad supports give (116) at every order in a narrower weak recoupling strip. The constant per component, rarity exponent and spatial rate are fixed. Compatibility supplies the incidence bound that prevents growth with the number of marks. This result supplies neither $u = 1$ nor the activity sum (112).

35.1 Joint sources and constants

Retain §33's reference and connectors. Fix any finite compatible family $H_1, \dots, H_r$ of nonempty connected edge supports: they are disjoint and no edge in one shares a face with an edge in another, as in §24.1. Put $h_i = |H_i|$, $M_I = \sum_{i \in I} h_i$ and take measurable F_i on y_{H_i} with $|F_i| \le 1_{B_{H_i}}$. Equation (125) supplies joint rarity on every union of these supports. No observable derivative is needed.

Fix $0 < \eta \le \gamma_{\mathrm{bad}} = \log(12/5)$ and set

$$\begin{aligned} \rho_2 &= \rho_0/256, & r_2 &= \rho_2/2, \\ L &= \log(1/p), & \alpha &= L/128, \\ R_i &= p^{-h_i/64}/1024, & d_i &= p^{h_i/256}/1024. \end{aligned} \tag{162}$$

Here ρ_0 is (154). Face sources v_p have radius ρ_2; marked sources t_i have radii R_i. Only the latter depend on component size and rarity. The common real base is $0 \le u \le r_2$. Translation $v_p = u + z_p$, $|z_p| \le r_2$, uses this polydisc without asserting convexity at arbitrary independent real face coefficients.

35.2 An exact composite polymer gas

Expand

$$\mathcal{Z}(v,t) = E_0\left[\prod_p (1+f_p(v_p))\prod_i (1+t_iF_i)\right].$$

The elementary entities are faces with support ∂p and marks with support H_i. Join entities with overlapping coordinate supports. A face has at most four face neighbors and at most **one** marked neighbor, because compatibility forbids two marks on the same face. Mark i has at most $2h_i$ face neighbors and no marked neighbors.

A composite polymer $\gamma = (\Gamma, I)$ is a nonempty connected set of entities. Its activity is

$$w_\gamma(v,t) = E_0\left[\prod_{p\in\Gamma} f_p(v_p)\prod_{i\in I} t_iF_i\right].$$

Different elementary components have disjoint coordinate supports, so their expectations factor under E_0. This is exactly the hard-core gas of composite polymers, incompatible when supports overlap. Each mark occurs once in the finite product; cluster expansions may repeat polymers, with marker multiplicities kept in their source powers.

With $m = |\Gamma|$, $a = 256KC\rho_2$, one Cauchy step on the joint bad event and (158) give

$$\|w_\gamma\|_{\rho_2,R} \le a^m \prod_{i\in I}(R_i p^{h_i/32}). \tag{163}$$

For $I = \varnothing$ use (153); for $m = 0$ use joint rarity directly. Thus the bound covers every polymer without repeatedly spending rarity. All on-site barriers remain inside E_0.

35.3 A weighted animal sum without order loss

Use size function $A(\gamma) = m + \alpha M_I$ and retain $e^{\eta m}$ in the activities. The resulting elementary positive weights are

$$s = ae^{1+\eta} = \frac{1}{256(10+8\mathrm{e})}, \qquad v_i = R_i p^{h_i/32} e^{\alpha h_i} = p^{h_i/128}/1024.$$

Let $\mathcal{A}_x$ sum the product weights over all connected elementary sets containing entity x. Then

$$\mathcal{A}_p \le b = 2s, \qquad \mathcal{A}_i \le d_i. \tag{164}$$

Choose one rooted spanning tree for each connected set through x. In the exponential rooted-tree recursion

$$T_x^{(0)} = v_x, \qquad T_x^{(n+1)} = v_x \exp\left(\sum_{y\sim x} T_y^{(n)}\right),$$

each tree with distinct entity labels contributes its full product weight: child permutations cancel the factorial in the exponential. Repeated labels add positive terms and overcount. A finite supersolution therefore bounds the animal sums. Choose b at faces and d_i at marks. Since $L \ge \log 2 \ge 2/3$,

$$b < 1/3328, \quad 2b < L/256, \quad d_i \le 1/1024,$$

and consequently

$$se^{4b+1/1024} \le 2s = b, \qquad v_i e^{2h_i b} \le p^{h_i/256}/1024 = d_i.$$

The first inequality uses $4b + 1/1024 < \log 2$; the second uses $2b \le L/256$. They prove (164) at every depth and then in the limit. No tree factorial or size factor has been discarded.

An incompatible polymer contains an entity overlapping an original constituent. For a face, these are its at most five closed face neighbors and its at most one marked neighbor, with rooted sum $5b + 1/1024 < 1$. For H_i, they are that mark and at most $2h_i$ incident faces, costing $d_i + 2h_i b \le \alpha h_i$: $d_i \le 1/1024 \le Lh_i/256$ and $2b \le L/256$. Union over constituents gives the KP hypothesis

$$\sum_{\gamma' \not\sim \gamma} \|w_{\gamma'}\|_{\rho_2,R} e^{A(\gamma')+\eta|\Gamma'|} \le m + \alpha M_I = A(\gamma). \tag{165}$$

The KP theorem of §34.3 supplies connected convergence, pointed bounds and zero-freeness on the joint polydisc, uniformly in marks and volume. Its size function depends on L; the face radius and resulting bounds do not deteriorate as p decreases.

35.4 Squarefree extraction and fixed geometric decay

Fix mark i. The absolute cluster sum containing a polymer carrying i is bounded by

$$\sum_{\gamma:\ i\in I} \|w_\gamma\|_{\rho_2,R} e^{A(\gamma)+\eta|\Gamma|} \le \mathcal{A}_i \le d_i.$$

This sums the pointed bound of §34.3 over possible pinned polymers; multiple pinning only overcounts. The coefficient of $t_1 \cdots t_r$ in $\log \mathcal{Z}(v,t) - \log \mathcal{Z}(v,0)$ is exactly $\kappa_v(F_1,\dots,F_r)$. The squarefree coefficients of $\prod_i (1 + t_i F_i)$ give the ordinary moment–cumulant identity. There is no $r!$ for these distinct labeled sources. Extraction costs

$$d_i \prod_j R_j^{-1} \le 1024^r p^{\sum_j h_j/64}.$$

Each contributing cluster connects all the marks geometrically. Let m' count distinct faces in its union, and $N \ge m'$ all face occurrences, including repetitions of polymers. For $r \ge 2$, choose an elementary spanning tree with E_{FF} face–face and E_{FM} face–mark edges. Replace each face node of degree d by a tree on its incident interface edges, costing at most $d-1$ unit steps, since a face has diameter one. Contract each H_i **separately**. The resulting metric network has length at most

$$2E_{FF} + E_{FM} - m' = m' + 2r - 2 - E_{FM} \le m' + r - 2 \le 2m'.$$

Here $E_{FM} \ge r$ and compatibility gives $E_{FM} \le m'$, hence $r \le m'$. Doubling the network and recording successive marked-support visits produces a connected labeled graph of total distance at most $4m'$. Choose a spanning tree inside it. Thus the minimum labeled-tree length obeys $\ell(H_1,\dots,H_r) \le 4N$, without assuming a triangle inequality for distances between sets. A single mark has length zero.

The reserve $e^{\eta N}$ pays $e^{(\eta/4)\ell}$, giving

$$\begin{aligned} |\kappa_u(F_1,\dots,F_r)| &\le 1024^r p^{\sum_i h_i/64} e^{-(\eta/4)\ell(H_1,\dots,H_r)} \\ &\le B_*^r p^{\sum_i h_i/64} \sum_{T\in\mathcal{T}_r} \prod_{ij\in T} e^{-\gamma_* d(H_i,H_j)}, \\ B_* &= \max\{1024, \sqrt{C_{\mathrm{bad}}}\}, \qquad \gamma_* = \eta/4. \end{aligned} \tag{166}$$

A minimizing labeled tree supplies a term of the last sum. These are the constants of (116), with $c_* = 1/64$. The inherited C_{bad} condition is a normalization convention; this construction itself costs only 1024 per mark.

Translation $v_p = u + z_p$, $u \le r_2$, $|z_p| \le r_2$, is dominated coefficientwise by radius ρ_2. Hidden u-faces stay in the network. The same argument bounds the ordered connector-cumulant sum, divided by its response factorial, at radius r_2 with weight $e^{(\eta/4)\ell(H_1,\dots,H_r,p_1,\dots,p_n)}$. Each sourced face can be represented by an interface edge in its support; repeated labels add no distance. This is a connector-source extension, not the moving boundary-background source estimate (113).

Joint moments dominate the barrier limits at fixed volume, as in §§33–34. Constants are dimensionless after the t scaling and uniform in the admitted small heat times. Transferring to $SU(3)$ requires its own analogues of (125) and the face Hessian majorant.

Internal scope check. The composite gas, animal recursion, KP constants and squarefree extraction were accepted. Extending §34's single-root traversal did not justify the proposed $\eta/2$ rate for several roots. The explicit interface-network calculation proves the sufficient fixed rate $\eta/4$ without an additional geometric hypothesis.

36. Exact dressed activities in the weak recoupling strip

The correlated insertion family (109) has an exact dressed hard-core representation in the strip below. Its activities satisfy both numerical parts of (112), with a uniform connector-source norm. Good connecting edges are paid by the face reserve, while bad edges carry rarity. This closes a weak-strip dressing construction; the physical recoupling and background-source obligations persist.

36.1 Reference, additional heat condition and sources

Use the $J = \varnothing$ singleton reference in (125), with **no** on-site barriers already in its density. Their insertion factors are $f_e = e^{-w_k} - 1$, $F_H = \prod_{e\in H} f_e$, as in (109). They remain undifferentiated. Keep §33's nonlinear face connectors and moment assumptions. Let H now range over all nonempty connected edge sets in the degree-six graph; selected bad families must be compatible.

Impose the additional condition $L = \log(1/p) \geq 640$. It holds, for example, if $x = \varepsilon^2/t \geq 2736$: then $n = \lfloor 3x/8 \rfloor - 1 \geq 1025$ and $L = n\log(16/7) - \log 2 \geq 1024\log 2 > 640$. Fix the face reserve $\eta_f = 32$ and put

$$\begin{aligned} \rho_d &= \rho_0(32)/256, & r_d &= \rho_d/2, \\ s &= \frac{1}{256(10+8\mathrm{e})}, & b &= 2s, \\ q &= p^{1/64}, & \zeta &= \frac{q}{1-36q}, \\ \delta_d &= 2b+\zeta, & \delta_d' &= e^{\delta_d}\delta_d. \end{aligned} \tag{167}$$

Here $\rho_0(32)$ is (154) with that reserve. The constants obey

$$q \leq e^{-10} < 1/16384, \quad \zeta < 1/16348, \quad b < 1/3328, \quad \delta_d < 1/1000, \quad \delta_d' < 1/999.$$

The last bound uses $e^x \leq (1-x)^{-1}$ for $0 \leq x < 1$. The common real strip is $0 \leq u \leq r_d$; complex connector sources translate as $v_p = u + z_p$, $|z_p| \leq r_d$. No moving boundary background is sourced in this statement.

36.2 All possible bad supports in an exact mixed gas

The ratio being dressed is exactly

$$\mathcal{R}(v) = \frac{E_0[e^{-\sum_p v_p K_p}\prod_e (1+f_e)]}{E_0[e^{-\sum_p v_p K_p}]}.$$

Decomposing each selected bad-edge set into its connected components turns the numerator into a sum over compatible families $\mathcal{H}$, with factors F_H. Also expand the ordinary face Mayer factors. Group the selected faces and bad supports by coordinate-support overlap. Each resulting composite polymer contains internally compatible bad supports. Its external incompatibilities are coordinate overlap **or** exclusion between bad supports at graph distance at most one. These external exclusions enforce precisely the global compatibility that was not part of the overlap-component decomposition. Product independence of E_0 makes this mixed gas exact.

For m faces and total bad size $M = \sum_{H\in\mathcal{H}} |H|$, joint rarity and (158) give the coefficient majorant

$$\|w_\gamma\|_{\rho_d} \le a^m p^{M/32}, \qquad a = 256KC\rho_d. \tag{168}$$

The joint rarity applies only to internally compatible, hence disjoint, bad families. The tree counts below may ignore this compatibility to overcount; no rarity estimate is applied to overlapping families.

Retain e^{32m+8M} in the activities and use KP size $A = m + M$. The elementary weighted nodes are a face with weight $s = ae^{33}$ and a bad support of size h with weight $v_H = p^{h/32}e^{9h}$. The rooted-tree proof of §35.3 now has the supersolution b at faces and q^h at bad supports. Indeed the number of size-h connected bad sets through an edge is at most 36^{h-1}, by (112)'s traversal count. Therefore the sum of q^h through an edge is at most ζ; through a face it is at most 4ζ. The recursion inequalities are

$$se^{4b+4\zeta} \le 2s = b, \qquad v_H e^{2hb} \le q^h. \tag{169}$$

They use $4b + 4\zeta < \log 2$ and $9 + 2b < L/64$.

A face constituent's incompatibility sum is at most $5b + 4\zeta < 1$. A bad H has at most $2h$ incident faces and at most $7h$ edge sites in its closed neighborhood, so its sum is at most $2hb + 7h\zeta < h$. Union over constituents gives the KP inequality with size $m + M$. This proves uniform mixed-gas convergence with the stated reserves. Marked fugacities of modulus at most one can be included in the same majorant; their physical value one is admitted under this heat condition. This is distinct from the face recoupling coefficient u, whose value one is still outside the strip.

36.3 Connected logarithms retain both good and bad support costs

Subtract the pure-face logarithm from the mixed-gas logarithm. Clusters containing no bad support cancel exactly. Group the remaining clusters by physical edge support C and write

$$\log \mathcal{R}(v) = \sum_C D_C(v).$$

Each C is connected: coordinate overlap connects constituent supports, and a bad-hard-core exclusion joins them at edge-graph distance at most one. If N_f counts all face occurrences and M_b all bad-support edge occurrences, including repeated polymers, then $|C| \le 4N_f + M_b$. Thus $e^{8|C|} \le e^{32N_f+8M_b}$.

Pinning a cluster at a constituent covering an edge e and using the KP pointed bound of §34.3 gives

$$\sup_e \sum_{C\ni e} \|D_C\|_{\rho_d} e^{8|C|} \le 2b + \zeta = \delta_d < 1/1000. \tag{170}$$

There are two faces through e and bad-support rooted sums at most ζ. Multiple pinning overcounts, as required for the absolute bound. Restricting to clusters containing bad supports only decreases it. In particular $\|D_C\|_{\rho_d} \le \delta_d$. The norm retains physical support, including good connecting edges; it has not assigned them bad-edge rarity.

36.4 Exponentiation produces actual hard-core activities

At finite volume there are finitely many connected supports C. Absolute convergence in (170) gives exactly

$$\mathcal{R} = \prod_C (1 + g_C), \qquad g_C = e^{D_C} - 1.$$

In each selected C-family, join supports that overlap or are adjacent in the edge graph, and take the connected unions H. Define

$$z_H = \sum_{\substack{\mathcal{C} \text{ connected} \\ \cup_{C\in\mathcal{C}} C = H}} \prod_{C\in\mathcal{C}} g_C.$$

Each support C is selected at most once. Connected components are unique and their unions are compatible, so this gives the exact hard-core representation

$$\mathcal{R} = \sum_{\mathcal{H} \text{ compatible}} \prod_{H\in\mathcal{H}} z_H. \tag{171}$$

It is not merely a representation of the connected logarithm.

For the norm bound, $\|g_C\| \le \|D_C\| e^{\|D_C\|}$ and (170) imply

$$\sup_e \sum_{C\ni e} \|g_C\|_{\rho_d} e^{7.5|C|} \le e^{\delta_d}\delta_d = \delta'_d.$$

Apply the rooted-tree recursion to support objects C, with base weights $\|g_C\|e^{7|C|}$ and supersolution $T_C = \|g_C\|e^{7.5|C|}$. Its neighboring sums cost at most $7|C|\delta'_d < |C|/2$, since adjacency is covered by the closed edge neighborhood. This proves the supersolution inequality. Root each selected connected family at a member containing e and use $e^{7|\cup C|} \le \prod_C e^{7|C|}$. The animal sum then gives

$$\sup_e \sum_{H\ni e} \|z_H\|_{\rho_d} e^{7|H|} \le \delta'_d < 1/999. \tag{172}$$

No size factor, factorial or volume factor has been suppressed.

Consequently $\|z_H\|_{\rho_d} \le e^{-7|H|}$, and (112) holds in this strip with $\eta = 1$, $b_* = e^{-7}$:

$$\sup_e \sum_{H\ni e} \|z_H\|_{\rho_d} e^{2|H|} < 1/7, \qquad e^2 b_* = e^{-5} < 1/43.$$

Translation to the real base costs half the connector radius, as above. The bounds are uniform in volume, the admitted heat times and barrier approximation. Moment domination and bounded insertions justify the barrier limit at fixed volume. Constants are dimensionless after the t scaling; no group-dependent premise has been transferred to $SU(3)$.

Internal scope check. Both exact gas representations, the new exclusion costs, connected support grouping, weighted tree supersolutions and both numerical parts of (112) were checked. Connector sources must be distinguished from background sources.

37. Moving means with a fixed barrier: an exact quadratic source test

The subtracted log barrier ratio satisfies (113)'s analytic source norm in the exact model below. Imaginary mean displacement is charged only to bad edges; good connecting edges have unchanged Gaussian moments. This supplies a mechanism for moving-background control, with explicit hypotheses that the nonlinear gauge extension has not yet supplied.

37.1 Model, source locality and the fixed integration coordinates

Use §36's plane, barriers, p, $x = \varepsilon^2/t$ and connector strip. Replace its singleton reference by independent three-component Gaussians

$$G_{\mu_e}(\xi_e) = \left(\frac{3}{\pi t}\right)^{3/2} e^{-3(\xi_e-\mu_e)\cdot(\xi_e-\mu_e)/t}, \qquad \mu_e(z) = \mu_e(z_0) + \varepsilon \sum_a M_{ea}(z_a - z_{0a}). \tag{173}$$

The dot product in this analytic density is bilinear. Sources z_a are dimensionless. At every admitted real background z_0 require $|\mu_e(z_0)| \le 6\varepsilon/7$, and fixed real matrices satisfying

$\sup_e \sum_a \|M_{ea}\| \le D_\mu$, $D_\mu \ge 1$. Each edge depends on at most J labels, each label on at most I edges. Source positions are at edge-graph distance at most one from every incident edge; distances use the projected plane, allowing several directions at one position. These are additional locality hypotheses.

Take source-independent real centered quadratic connectors $K_p(\xi-\mu)$ between distinct edges of a face, with $|K_p(y)| \le K\sum_{e\in\partial p}|y_e|^2/t$, $K \le 7$, also for complex y with the Euclidean Hermitian norm. The real parent at common weak coefficient u is $3\sum_e|\xi_e-\mu_e|^2/t + u\sum_p K_p$. Its saddle is exactly μ; its quadratic form is positive in the stated small strip. No nonlinear cutoff or group conclusion is assumed. Keep $f_e(\xi_e) = e^{-w_k(\xi_e)}-1$ fixed in physical ξ_e, with $|f_e| \le 1_{\{|\xi_e|\ge 2\varepsilon-\rho_k\}}$ and $\rho_k \le \varepsilon/7$. Define $\mathcal{R}(v,z)$ by §36.2 using this reference and these connectors. The barrier is never sourced or differentiated.

The background polydisc radius is $R = 1/(7D_\mu)$. On it write $\mu = a+ib$; then $|a| \le \varepsilon$, $|b| \le \varepsilon/7$. In each mixed activity let B be its selected bad edges. Keep their integration contours real, and shift every other contour to $\xi_e = \mu_e + y_e$, $y_e \in \mathbb{R}^3$. The connector factors are entire and their quadratic growth is dominated by the Gaussian at §36's radius, so the contour shifts and their vanishing end integrals are justified. No f_e lies on a shifted contour. Every good edge now has exactly the centered real Gaussian law.

37.2 A source cost only on the bad edges

On a bad edge put $Y = \xi - a$ and $\zeta = |\xi-\mu|^2/t$. Then $|Y| \ge 6\varepsilon/7$, $\zeta = (|Y|^2+|b|^2)/t$, and $|G_\mu| = e^{3|b|^2/t}G_a$. For $x \ge 16$, exponential domination of the bad indicator gives

$$\int_{\text{bad}} |G_\mu| e^{\zeta/2} d\xi \le 2\sqrt{2}\exp[-65x/98] \le e^{-x/2}. \tag{174}$$

Indeed use $1_{\text{bad}} \le e^{|Y|^2/t - 36x/49}$; the remaining centered Gaussian exponential moment at coefficient $3/2$ is $2\sqrt{2}$, while the imaginary displacement costs at most $x/14$. Since $\zeta^n \le n!2^n e^{\zeta/2}$, the bad-edge moments obey $\int_{\text{bad}}|G_\mu|\zeta^n \le p\,n!C^n$, with $C = 4\mathrm{e}/3$: $\log(1/p) \le 3x/8$ gives $e^{-x/2} \le p$. Good moments are at most $n!(1/2)^n \le n!C^n$. Expanding the exponential therefore proves (152) on each contour, with an extra factor p on each bad edge.

For m distinct faces and $M = |B|$, §33.2's same assignments yield

$$\sup_{|z-z_0|_\infty \le R} \|w_{\Gamma,B}(z,\cdot)\|_{\rho_d} \le a^m p^M, \qquad a = 256KC\rho_d. \tag{175}$$

Here the norm is the absolute connector Taylor norm; background sources use an analytic supremum norm. Bad edges outside the face support use zero exponential coefficient and cost only p, avoiding an extra factor per such edge. The coefficient bound remains uniform in k. All background dependence is through means on B: after shifting, good-edge means disappear from the integrand.

37.3 Keep the bad-source support while grouping the logarithm

Retain both connected physical support C and union $B \subseteq C$ of bad edges in the logarithm, writing $\log\mathcal{R} = \sum_{C,B\ne\varnothing} D_{C,B}$. Its source set A_B is the union of labels incident on B, so $|A_B| \le J|B|$. Grouping only by C would enlarge source sets across different bad placements and discard this useful bound. The connected edge support and its distance-one source attachments give a network of length at most $|C|-1+|A_B|$. Doubling this network gives the safe source-MST estimate $\ell(A_B) \le 2|C| + 2J|B|$; it also bounds a minimum connecting-tree convention for ℓ.

Set $\tau = J[2+\log(1+7D_\mu)]$ and strengthen the heat condition to $L = \log(1/p) \ge \max\{640, 4(10+\tau)\}$. Repeat §36's mixed-gas proof, now retaining $e^{32m+(8+\tau)M}p^{-M/2}$ before its KP size $m+M$. The bad elementary weight is $p^{h/2}e^{(9+\tau)h}$. The same supersolution $q^h = p^{h/64}$

works because $(31/64)L \geq 9 + \tau + 2b$; the face supersolution and all incompatibility bounds are unchanged. Pointing at an edge gives

$$\sup_e \sum_{\substack{C \ni e \\ B \neq \varnothing}} \|D_{C,B}\|_{R,\rho_d} e^{8|C|+\tau|B|} p^{-|B|/2} \leq \delta_d < 1/1000. \tag{176}$$

For cluster occurrences $|C| \leq 4N_f + M_b$ and $|B| \leq M_b$, so the retained occurrence weights dominate those in (176). The norm includes both the background supremum and the connector coefficient norm. Holomorphic dominated integrals and this locally uniform expansion justify analyticity at all orders.

37.4 The subtracted barrier part of the source target

For an admitted real reference z_{ref} define $r_{\mathrm{bad}}(z) = -\log \mathcal{R}(v,z) + \log \mathcal{R}(v,z_{\mathrm{ref}})$ and $R_{C,B} = -D_{C,B}(z) + D_{C,B}(z_{\mathrm{ref}})$. Use the uniform rooted bound (176) separately at z and at the reference; the reference need not belong to the particular local polydisc. Root a source at one of its at most I incident bad edges. Since B is nonempty, (176) and the spatial estimate imply

$$\begin{aligned} \sup_a \sum_{(C,B):a\in A_B} e^{\ell(A_B)} \|R_{C,B}\|_{R,\rho_d} (1+1/R)^{|A_B|} &\leq 2I\delta_d p^{1/2} \\ &\leq \frac{2I}{1000} e^{-x/16} \leq A t^{\alpha'}, \\ A = \frac{2I}{1000}\left[\frac{8\alpha'}{\delta \mathrm{e}}\right]^{\alpha'/(2\delta)}, &\qquad \alpha' = 1/2 - 3\delta > 0. \end{aligned} \tag{177}$$

The inequality $p^{1/2} \leq e^{-x/16}$ uses its actual definition: $n \geq 3x/8 - 2$ gives $L \geq x/4 - 7/3 \geq x/8$ for $x \geq 24$. The final bound maximizes $x^{\alpha'/(2\delta)} e^{-x/16}$. Thus (113) holds for this part with $u = 1$, $\eta = 1$ and $q = u^{-1} = 1 < \mathrm{e} - 1$. Translate the connector radius to the real weak base as in §36; the background radius is independent of t, plane size and barrier approximation. Fixed-volume dominated convergence also admits the barrier limit.

Internal scope check. The bad moments, strengthened KP budget, source rooting, subtraction and heat-power conversion were checked. A constructed source network must be doubled for the labeled-terminal MST convention.

38. Consequence for STATE

The centered quadratic test supplies an all-order moving-mean source bound for the subtracted log barrier ratio, with a fixed radius and no amplification on good connecting edges. It does not establish full (113): its Gaussian reference, source-independent coefficients and finite local mean dependence are explicit model premises. Nonlinear transfer requires replacing those premises by a complex centered estimate for the actual nonlinear local factors and a source expansion of the moving saddle, allowing its decaying nonlocal dependence. The real convexity and moment bounds alone do not supply this; the smooth cutoff in §15 cannot itself be shifted holomorphically. Abandon a transfer that differentiates the barrier, charges e^{cx} to every good connector, or assumes contact cancellation. The deterministic and Gaussian subtractions, target-covering paths, physical recoupling, group-integral comparison, $SU(3)$ constants and iteration stay open. New research remains suspended during consolidation.